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## GEOMETRICAL OPTICS

### KEY CONCEPTS

#### INTRODUCTION

The branch of Physics called optics deals with the behavior of light and other electromagnetic waves. Under many circumstances, the wavelength of light is negligible compared with the dimensions of the device as in the case of ordinary mirrors and lenses. A light beam can then be treated as a ray whose propagation is governed by simple geometric rules. The part of optics that deals with such phenomenon is known as geometric optics.

#### PROPAGATION OF LIGHT

Light travels along straight line in a medium or in vacuum. The path of light changes only when there is an object in its path or where the medium changes. We call this rectilinear (straight-line) propagation of light. Light that starts from a point A and passes through another point B in the same medium actually passes through all the points on the straight line AB. Such a straight line path of light is called a ray of light. Light rays start from each point of a source and travel along straight lines till they fall on an object or a surface separating two media (mediums). A bundle of light rays is called a beam of light.

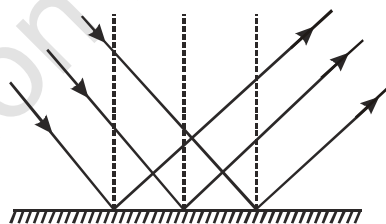
Apart from vacuum and gases, light can travel through some liquids and solids. A medium in which light can travel freely over large distances is called a transparent medium. Water, glycerine, glass and clear plastics are transparent. A medium in which light cannot travel is called opaque. Wood, metals, bricks, etc., are opaque. In materials like oil, light can travel some distance, but its intensity reduces rapidly. Such materials are called translucent.

#### REFLECTION OF LIGHT

When light rays strike the boundary of two media such as air and glass, a part of light is bounced back into the same medium. This is called **Reflection of light**.

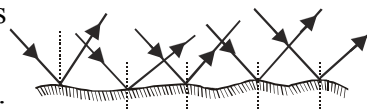
##### (i) Regular / Specular reflection :

When the reflection takes place from a perfect plane surface then after reflection rays remain parallel. It is called **Regular reflection**.



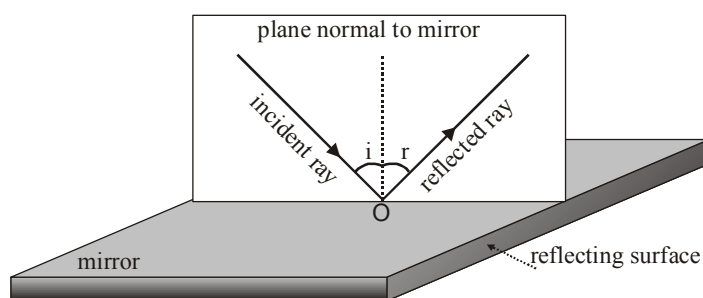
##### (ii) Diffused reflection

When the surface is rough, light is reflected from the surface from bits of its plane surfaces in irregular directions. This is called **diffused reflection**. This process enables us to see an object from any position.

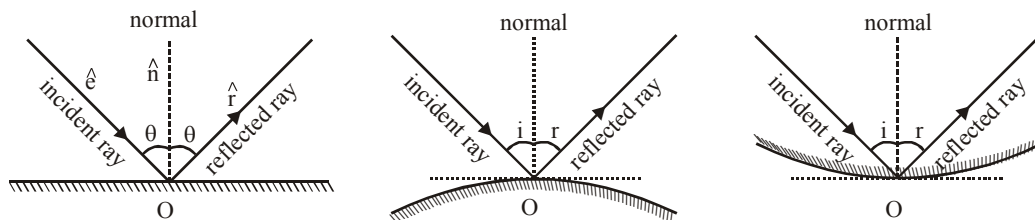


#### LAWS OF REFLECTION

- (1) Incident ray, reflected ray and normal lies in the same plane.



- (2) The angle of reflection is equal to the angle of incident i.e.  $\angle i = \angle r$ .

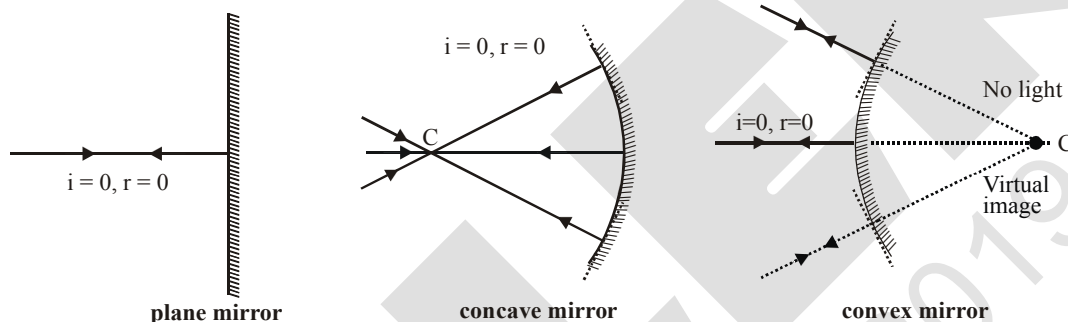


In vector form

$$\hat{r} = \hat{e} - 2(\hat{e} \cdot \hat{n})\hat{n}$$

### KEY POINTS

- **Rectilinear propagation of light** : In a homogeneous transparent medium light travels in straight line.
- When a ray is incident normally on a boundary after reflection it retraces its path.



- The frequency, wavelength and speed does not change on reflection.
- Eye is mostly sensitive for yellow colour and least sensitive for violet and red colour.

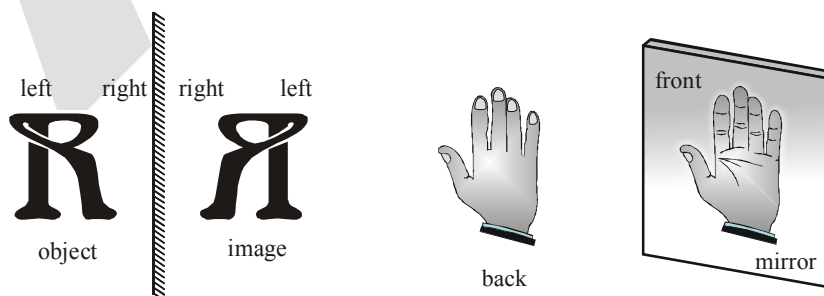
Due to this reason :

- Commercial vehicle's are painted with yellow colour.
- Sodium lamps (yellow colour) are used in road lights.

### REFLECTION FROM PLANE MIRROR

Plane mirror is the perpendicular bisector of the line joining object and image.

- The image formed by a plane mirror suffers **lateral-inversion**, i.e., in the image formed by a plane mirror left is turned into right and vice-versa with respect to object.



When a watch is placed in front of a plane mirror then watch is object and its time is object time and image of watch observed by a person standing in front of mirror then time seen by person.

(i) Object Time =  $A^H$

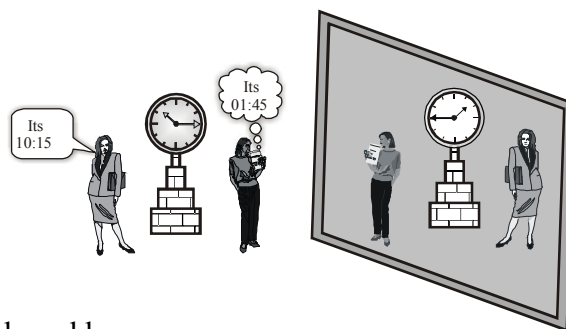
Image Time =  $12 - A^H$

(ii) Object Time =  $A^H B^M$

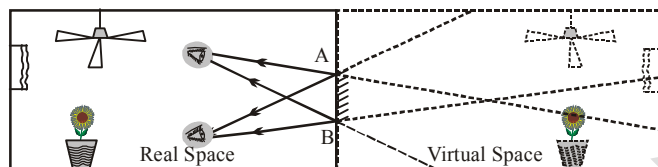
Image Time =  $11 - 60' - A^H B^M$

(iii) Object Time =  $A^H B^M C^S$

Image Time =  $11 - 59' - 60'' - A^H B^M C^S$



- A plane mirror behaves like a window to virtual world.



AB → Plane mirror

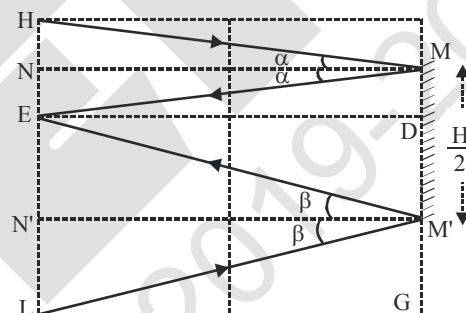
- To see complete image in a plane mirror the minimum length of plane mirror should be half the height of a person.

From figure.  $\triangle HNM$  and  $\triangle ENM$  are congruent

$$\therefore EN = HN \quad \therefore MD = EN = \frac{1}{2} HE$$

Similarly  $\triangle EN' M'$  and  $\triangle LN' M'$  are congruent

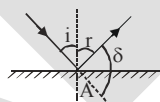
$$\begin{aligned} \text{Length of the mirror } MM' &= MD + M'D = \frac{1}{2} HE + \frac{1}{2} EL \\ &= \frac{1}{2} (HE + EL) = \frac{1}{2} HL \end{aligned}$$



$\therefore$  Minimum of length of mirror is just half of the person.

- This result does not depend on position of eye (height of the eye from ground).
- This result is independent of distance of person in front of mirror.

Deviation for a single mirror



$$\delta = 180 - (i + r); \angle i = \angle r; \delta = 180 - 2i$$

- Total deviation produced by the combination of two plane mirrors which are inclined at an angle  $\theta$  from each other.

$$\delta = \delta_1 + \delta_2 = 180 - 2\alpha + 180 - 2\beta = 360 - 2(\alpha + \beta) \dots (i)$$

$$\text{From } \triangle QAB, \theta + 90 - \alpha + 90 - \beta = 180 \Rightarrow \theta = \alpha + \beta \dots (ii)$$

Putting the value of  $\theta$  in (i) from (ii),  $\delta = 360 - 2\theta$

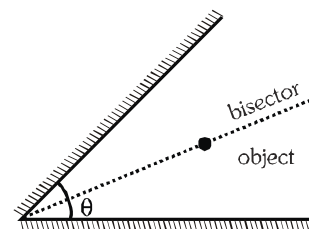
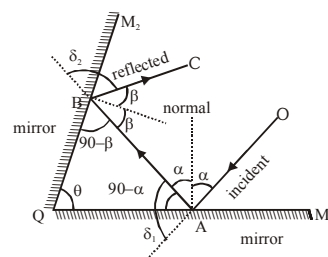
- If there are two plane mirror inclined to each other at an angle  $\theta$  the number of image (n) of a point object formed are determined as follows.

(a) If  $\frac{360^\circ}{\theta} = m$  is even then number of images  $n = m - 1$

(b) If  $\frac{360^\circ}{\theta} = m$  is odd. There will be two case.

(i) When object is not on bisector, then number of images  $n = m$

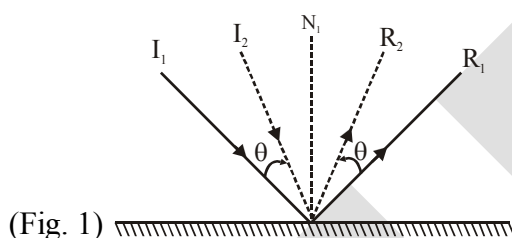
(ii) When object is at bisector, then number of images  $n = m - 1$



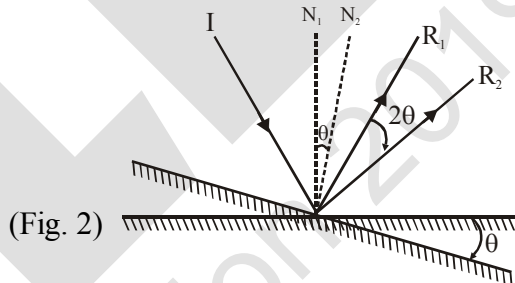
- (c) If  $\frac{360^\circ}{\theta} = m$  is a fraction, and the object is placed symmetrically then no. of images  $n = \text{nearest even integer}$

| S.No. | $\theta$ in degree | $m = \frac{360^\circ}{\theta}$ | Number of images formed if object is placed |               |
|-------|--------------------|--------------------------------|---------------------------------------------|---------------|
|       |                    |                                | asymmetrically                              | symmetrically |
| 1.    | 0                  | $\infty$                       | $\infty$                                    | $\infty$      |
| 2.    | 30                 | 12                             | 11                                          | 11            |
| 3.    | 45                 | 8                              | 7                                           | 7             |
| 4.    | 60                 | 6                              | 5                                           | 5             |
| 5.    | 72                 | 5                              | 5                                           | 4             |
| 6.    | 75                 | 4.8                            | —                                           | 4             |
| 7.    | 90                 | 4                              | 3                                           | 3             |
| 8.    | 112.5              | 3.2                            | —                                           | 4             |
| 9.    | 120                | 3                              | 3                                           | 2             |

- If the object is placed between two plane mirrors then images are formed due to multiple reflections. At each reflection, a part of light energy is absorbed. Therefore, distant images get fainter.
- Keeping the mirror fixed if the incident ray is rotated by some angle, the reflected ray is also rotated by the same angle but in opposite sense. (See Fig. 1)

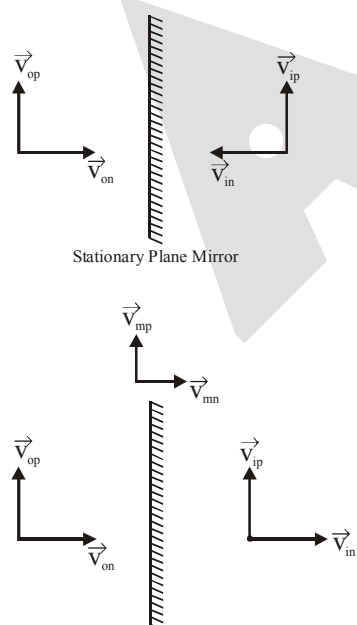


(Fig. 1)



(Fig. 2)

- Keeping the incident ray fixed, if the mirror is rotated by some angle, then the reflected ray rotates by double the angle in the same sense. (See Fig. 2)



Stationary Plane Mirror

$$\vec{v}_{on} = -\vec{v}_{in}, \quad \vec{v}_{op} = \vec{v}_{ip}$$

though speed of object and image are the same

$v_{op}$  = component of velocity of object along parallel to mirror.

$v_{on}$  = component of velocity of object along normal to mirror.

$v_{ip}$  = component of velocity of image along parallel to mirror.

$v_{in}$  = component of velocity of image along normal to mirror.

If mirror is moving  $\vec{v}_{ip} = \vec{v}_{op}$  and  $(\vec{v}_{im})_n = -(\vec{v}_{om})_n$

$$\Rightarrow \vec{v}_{in} - \vec{v}_{mn} = -(\vec{v}_{on} - \vec{v}_{mn}) \Rightarrow \vec{v}_{in} = 2\vec{v}_{mn} - \vec{v}_{on}$$

$\vec{v}_{mn}$  = component of velocity of mirror along normal.

$\vec{v}_{op}$  = component of velocity of object along mirror.

$\vec{v}_{on}$  = component of velocity of object along normal

$\vec{v}_{ip}$  = component of velocity of image along mirror.

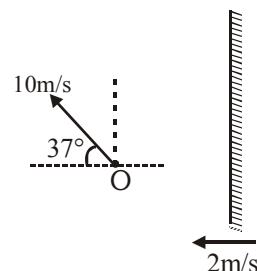
$\vec{v}_{in}$  = component of velocity of image along normal.

**Ex.** Find the velocity of the image.

**Sol.**  $\vec{v}_{ox} = (-10 \cos 37^\circ) \hat{i} = -8\hat{i}$  and

$$\vec{v}_{oy} = (10 \sin 37^\circ) \hat{j} = 6\hat{j}$$

$$\vec{v}_{ix} = 2\vec{v}_{mx} - \vec{v}_{ox} = 2(-2\hat{i}) - (-8\hat{i}) = 4\hat{i} \text{ and } \vec{v}_{iy} = \vec{v}_{oy} = 6\hat{j}$$

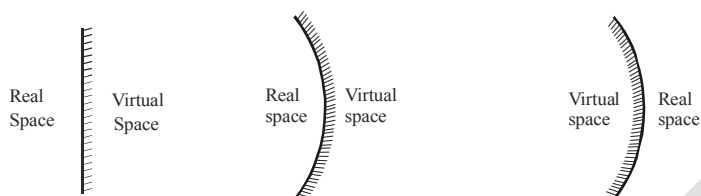


## REAL AND VIRTUAL SPACES

A mirror, plane or spherical divides the space into two ;

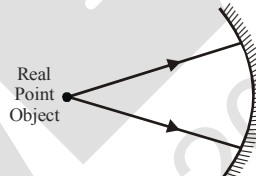
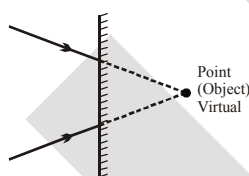
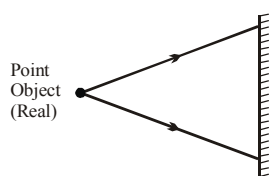
(a) Real space, a side where the reflected rays exist.

(b) Virtual space is on the other side where the reflected rays do not exist.



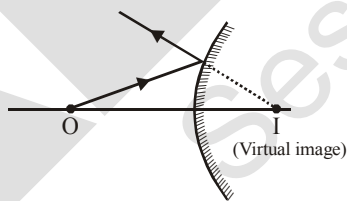
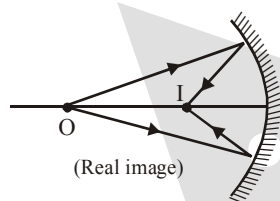
## OBJECT

Object is decided by incident rays only. The point object is that point from which the incident rays actually diverge (Real object) or towards which the incident rays appear to converge (virtual object).



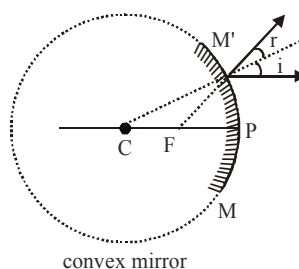
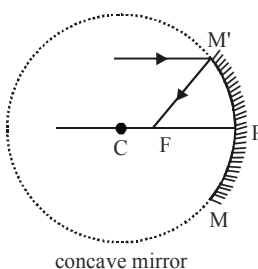
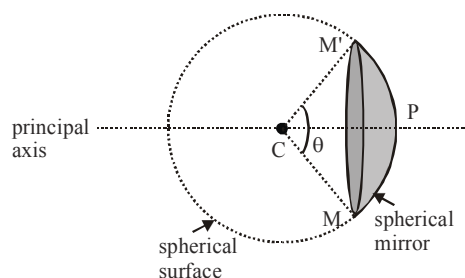
## IMAGE

Image is decided by reflected or refracted rays only. The point image is that point at which the refracted / reflected rays reflected from the mirror, actually converge (real image) or from which the refracted /reflected rays appear to diverge (virtual image).



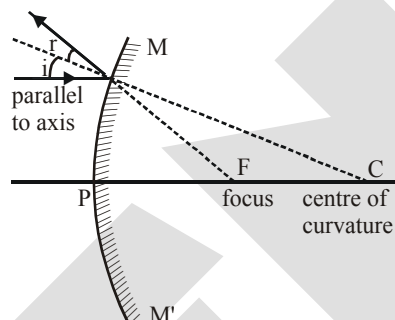
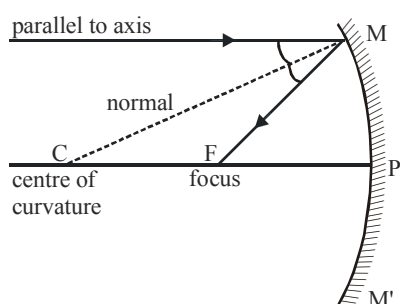
## SPHERICAL (CURVED) MIRROR

Curved mirror is part of a hollow sphere. If reflection takes place from the inner surface then The mirror is called concave and if its outer surface acts as reflector it is convex.

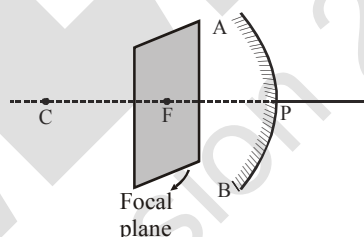
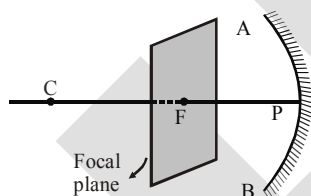


## DEFINITIONS FOR THIN SPHERICAL MIRRORS

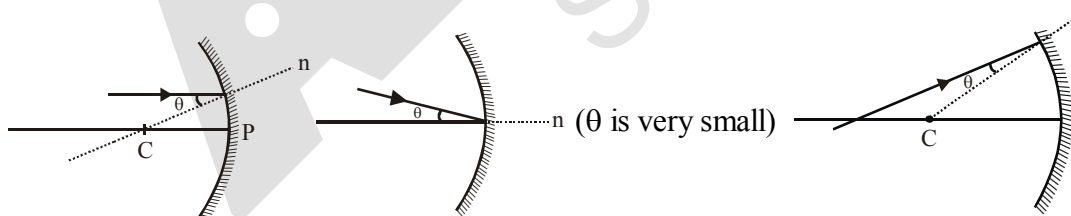
- (i) **Pole** is any point on the reflecting surface of the mirror. For convenience we take it as the midpoint P of the mirror (as shown).
- (ii) **Principal-section** is any section of the mirror such as MM' passing through the pole is called principal-section.
- (iii) **Centre of curvature** is the centre C of the sphere of which the mirror is a part.
- (iv) **Radius of curvature** is the radius R of the sphere of which the mirror is a part.
- (v) **Principal-axis** is the line CP, joining the pole and centre of curvature of the mirror.
- (vi) **Principal-focus** is an image point F on principal axis for which object is at infinity.



- (vii) **Focal-length** is the distance PF between pole P and focus F along principal axis.
- (viii) **Aperture**, in reference to a mirror, means the effective diameter of the light reflecting area of the mirror.
- (ix) **Focal Plane** is the plane passing through focus and perpendicular to principal axis.

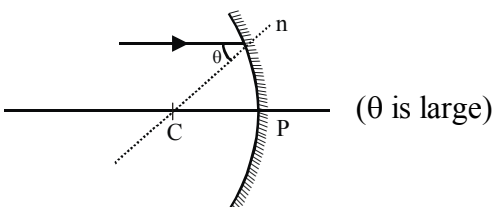


- (x) **Paraxial Rays** Those rays which make small angle with normal at point of incidence and hence are close to principal axis.

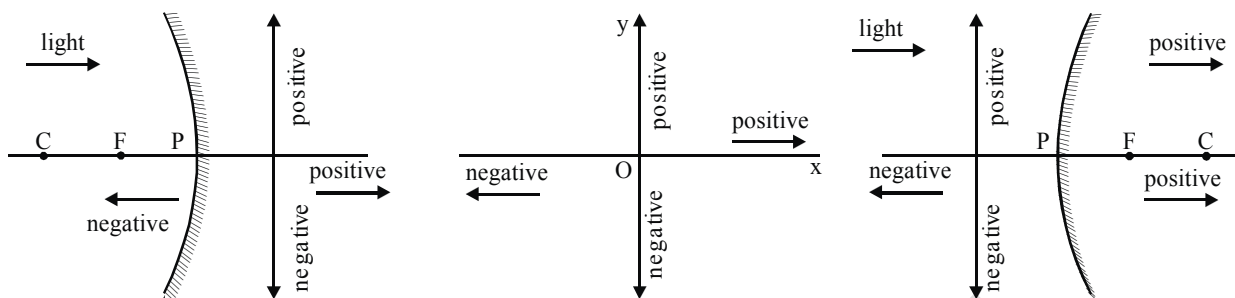


- (xi) **Marginal rays :**

Rays having a large angle of incidence



## SIGN- CONVENTION

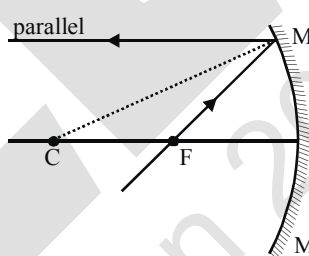
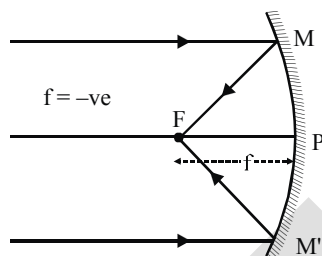


- Along principal axis, distances are measured from the pole (pole is taken as the origin).
- Distance in the direction of light are taken to be positive while opposite to be negative.
- The distances above principal axis are taken to be positive while below it negative.
- Whenever and wherever possible the ray of light is taken to travel from left to right.

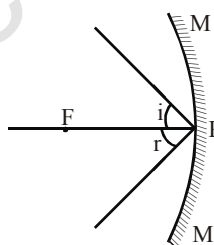
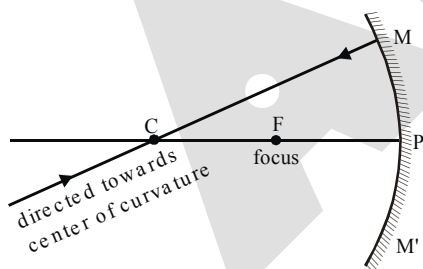
## RULES FOR IMAGE FORMATION (FOR PARAXIAL RAYS ONLY)

(These rules are based on the laws of reflection  $\angle i = \angle r$ )

- A ray parallel to principal axis after reflection from the mirror passes or appears to pass through its focus (by definition of focus).
- A ray passing through or directed towards focus, after reflection from the mirror, becomes parallel to the principal axis.



- A ray passing through or directed towards centre of curvature, after reflection from the mirror, retraces its path (as for it  $\angle i = 0$  and so  $\angle r = 0$ ).
- Incident and reflected rays at the pole of a mirror are symmetrical about the principal axis  $\angle i = \angle r$ .



## RELATIONS FOR SPHERICAL MIRRORS

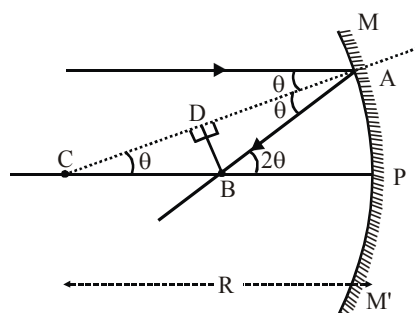
Relation between  $f$  and  $R$  for the spherical mirror

- For Marginal rays

In  $\triangle ABC$ ,  $AB = BC$

$$AC = CD + DA = 2BC \cos \theta \Rightarrow R = 2BC \cos \theta$$

$$\Rightarrow BC = \frac{R}{2 \cos \theta} \text{ and } BP = PC - BC = R - \frac{R}{2 \cos \theta}$$



**Note :** B is not the focus ; it is just a point where a marginal ray after reflection meets.

- **For paraxial rays (parallel to principal axis)**

$$(\theta \text{ small so } \sin\theta \approx \theta, \cos\theta \approx 1, \tan\theta \approx \theta) \text{ Hence } BC = \frac{R}{2} \text{ and } BP = \frac{R}{2}$$

Thus, point B is the midpoint of PC (i.e. radius of curvature) and is defined as FOCUS so  $BP = f = \frac{R}{2}$

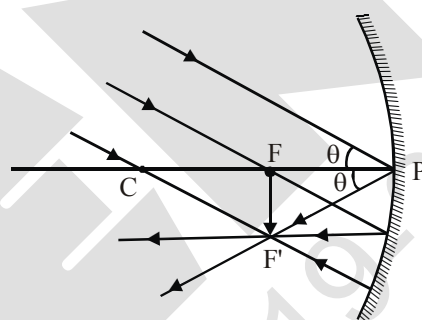
**(Definition :** Paraxial rays parallel to the principal axis after reflection from the mirror meet the principal axis at focus)

- BP is the focal length (f)  $\boxed{f = \frac{R}{2}}$

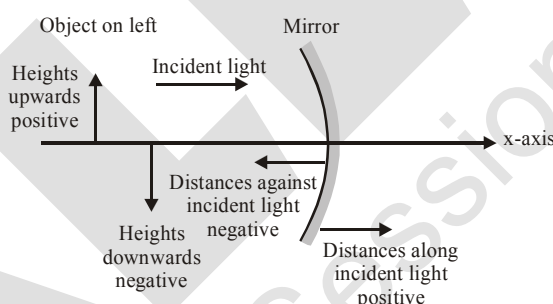
- **Paraxial rays (not parallel to principal axis)**

Such rays after reflection meet at a point in the focal plane ( $F'$ ), such that

$$\frac{FF'}{FP} = \tan\theta \approx \theta \Rightarrow \frac{FF'}{f} = \theta \Rightarrow FF' = f\theta$$



### Sign convention



### Relation between u, v and f for curved mirror

An object is placed at a distance u from the pole of a mirror and its image is formed at a distance v (from the pole)

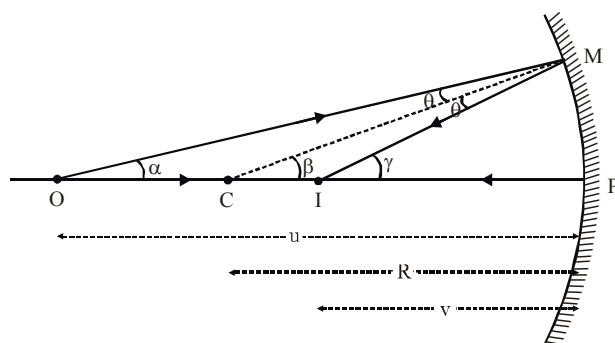
$$\text{If angle is very small : } \alpha = \frac{MP}{u}, \beta = \frac{MP}{R}, \gamma = \frac{MP}{v}$$

$$\text{from } \triangle CMO, \beta = \alpha + \theta \Rightarrow \theta = \beta - \alpha$$

$$\text{from } \triangle CMI, \gamma = \beta + \theta \Rightarrow \theta = \gamma - \beta$$

$$\text{so we can write } \beta - \alpha = \gamma - \beta \Rightarrow 2\beta = \gamma + \alpha$$

$$\therefore \frac{2}{R} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$



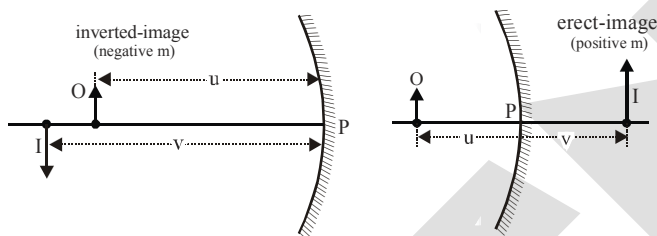
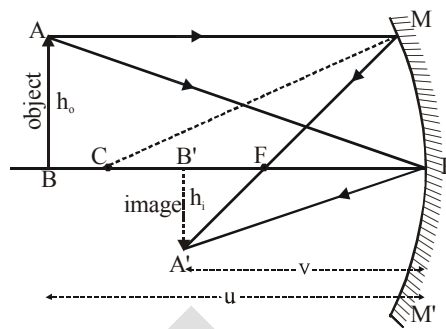
## MAGNIFICATION

### Transverse or lateral magnification

Linear magnification  $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o}$

$\triangle ABP$  and  $\triangle A'B'P$  are similar so  $\frac{-h_i}{h_o} = \frac{-v}{-u} \Rightarrow \frac{h_i}{h_o} = -\frac{v}{u}$

Magnification  $m = -\frac{v}{u}$ ;  $m = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f} = \frac{h_i}{h_o}$



If one dimensional object is placed perpendicular to the principal axis then linear magnification is

called transverse or lateral magnification.

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

### Magnification

$|m| > 1$

$m < 0$

### Image

enlarged

inverted

### Magnification

$|m| < 1$

$m > 0$

### Image

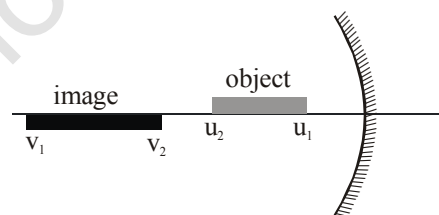
diminished

erect

### Longitudinal magnification

If one dimensional object is placed with its length along the principal axis then linear magnification is called longitudinal magnification.

Longitudinal magnification  $m_L = \frac{\text{length of image}}{\text{length of object}} = \frac{|v_2 - v_1|}{|u_2 - u_1|}$



For small objects only :  $m_L = -\frac{dv}{du}$

differentiation of  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  gives us  $-\frac{dv}{v^2} - \frac{du}{u^2} = 0 \Rightarrow -\frac{dv}{du} = \left[\frac{v}{u}\right]^2$  so  $m_L = -\frac{dv}{du} = \left[\frac{v}{u}\right]^2 = m^2$

### Superficial magnification

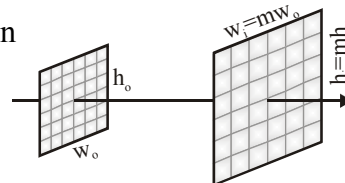
If two dimensional object placed with its plane perpendicular to principal axis its magnification is known as superficial magnification

Linear magnification  $m = \frac{h_i}{h_o} = \frac{w_i}{w_o}$

$h_i = mh_o$ ,  $w_i = mw_o$  and  $A_{obj} = h_o \times w_o$

Area of image :  $A_{image} = h_i \times w_i = mh_o \times mw_o = m^2 A_{obj}$

superficial magnification  $m_s = \frac{\text{area of image}}{\text{area of object}} = \frac{(ma) \times (mb)}{(a \times b)} = m^2$

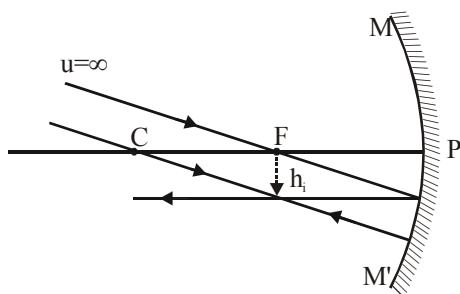


## IMAGE FORMATION BY SPHERICAL MIRRORS

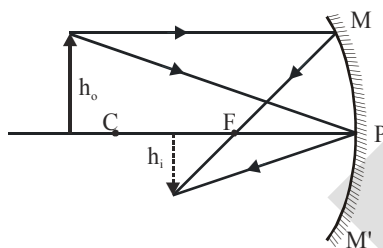
## Concave mirror

(i) **Object** : Placed at infinity**Image** : real, inverted, diminished at F

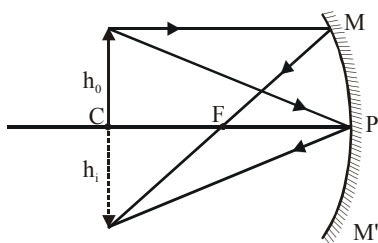
$|m| \ll 1 \text{ \& } m < 0$

(ii) **Object** : Placed in between infinity and C**Image** : real, inverted, diminished in between C and F

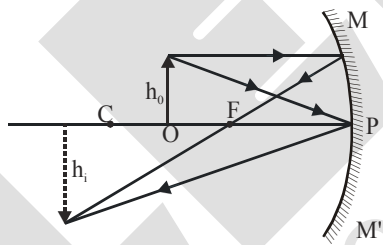
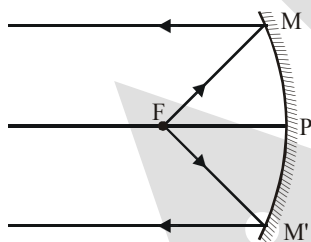
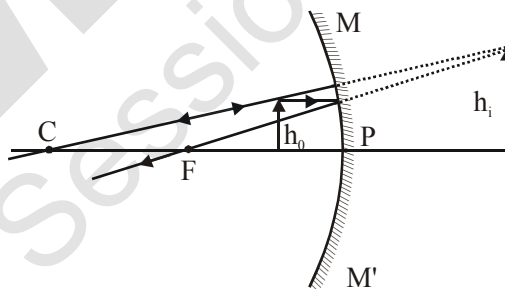
$|m| < 1 \text{ \& } m < 0$

(iii) **Object** : Placed at C**Image** : real, inverted, equal at C

$(m = -1)$

(iv) **Object** : Placed in between F and C**Image** : real, inverted, enlarged beyond C

$|m| > 1 \text{ \& } m < 0$

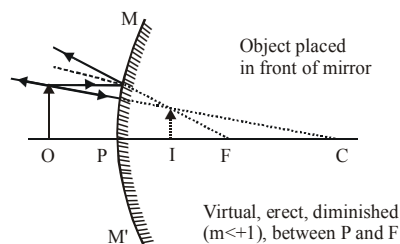
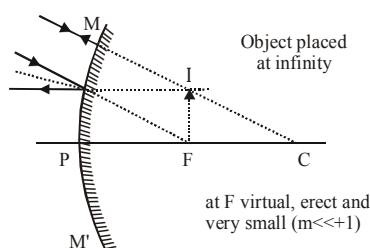
(v) **Object** : Placed at F**Image** : real, inverted, very large (assumed) at infinity ( $m \ll -1$ )(vi) **Object** : Placed between F and P**Image** : virtual, erect, enlarged and behind the mirror ( $m > +1$ )

## For concave mirror

| Object                  | Image         | Magnification                 |
|-------------------------|---------------|-------------------------------|
| $-\infty$               | F             | $ m  \ll 1 \text{ \& } m < 0$ |
| $-\infty - C$           | C - F         | $ m  < 1 \text{ \& } m < 0$   |
| C                       | C             | $m = -1$                      |
| C - F                   | $-\infty - C$ | $ m  > 1 \text{ \& } m < 0$   |
| Just before F towards C | $-\infty$     | $m \ll -1$                    |
| Just after F towards P  | $+\infty$     | $m \gg 1$                     |

## Convex mirror

Image is always virtual and erect, whatever be the position of the object and  $m$  is always positive.



## KEY POINTS

### Differences in real & virtual image for spherical mirror

#### Real Image

- (i) Inverted w.r.t. object
- (ii) Can be obtained on screen
- (iii) Its magnification is negative
- (iv) Forms in front of mirror

#### Virtual Image

- (i) Erect w.r.t. object
- (ii) Can not be obtained on screen
- (iii) Its magnification is positive
- (iv) Forms behind the mirror

- For real extended object, if the image formed by a single mirror is erect it is always virtual (i.e.,  $m$  is +ve) and in this situation if the size of image is :

| Smaller than object the mirror is convex | Equal to object the mirror is plane | Larger than object the mirror is concave |
|------------------------------------------|-------------------------------------|------------------------------------------|
| $m < +1$<br>                             | $m = +1$<br>                        | $m > +1$<br>                             |

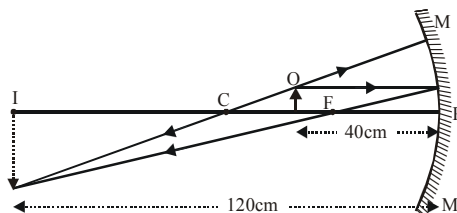
**Ex.** The focal length of a concave mirror is 30cm. Find the position of the object in front of the mirror, so that the image is three times the size of the object.

**Sol.** As the object is in front of the mirror it is real and for real object the magnified image formed by concave mirror can be inverted (i.e., real) or erect (i.e., virtual), so there are two possibilities.

(a) If the image is inverted (i.e., real)

$$m = \frac{f}{f - u} \Rightarrow -3 = \frac{-30}{-30 - u} \Rightarrow u = -40 \text{ cm}$$

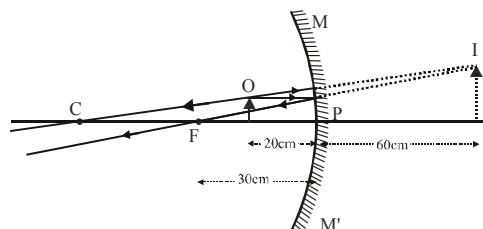
Object must be at a distance of 40 cm in front of the mirror (in between C and F).



(b) If the image is erect (i.e., virtual)

$$m = \frac{f}{f - u} \Rightarrow 3 = \frac{-30}{-30 - u} \Rightarrow u = -20 \text{ cm}$$

Object must be at a distance of 20 cm in front of the mirror (in between F and P).



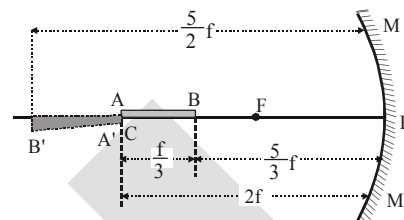
**Ex.** A thin rod of length  $\frac{f}{3}$  is placed along the principal axis of a concave mirror of focal length  $f$  such that its image which is real and elongated, just touches the rod. What is magnification?

**Sol.** Image is real and enlarged, the object must be between  $C$  and  $F$ . One end  $A'$  of the image coincides with the end  $A$  of rod itself, so

$$v_A = u_A, \frac{1}{v_A} + \frac{1}{v_A} = \frac{1}{-f} \text{ i.e., } v_A = u_A = -2f$$

so it clear that the end  $A$  is at  $C$ .  $\therefore$  the length of rod is  $\frac{f}{3}$

$\therefore$  distance of the other end  $B$  from  $P$  is  $u_B = 2f - \frac{f}{3} = \frac{5f}{3}$



if the distance of image of end  $B$  from  $P$  is  $v_B$  then  $\frac{1}{v_B} + \frac{1}{-\frac{5f}{3}} = \frac{1}{-f} \Rightarrow v_B = -\frac{5f}{2}$

$\therefore$  the length of the image  $|v_B| - |v_A| = \frac{5f}{2} - 2f = \frac{1}{2}f$  and magnification  $m = \frac{|v_B| - |v_A|}{|u_B| - |u_A|} = \frac{\frac{1}{2}f}{-\frac{1}{3}f} = -\frac{3}{2}$

Negative sign implies that image is inverted with respect to object and so it is real.

**Ex** A concave mirror of focal length 10 cm and convex mirror of focal length 15 cm are placed facing each other 40 cm apart. A point object is placed between the mirror on their common axis and 15 cm from the concave mirror. Find the position of image produced by the reflection first at concave mirror and then at convex mirror.

**Sol.** For  $M_1$  mirror  $O$  act as a object, let its image is  $I_1$  then,

$$u = -15 \text{ cm}, f = -10 \text{ cm} \Rightarrow \frac{1}{v} + \frac{1}{-15} = \frac{1}{-10} \Rightarrow v = -30 \text{ cm}$$

Image  $I_1$  will act as a object for mirror  $M_2$  its distance from mirror  $M_2$ .

$$u_1 = -(40 - 30) \text{ cm} = -10 \text{ cm}$$

$$\text{so } \frac{1}{v_1} + \frac{1}{u_1} = \frac{1}{f} \Rightarrow \frac{1}{v_1} - \frac{1}{10} = \frac{1}{15} \Rightarrow v_1 = +6 \text{ cm}$$

So final image  $I_2$  is formed at a distance 6 cm behind the convex mirror and is virtual.

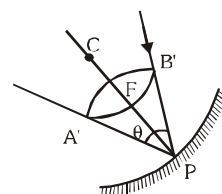
**Ex.** The sun subtends an angle  $\theta$  radians at the pole of a concave mirror of focal length  $f$ . What is the diameter of the image of the sun formed by the mirror.

**Sol.** Since the sun is at large distance very distant,  $u$  is very large and so  $\frac{1}{u} \approx 0$

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = -\frac{1}{f} \Rightarrow v = -f$$

The image of sun will be formed at the focus and will be real, inverted and diminished

$$A'B' = \text{height of image and } \theta = \frac{\text{Arc}}{\text{Radius}} = \frac{A'B'}{FP} \Rightarrow \theta = \frac{d}{f} \Rightarrow d = f\theta$$



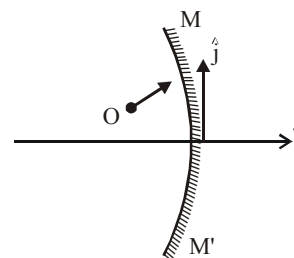
## VELOCITY OF IMAGE OF MOVING OBJECT (SPHERICAL MIRROR)

### (a) Velocity component along axis (Longitudinal velocity)

When an object is coming from infinite towards the focus of concave mirror

$$\therefore -\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{v^2} \frac{dv}{dt} - \frac{1}{u^2} \frac{du}{dt} = 0 \Rightarrow \vec{v}_{ix} = -\frac{v^2}{u^2} \vec{v}_{ox} = -m^2 \vec{v}_{ox}$$

$$\left[ v_{ix} = \frac{dv}{dt} = \text{velocity of image along principal-axis; } v_{ox} = \frac{du}{dt} = \text{velocity of object along principal-axis} \right]$$



### (b) Velocity component perpendicular to axis (Transverse velocity)

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f}{f-u} \Rightarrow h_i = \left( \frac{f}{f-u} \right) h_o$$

$$\frac{dh_i}{dt} = \left( \frac{f}{f-u} \right) \frac{dh_o}{dt} + \frac{f h_o}{(f-u)^2} \frac{du}{dt}; \vec{v}_{iy} = \left[ m \vec{v}_{oy} + \frac{m^2 h_o}{f} \vec{v}_{ox} \right] \hat{j}$$

$$\left[ \begin{array}{l} \frac{dh_i}{dt} = \text{velocity of image } \perp \text{ to principal-axis} \\ \frac{dh_o}{dt} = \text{velocity of object } \perp \text{ to principal-axis} \end{array} \right]$$

**Note :** Here principal axis has been taken to be along x-axis.

## POWER OF A MIRROR

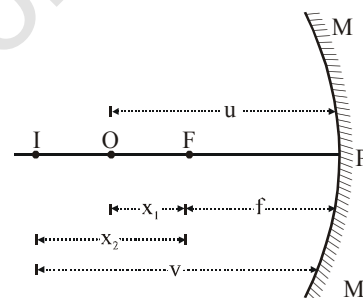
The power of a mirror is defined as  $P = -\frac{1}{f(\text{m})} = -\frac{100}{f(\text{cm})}$

## NEWTON'S FORMULA

In case if spherical mirrors if object distance ( $x_1$ ) and image distance ( $x_2$ ) are measured from focus instead of pole,  
 $u = -(f + x_1)$  and  $v = -(f + x_2)$ ,

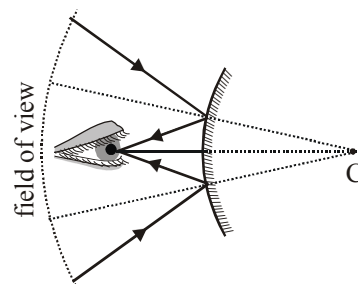
$$\text{by } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{(f+x_2)} - \frac{1}{(f+x_1)} = -\frac{1}{f}$$

on solving  $x_1 x_2 = f^2$  This is Newton's formula.



## POINTS

- Convex mirrors** gives erect, virtual and diminished image. In convex mirror the field of view is increased as compared to plane mirror. It is used as rear-view mirror in vehicles.
- Concave mirrors** give enlarged, erect and virtual image, so these are used by dentists for examining teeth. Due to their converging property concave mirrors are also used as reflectors in automobile head lights and search lights
- As focal length of a spherical mirror  $f = R/2$  depends only on the radius of mirror and is independent of wavelength of light and refractive index of medium so the focal length of a spherical mirror in air or water and for red or blue light is same.



## REFRACTION

Refraction is the phenomenon in which direction of propagation of light changes at the boundary when it passes from one medium to the other. In case of refraction frequency does not change.

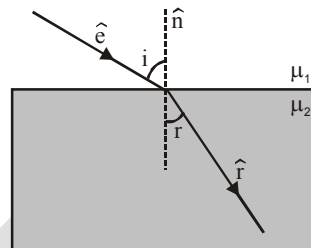
⊙ **Laws of Refraction**

- (i) Incident ray, refracted ray and normal always lie in the same plane.

In vector form  $(\hat{e} \times \hat{n}) \cdot \hat{r} = 0$

- (ii) The product of refractive index and sine of angle of incidence at a point in a medium is constant.  $\mu_1 \sin i = \mu_2 \sin r$  (Snell's law)

In vector form  $\mu_1 |\hat{e} \times \hat{n}| = \mu_2 |\hat{r} \times \hat{n}|$



**Absolute refractive index**

It is defined as the ratio of speed of light in free space 'c' to that in a given medium v.  $\mu$  or  $n = \frac{c}{v}$

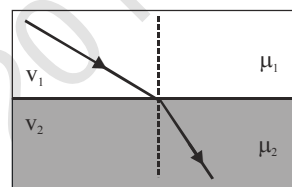
Denser is the medium, lesser will be the speed of light and so greater will be the refractive index,

$$\because v_{\text{glass}} < v_{\text{water}} \quad \therefore \mu_{\text{G}} > \mu_{\text{W}}$$

**Relative refractive index**

When light passes from one medium to the other, the refractive index of medium 2 relative to 1 is written as  ${}_1\mu_2$  and is defined as

$${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{(c/v_2)}{(c/v_1)} = \frac{v_1}{v_2}$$



⊙ **Bending of light ray**

According to Snell's law,  $\mu_1 \sin i = \mu_2 \sin r$

- (i) If light passes from rarer to denser medium  $\mu_1 = \mu_R$  and  $\mu_2 = \mu_D$

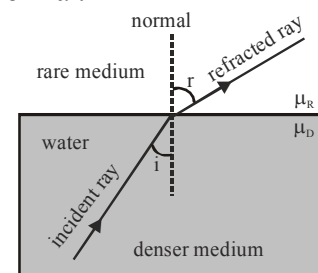
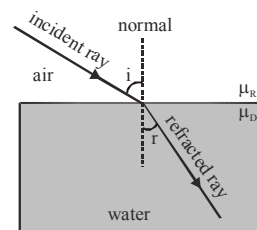
so that  $\frac{\sin i}{\sin r} = \frac{\mu_D}{\mu_R} > 1 \Rightarrow \angle i > \angle r$

In passing from rarer to denser medium, the ray bends towards the normal.

- (ii) If light passes from denser to rarer medium  $\mu_1 = \mu_D$  and  $\mu_2 = \mu_R$

$$\frac{\sin i}{\sin r} = \frac{\mu_R}{\mu_D} < 1 \Rightarrow \angle i < \angle r$$

In passing from denser to rarer medium, the ray bends away from the normal



### APPARENT DEPTH AND NORMAL SHIFT

If a point object in denser medium is observed from rarer medium and boundary is plane, then from Snell's law we have  $\mu_D \sin i = \mu_R \sin r$ ...(i)  
If the rays OA and OB are close enough to reach the eye.

$$\sin i \approx \tan i = \frac{p}{d_{ac}} \text{ and } \sin r \approx \tan r = \frac{p}{d_{ap}}$$

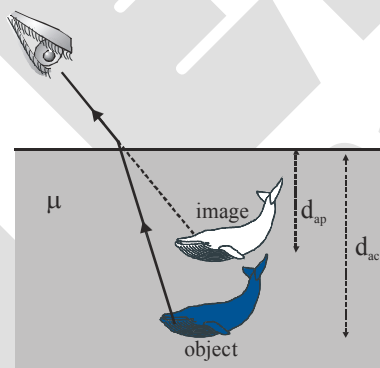
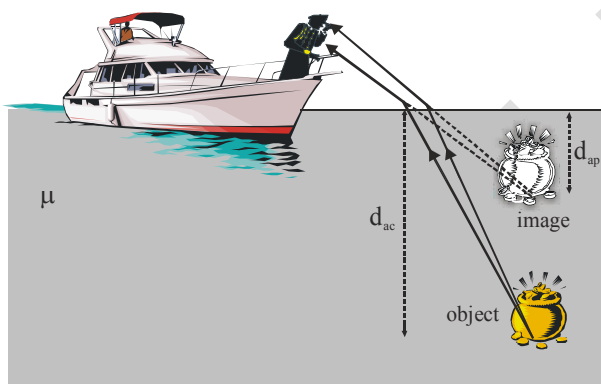
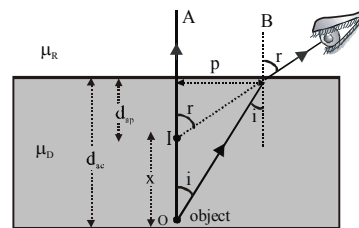
here  $d_{ac}$  = actual depth,  $d_{ap}$  = apparent depth

$$\text{So that equation (i) becomes } \mu_D = \frac{p}{d_{ac}} = \mu_R \frac{p}{d_{ap}} \Rightarrow \frac{d_{ac}}{d_{ap}} = \frac{\mu_D}{\mu_R} = \frac{\mu_1}{\mu_2}$$

$$(\text{If } \mu_R = 1, \mu_D = \mu) \text{ then } d_{ap} = \frac{d_{ac}}{\mu} \text{ so } d_{ap} < d_{ac} \quad \dots(\text{ii})$$

The distance between object and its image, called normal shift (x)

$$x = d_{ac} - d_{ap} \left[ \because d_{ap} = \frac{d_{ac}}{\mu} \right]; x = d_{ac} - \frac{d_{ac}}{\mu} = d_{ac} \left[ 1 - \frac{1}{\mu} \right] \dots(\text{iii}) \quad \text{If } d_{ac} = d \text{ then } x = d \left[ 1 - \frac{1}{\mu} \right]$$



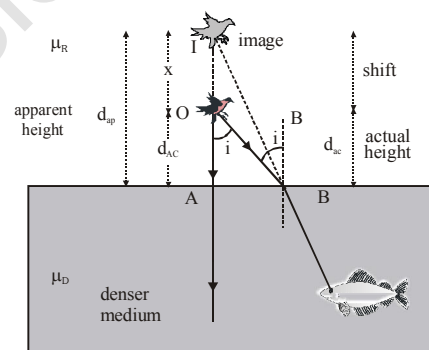
### Object in a rarer medium is seen from a denser medium

$$\frac{d_{ac}}{d_{ap}} = \frac{\mu_1}{\mu_2} = \frac{\mu_R}{\mu_D} = \frac{1}{\mu} (< 1)$$

$$d_{ap} = \mu d_{ac} \text{ i.e., } d_{ap} > d_{ac}$$

A high flying object appears to be higher than in reality.

$$x = d_{ap} - d_{ac} \Rightarrow x = [\mu - 1] d_{ac}$$



### LATERAL SHIFT

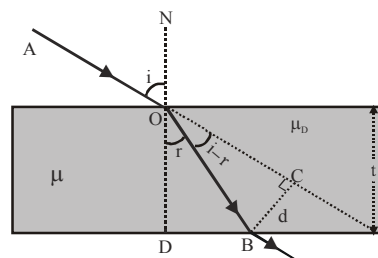
The perpendicular distance between incident and emergent ray is known as lateral shift.

Lateral shift  $d = BC$  and  $t$  = thickness of slab

$$\text{In } \Delta BOC \sin(i-r) \frac{BC}{OB} = \frac{d}{OB} \Rightarrow d = OB \sin(i-r) \dots(\text{i})$$

$$\text{In } \Delta OBD \cos r = \frac{OD}{OB} = \frac{t}{OB} \Rightarrow OB = \frac{t}{\cos r} \dots(\text{ii})$$

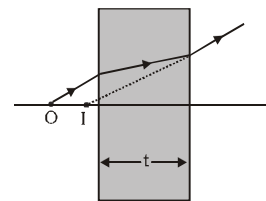
$$\text{From (i) and (ii) } d = \frac{t}{\cos r} \sin(i-r)$$



**TRANSPARENT GLASS SLAB (Normal shift)**

When an object is placed in front of a glass slab, it shifts the object in the direction of incident light and forms an image at a distance  $x$ .

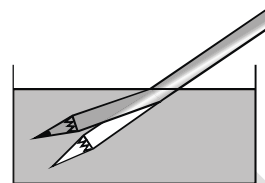
$$x = t \left[ 1 - \frac{1}{\mu} \right]$$

**SOME ILLUSTRATIONS OF REFRACTION**

- Bending of an object**

When a point object in a denser medium is seen

from a rarer medium it appears to bend by  $\frac{d}{\mu}$



- Twinkling of stars**

Due to fluctuations in refractive index of atmosphere the refraction becomes irregular and the light sometimes reaches the eye and sometimes it does not. This gives rise to twinkling of stars.

**KEY POINTS**

- $\mu$  is a scalar and has no units and dimensions.
- If  $\epsilon_0$  and  $\mu_0$  are electric permittivity and magnetic permeability respectively of free space while  $\epsilon$  and  $\mu$  those of a given medium, then according to electromagnetic theory.

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \text{ and } v_m = \frac{1}{\sqrt{\epsilon \mu}} \Rightarrow n_m = \frac{c}{v} = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \cdot \sqrt{\epsilon \mu} = \sqrt{\epsilon_r \mu_r}$$

- As in vacuum or free space, speed of light of all wavelengths is maximum and equal to  $c$  so for all wavelengths the refractive index of free space is minimum and is  $\mu = \frac{c}{v_m} = \frac{c}{c} = 1$

**Ex.** A ray of light is incident on a transparent glass slab of refractive index 1.62. If the reflected and refracted rays are mutually perpendicular, what is the angle of incidence? [ $\tan^{-1}(1.62) = 58.3^\circ$ ]

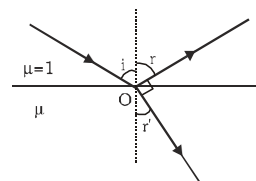
**Sol.** According to given problem :  $r + 90^\circ + r' = 180^\circ$

i.e.,  $r' = 90^\circ - r \Rightarrow r' = (90^\circ - i)$  [ $\because \angle i = \angle r$ ]

And as according to Snell's law:  $1 \sin i = \mu \sin r'$

$\sin i = \mu \sin (90 - i) \Rightarrow \sin i = \mu \cos i$  [ $\because \sin (90 - i) = \cos i$ ]

$\Rightarrow \tan i = \mu \Rightarrow i = \tan^{-1} \mu = \tan^{-1}(1.62) = 58.3^\circ$



**Ex.** A 20 cm thick glass slab of refractive index 1.5 is kept in front of a plane mirror. An object is kept in air at a distance 40 cm from the mirror. Find the position of image w.r.t an observer near the object. What is effect of separation between glass slab and the mirror on image.

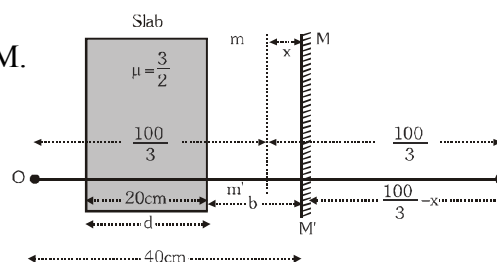
**Sol.** Shifting in object due to glass slab  $x = d \left( 1 - \frac{1}{\mu} \right) = 20 \left[ 1 - \frac{1}{1.5} \right] = \frac{20}{3}$  cm

Distance of object from mirror (as seen by mirror)  $= 40 - \frac{20}{3} = \frac{100}{3}$  cm

Image will be formed at a distance  $\frac{100}{3}$  cm from mirror M.

Shifting in image due to glass slab  $= \frac{20}{3}$  cm

So distance of image from mirror  $= \frac{100}{3} - \frac{20}{3} = \frac{80}{3}$  cm



Distance of image from the actual plane mirror is independent of separation  $b$  between glass slab and the mirror. If the distance is more then brightness of image will be less.

**Ex.** If one face of a prism angle  $30^\circ$  and  $\mu = \sqrt{2}$  is silvered, the incident ray retraces its initial path. What is the angle of incidence ?

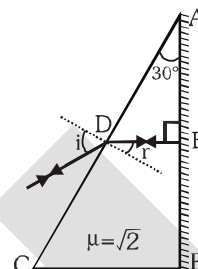
**Sol.** As incident ray retraces its path the ray is incident normally on the silvered face of the prism as shown in figure.

Further, as in  $\triangle AED$   $30^\circ + 90^\circ + \angle D = 180^\circ \Rightarrow \angle D = 60^\circ$

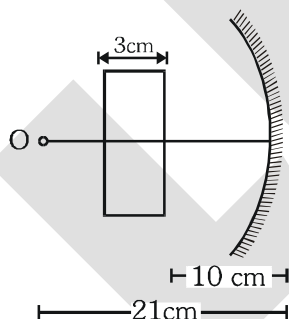
Now as by construction,  $\angle D + \angle r = 90^\circ \Rightarrow \angle r = 90^\circ - 60^\circ = 30^\circ$

$\therefore$  from Snell's law at surface AC,  $1 \sin i = \sqrt{2} \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$

$\therefore \sin i = \frac{1}{\sqrt{2}}$  so  $i = 45^\circ$



**Ex.** An object is placed 21 cm in front of a concave mirror of radius of curvature 20 cm. A glass slab of thickness 3 cm and refractive index 1.5 is placed closed to the mirror in space between the object and the mirror. Find the position of final image formed if distance of nearer surface of the slab from the mirror is 10 cm.

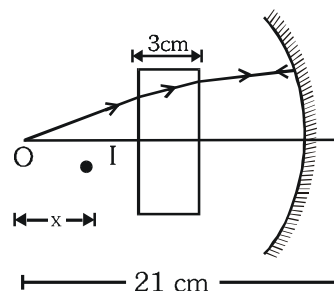


**Sol.** shift by slab  $x = d \left( 1 - \frac{1}{\mu} \right) = 3 \left( 1 - \frac{1}{1.5} \right) = 1 \text{ cm}$

for image formed by mirror  $u = -(21 - 1) \text{ cm} = -20 \text{ cm}$ .

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{-20} + \frac{1}{v} = \frac{1}{-10} \Rightarrow v = -20 \text{ cm}$$

shift in the direction of light  $v = -(20 + 1) = -21 \text{ cm}$ .

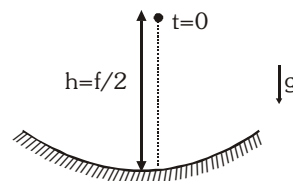


**Ex.** A particle is dropped along the axis from a height  $\frac{f}{2}$  on a concave mirror of focal length  $f$  as shown in figure. Find the maximum speed of image.

**Sol.**  $v_{IM} = -m^2 v_{OM} = -m^2 (gt)$  where

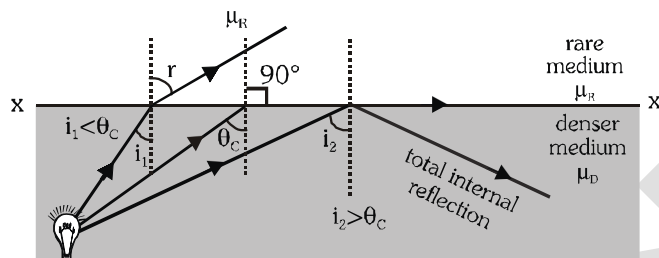
$$m = \frac{f}{f - u} = \frac{-f}{-f + \left( \frac{f}{2} - \frac{gt^2}{2} \right)} = \frac{2f}{f + gt^2} \Rightarrow v_1 = - \left( \frac{2f}{f + gt^2} \right)^2 (gt) = \frac{-4f^2 gt}{(f + gt^2)^2}$$

For maximum speed  $\frac{dv_1}{dt} = 0 \Rightarrow t = \sqrt{\frac{f}{3g}} \Rightarrow v_{\text{max}} = \frac{3}{4} \sqrt{3fg}$



## TOTAL INTERNAL REFLECTION

When light ray travel from denser to rarer medium it bend away from the normal if the angle of incident is increased, angle of refraction will also increased. At a particular value of angle the refracted ray subtend  $90^\circ$  angle with the normal, this angle of incident is known as critical angle ( $\theta_c$ ). If angle of incident further increase the ray come back in the same medium this phenomenon is known as total internal reflection.



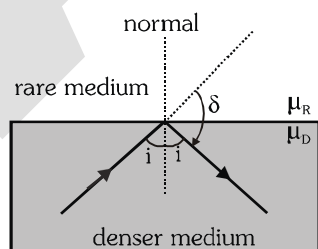
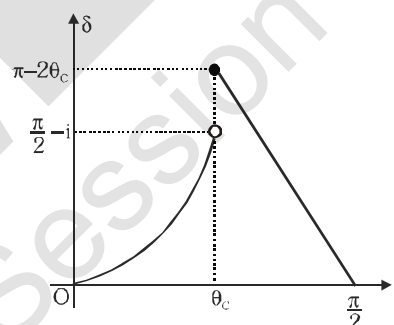
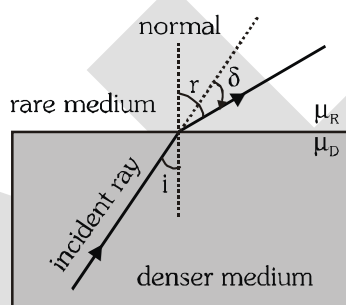
## CONDITIONS

- Angle of incident  $>$  critical angle [ $i > \theta_c$ ]
- Light should travel from denser to rare medium  $\Rightarrow$  Glass to air, water to air, Glass to water

Snell's Law at boundary  $xx'$ ,  $\mu_D \sin \theta_c = \mu_R \sin 90^\circ \Rightarrow \sin \theta_c = \frac{\mu_R}{\mu_D}$

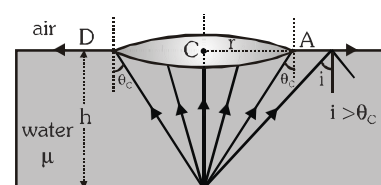
**Graph between angle of deviation ( $\delta$ ) and angle of incidence ( $i$ ) as rays goes from denser to rare medium**

- If  $i < \theta_c$ ,  $\mu_D \sin i = \mu_R \sin r$ ;  $r = \sin^{-1}\left(\frac{\mu_D}{\mu_R} \sin i\right)$  so  $\delta = r - i = \sin^{-1}\left(\frac{\mu_D}{\mu_R} \sin i\right) - i$



- If  $i > \theta_c$ ;  $\delta = \pi - 2i$

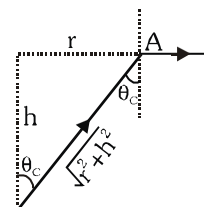
- A point object is situated at the bottom of tank filled with a liquid of refractive index  $\mu$  upto height  $h$ . It is found light from the source come out of liquid surface through a circular portion above the object



$$\sin \theta_c = \frac{r}{\sqrt{r^2 + h^2}} \quad \& \quad \sin \theta_c = \frac{1}{\mu} \Rightarrow \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + h^2}} \Rightarrow \frac{1}{\mu^2} = \frac{r^2}{r^2 + h^2}$$

$$\Rightarrow \mu^2 r^2 = r^2 + h^2 \Rightarrow (\mu^2 - 1)r^2 = h^2 \Rightarrow \text{radius of circular portion}$$

$$r = \frac{h}{\sqrt{\mu^2 - 1}} \quad \text{and} \quad \text{area} = \pi r^2$$



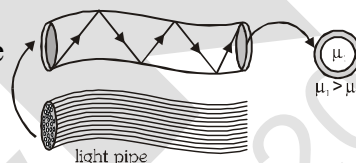
### SOME ILLUSTRATIONS OF TOTAL INTERNAL REFLECTION

- Sparkling of diamond**

The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so  $\theta_c = 24^\circ$ . Now the cutting of diamond are such that  $i > \theta_c$ . So TIR will take place again and again inside it. The light which beams out from a few places in some specific directions makes it sparkle.

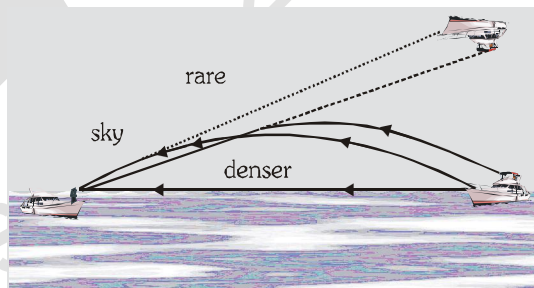
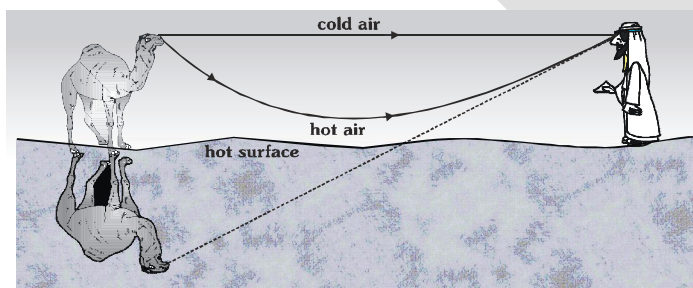
- Optical Fibre**

In it light through multiple total internal reflections is propagated along the axis of a glass fibre of radius of few microns in which index of refraction of core is greater than that of surroundings.



- Mirage and looming**

Mirage is caused by total internal reflection in deserts where due to heating of the earth, refractive index of air near the surface of earth becomes lesser than above it. Light from distant objects reaches the surface of earth with  $i > \theta_c$  so that TIR will take place and we see the image of an object along with the object as shown in figure.



Similar to 'mirage' in deserts, in polar regions 'looming' takes place due to TIR. Here  $\mu$  decreases with height and so the image of an object is formed in air if ( $i > \theta_c$ ) as shown in figure.

### KEY POINTS

- A diver in water at a depth  $d$  sees the world outside through a horizontal circle of radius.  $r = d \tan \theta_c$ .
- In case of total internal reflection, as all (i.e. 100%) incident light is reflected back into the same medium there is no loss of intensity while in case of reflection from mirror or refraction from lenses there is some loss of intensity as all light can never be reflected or refracted. This is why images formed by TIR are much brighter than formed by mirrors or lenses.

**Ex.** A rectangular block of glass is placed on a printed page laying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.

**Sol.** The situation is depicted in figure. Light will not emerge out from the vertical face BC if at it

$$i > \theta_c \text{ or } \sin i > \sin \theta_c \Rightarrow \sin i > \frac{1}{\mu} \left[ \text{as } \sin \theta_c = \frac{1}{\mu} \right] \dots (i)$$

But from Snell's law at O  $1 \times \sin \theta = \mu \sin r$

$$\text{And in } \Delta OPR, r + 90 + i = 180 \Rightarrow r + i = 90^\circ \Rightarrow r = 90 - i$$

$$\text{So } \sin \theta = \mu \sin (90 - i) = \mu \cos i \Rightarrow \cos i = \frac{\sin \theta}{\mu}$$

$$\text{so } \sin i = \sqrt{1 - \cos^2 i} = \sqrt{1 - \left[ \frac{\sin \theta}{\mu} \right]^2} \dots (ii)$$

so substituting the value of  $\sin i$  from equation (ii) in (i),

$$\sqrt{1 - \frac{\sin^2 \theta}{\mu^2}} > \frac{1}{\mu} \text{ i.e., } \mu^2 > 1 + \sin^2 \theta \because (\sin^2 \theta)_{\max} = 1 \therefore \mu^2 > 2 \Rightarrow \mu > \sqrt{2} \therefore$$

$$\mu_{\min} = \sqrt{2}$$

### REFRACTION AT TRANSPARENT CURVED SURFACE

$\mu_1$  = refractive index of the medium in which actual incident ray lies.

$\mu_2$  = refractive index of the medium in which actual refractive ray lies.

O = Object

P = pole

C = centre of curvature

R = PC = radius of curvature

#### Refraction from curved surface

$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

if angle is very small :  $\mu_1 \theta_1 = \mu_2 \theta_2 \dots (i)$

$$\text{But } \theta_1 = \alpha + \beta \dots (ii)$$

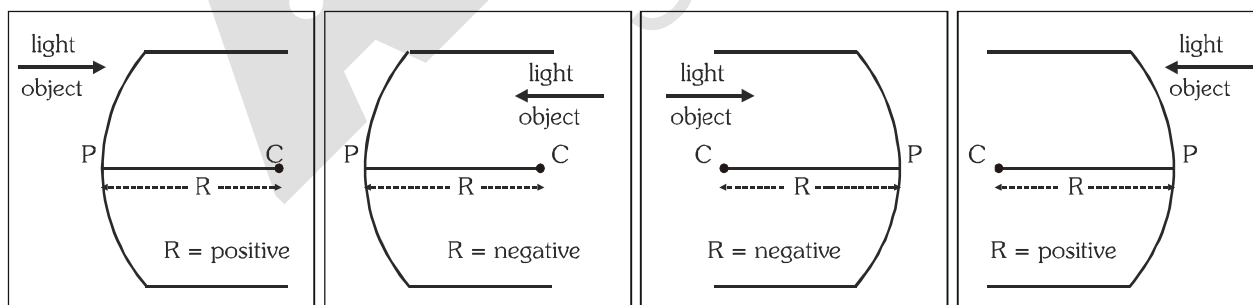
$$\beta = \theta_2 + \gamma \dots (iii)$$

from (i), (ii) and (iii)  $\mu_1(\alpha + \beta) = \mu_2(\beta - \gamma)$

$$\Rightarrow \mu_1 \alpha + \mu_1 \beta = \mu_2 \beta - \mu_2 \gamma \Rightarrow \mu_1 \alpha + \mu_2 \gamma = (\mu_2 - \mu_1) \beta$$

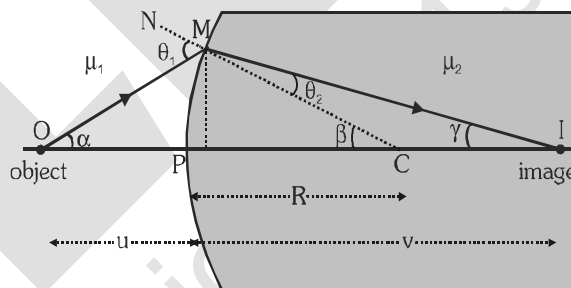
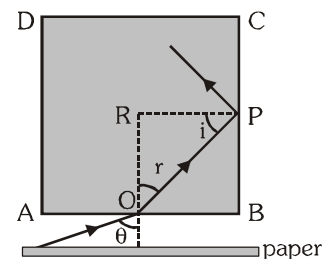
$$\Rightarrow \frac{\mu_1 PM}{-u} + \frac{\mu_2 PM}{v} = \frac{(\mu_2 - \mu_1) PM}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

#### SIGN CONVENTION FOR RADIUS OF CURVATURE



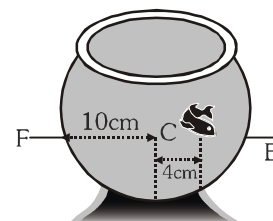
These are valid for all single refraction surfaces – convex, concave or plane. In case of plane refracting

$$\text{surface } R \rightarrow \infty, \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = 0 \text{ i.e. } \frac{u}{v} = \frac{\mu_1}{\mu_2} \text{ or } \frac{d_{Ac}}{d_{Ap}} = \frac{\mu_1}{\mu_2}$$





**Ex.** In a thin spherical fish bowl of radius 10 cm filled with water of refractive index  $(4/3)$ , there is a small fish at a distance 4 cm from the centre C as shown in Figure. Where will the fish appear to be, if seen from (a) E and (b) F (neglect the thickness of glass) ?



**Sol.** In the case of refraction from curved surface  $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$

(a) Seen from E  $\mu_1 = \frac{4}{3}$ ,  $\mu_2 = 1$ ,  $R = -10$  cm &  $u = -(10 - 4) = -6$  cm

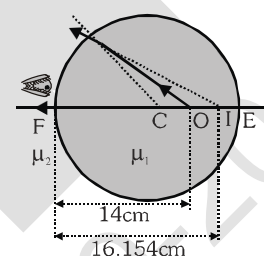
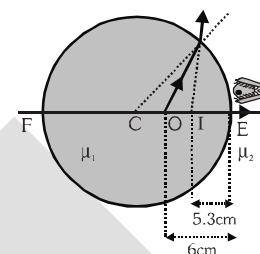
$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-6} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{90}{17} = -5.3 \text{ cm}$$

i.e., fish will appear at a distance 5.3 cm from E towards F (lesser than actual distance, i.e., 6 cm)

(b) Seen from F  $\mu_1 = \frac{4}{3}$ ,  $\mu_2 = 1$ ,  $R = -10$  cm and  $u = -(10 + 4) = -14$  cm

$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-14} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{-210}{13} = -16.154 \text{ cm}$$

so fish will appear at a distance 16.154 cm from F toward E (more than actual distance, i.e., 14 cm)

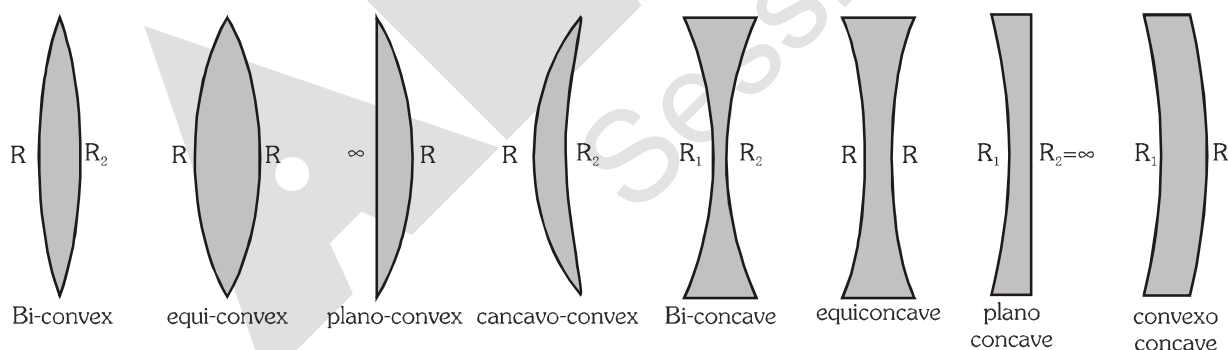


## LENS

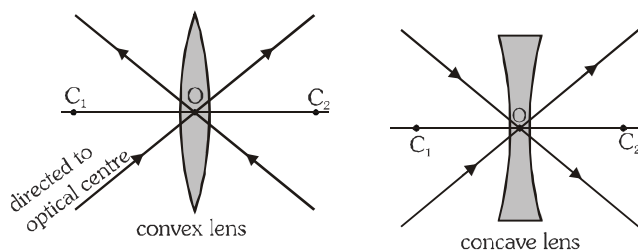
A lens is a piece of transparent material with two refracting surfaces such that at least one is curved and refractive index of its material is different from that of the surroundings.

A thin spherical lens with refractive index greater than that of surroundings behaves as a convergent or convex lens, i.e., converges parallel rays if its central (i.e. paraxial) portion is thicker than marginal one.

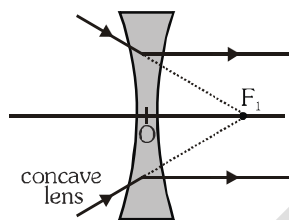
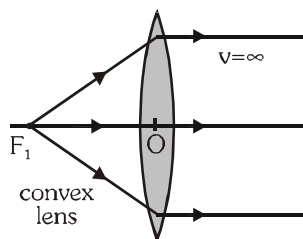
However if the central portion of a lens is thinner than marginal, it diverges parallel rays and behaves as divergent or concave lens. This is how we can classify and identify convergent and divergent lenses.



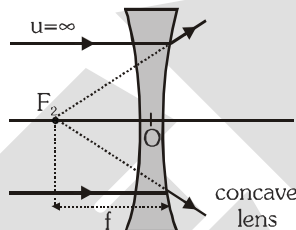
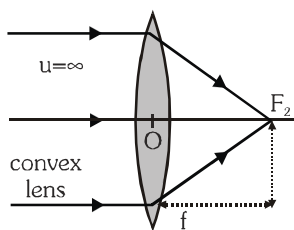
- **Optical Centre :** O is a point for a given lens through which any ray passes undeviated



- **Principal Axis :**  $C_1 C_2$  is a line passing through optical centre and perpendicular to the lens.
- **Principal Focus :** A lens has two surfaces and hence two focal points. First focal point is an object point on the principal axis for which image is formed at infinity.



While second focal point is an image point on the principal axis for which object lies at infinity



- **Focal Length  $f$**  is defined as the distance between optical centre of a lens and the point where the parallel beam of light converges or appears to converge.
- **Aperture :** In reference to a lens, aperture means the effective diameter. Intensity of image formed by a lens which depends on the light passing through the lens will depend on the square of aperture, i.e., Intensity  $\propto$  (Aperture)<sup>2</sup>

### LENS-MAKER'S FORMULA

#### In case of image formation by a lens

Image formed by first surface acts as object for the second.

So, from the formula of refraction at curved surface.

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

For first surface A  $\frac{\mu_L}{v_1} - \frac{\mu_M}{u} = \frac{\mu_L - \mu_M}{R_1}$  ... (i) [ $\because \mu_2 = \mu_L, \mu_1 = \mu_M$ ]

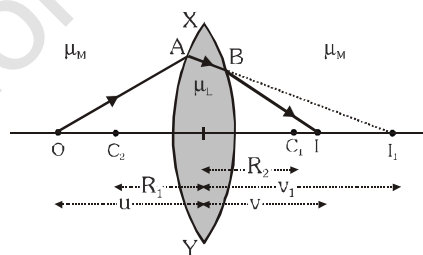
For second surface B  $\frac{\mu_M}{v} - \frac{\mu_L}{v_1} = \frac{\mu_M - \mu_L}{R_2} = -\frac{\mu_L - \mu_M}{R_2}$  ... (ii) [ $\because \mu_2 = \mu_M, \mu_1 = \mu_L, \mu_1 = \mu_2, u \rightarrow v_1$ ]

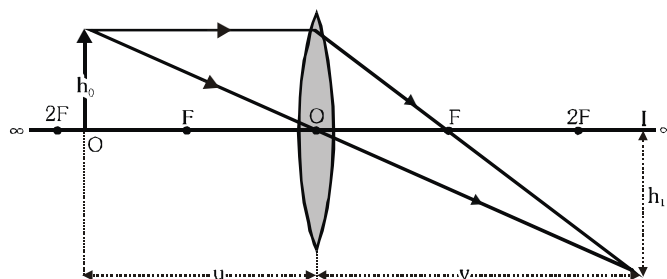
By adding (i) and (ii)

$$\mu_M \left[ \frac{1}{v} - \frac{1}{u} \right] = (\mu_L - \mu_M) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{\mu_L - \mu_M}{\mu_M} \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \dots \text{(iii)} \quad \left( \because \mu = \frac{\mu_L}{\mu_M} \right)$$

Now if object is at infinity, Image will be formed at the focus,  $u = -\infty, v = f$

$$\text{So } \frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right] \dots \text{(iv)}$$





This is known as lens makers formula by equating (iii) and (iv)  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

this is known as lens formula

**Magnification** :  $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_1}{h_0} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$

### RULES FOR IMAGE FORMATION

- A ray passing through optical centre proceeds undeviated through the lens
- A ray passing through first focus or directed towards it, after refraction from the lens, becomes parallel to the principal axis.
- A ray passing parallel to the principal axis after refraction through the lens passes or appears to pass through  $F_2$

#### For Convergent or convex lens

| Object                  | Image            | Magnification         |
|-------------------------|------------------|-----------------------|
| $-\infty$               | F                | $ m  \ll 1$ & $m < 0$ |
| $-\infty - 2F$          | $F-2F$           | $ m  < 1$ & $m < 0$   |
| $2F$                    | $2F$             | $m = -1$              |
| $F - 2F$                | $\infty-2F$      | $ m  > 1$ & $m < 0$   |
| Just before F towards C | $+\infty$        | $m \ll -1$            |
| Just before F towards P | $-\infty$        | $m \gg 1$             |
| $F - O$                 | In front of lens | $m > 1$               |

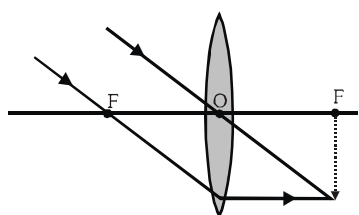
### IMAGE FORMATION FOR CONVEX LENS (CONVERGENT LENS)

- (i) Object is placed at infinity

**Image :**

at F real inverted very small in size

$$|m| \ll 1 \text{ \& } m < 0$$

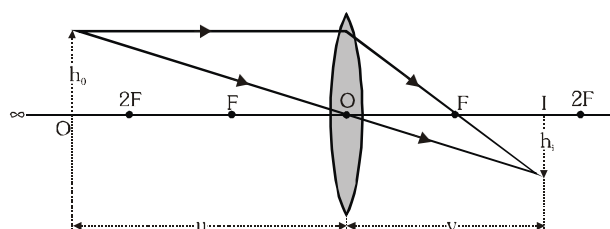


- (ii) Object is placed in between  $\infty - 2F$

**Image :**

real ( $F - 2F$ ) inverted small in size (diminished)

$$|m| < 1 \text{ \& } m < 0$$

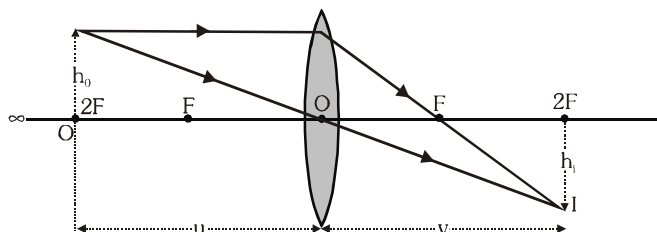


(iii) Object is placed at  $2F$

**Image :**

real (at  $2F$ ) inverted equal (of same size)

( $m = -1$ )

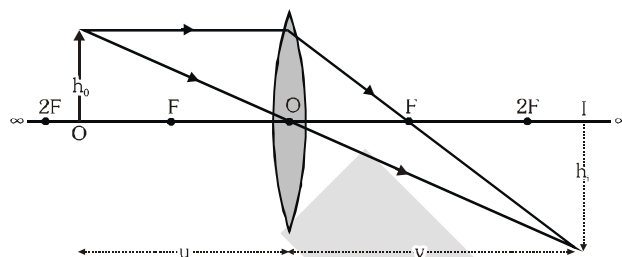


(iv) Object is placed in between  $2F - F$

**Image :**

real ( $2F - \infty$ ) inverted enlarged

$|m| > 1$  &  $m < 0$

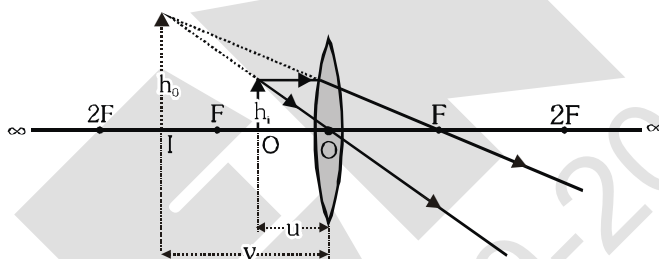


(v) Object is placed in between  $F - O$

**Image :**

virtual (in front of lens) erected enlarge

( $m > +1$ )



### IMAGE FORMATION FOR CONCAVE LENS (DIVERGENT LENS)

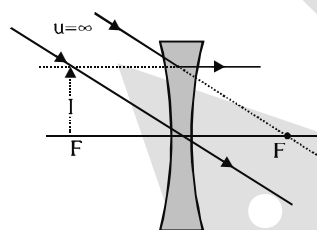
Image is virtual, diminished, erect, towards the object,  $m = +ve$

(i) Object is placed at infinity

**Image :**

At  $F$  virtual erected

diminished ( $m \ll +1$ )

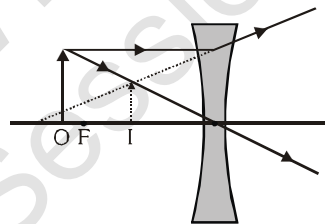


(ii) Object is placed in front of lens

**Image :**

between  $F$  and optical centre

virtual erected diminished ( $m < +1$ )



### Sign convention for object/image for lens

|                |          |
|----------------|----------|
| Real object    | $u - ve$ |
| Real image     | $v + ve$ |
| Virtual object | $u + ve$ |
| Virtual image  | $v - ve$ |

### POWER OF LENS

Reciprocal of focal length in meter is known as power of lens.

**SI UNIT :** dioptre (D)

**Power of lens :**  $P = \frac{1}{f(m)} = \frac{100}{f(cm)}$  dioptre [in air]

## COMBINATION OF LENSES

**Two thin lens are placed in contact with each other**

power of combination,  $P = P_1 + P_2 \Rightarrow \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Use sign convention when solve numericals

**Two thin lens are placed in at a small distance d**  
(provided incident rays are parallel to principal axis).

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \Rightarrow P = P_1 + P_2 - d P_1 P_2$$

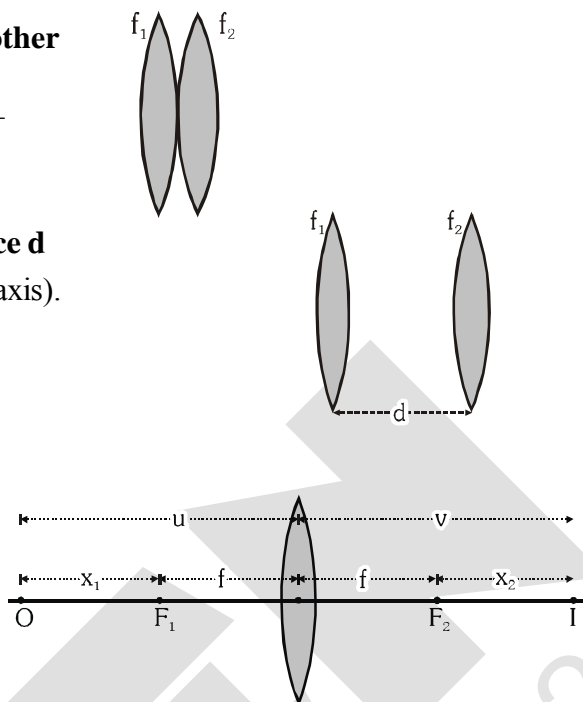
Use sign convention when solving numericals

• **Newton's Formula**

$$f = \sqrt{x_1 x_2}$$

$x_1$  = distance of object from focus.

$x_2$  = distance of image from focus.



## SOME SPECIAL CASES

(i) The focal length of equiconvex lens placed in air

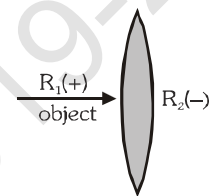
refractive index of lens  $\mu_L = \mu$

$$R_1 = +R,$$

refractive index of medium  $\mu_M = 1$

$$R_2 = -R$$

$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \left( -\frac{1}{R} \right) \right] \Rightarrow \text{Focal length } f = \frac{R}{2(\mu - 1)}$$



(ii) Focal length of planoconvex lens placed in air

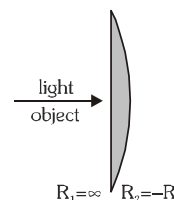
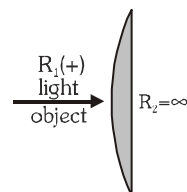
$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \frac{1}{\infty} \right]$$

$$\Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$

If object is placed towards plane surface

$$\frac{1}{f} = (\mu - 1) \left[ \frac{1}{\infty} - \left( -\frac{1}{R} \right) \right]$$

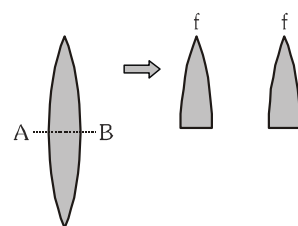
$$\Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$



(iii) If an equiconvex lens of focal length  $f$  is cut into equal parts by a horizontal plane AB then the focal length of each part will be equal to that of initial lens. Because  $\mu$ ,  $R_1$  and  $R_2$  will remain unchanged. Only intensity will be reduced.

$\therefore$  intensity  $I \propto (\text{apertures})^2$

$\therefore$  intensity through a single part will be reduced



- (iv) If the same lens is cut into equal parts by a vertical plane CD the focal length of each part will be double of initial value but intensity will remain unchanged.

For equiconvex lens  $\frac{1}{f} = \frac{(\mu - 1)2}{R}$  For plano convex lens  $\frac{1}{f_1} = \frac{\mu - 1}{R}$

So  $\frac{1}{f} = \frac{2}{f_1} \Rightarrow f_1 = 2f \Rightarrow$  Focal length of each part = 2

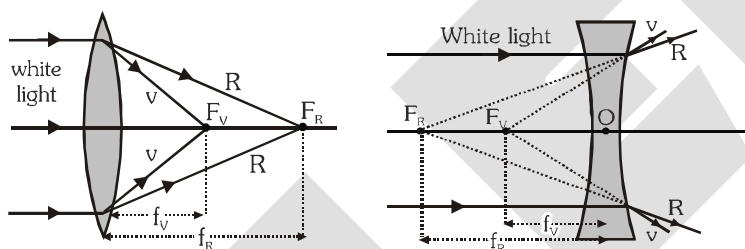
(focal length of original lens)

- (v) If a lens is made of number of layers of different refractive index for a given wavelength then no. of images is equal to number of refractive index, as

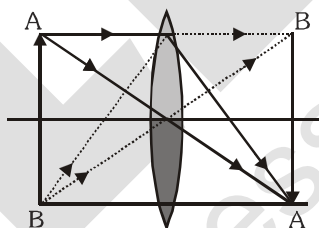
$\frac{1}{f} \propto (\mu - 1)$

In figure number of images = 2

- (vi) Focal length of lens depends on wavelength.  $\therefore \frac{1}{f} \propto (\mu - 1) \propto \frac{1}{\lambda} [f \propto \lambda] \quad f_R > f_V$



- (vii) If half portion of lens is covered by black paper then intensity of image will be reduced but complete image will be formed.

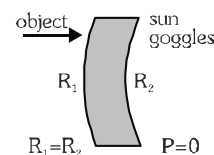


- (viii) Sun-goggles :

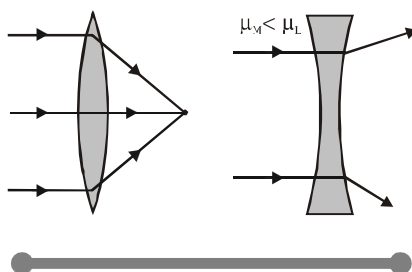
radius of curvature of two surfaces is equal with centre on the same side

$R_1 = R_2 = +R \quad \text{so} \quad \frac{1}{f} = (\mu - 1) \left[ \frac{1}{R} - \frac{1}{R} \right]$

$\Rightarrow \frac{1}{f} = 0 \Rightarrow f = \infty \quad \text{and} \quad P = 0 \Rightarrow$  sun goggles have no power



- (ix) If refractive index of medium < Refractive index of lens



If  $\mu_M < \mu_L$  then  $f = +ve$

Convex lens behave as convex lens. While concave lens behave as concave lens.

- (x) Refractive index of medium = Refractive index of lens ( $\mu_M = \mu_L$ )

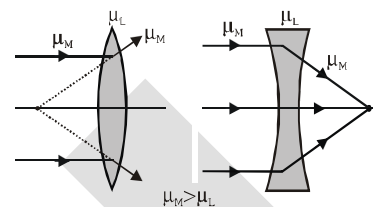
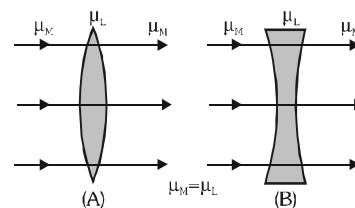
$$\frac{1}{f} = \left( \frac{\mu_L}{\mu_M} - 1 \right) \left( \frac{1}{R_1} - \frac{1}{R_2} \right); \frac{1}{f} = 0 \Rightarrow f = \infty \text{ \& } P = 0$$

Lens will behave as plane glass plate

- (xi) Refractive index of medium > Refractive index of lens

$$\mu_M > \mu_L \Rightarrow \frac{\mu_L}{\mu_M} < 1 \text{ [ } f \text{ will be negative ]}$$

convex lens will behave as concave lens and concave lens will behave as convex lens. If a air bubble is formed in water it behaves as concave lens.

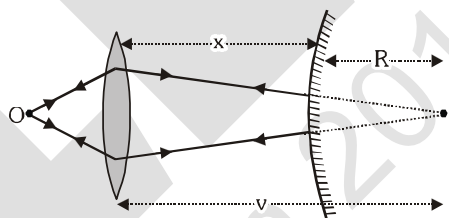
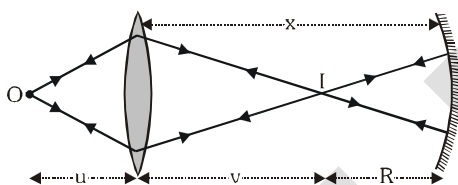


- Ex.** A point source S is placed at distance of 15 cm from a converging lens of focal length 10 cm. Where should a (i) concave mirror (ii) convex mirror of focal length 12 cm be placed so that real image is formed on object itself.

**Sol.**  $u = -15\text{cm}, f = +10\text{cm}; \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{(-15)} = \frac{1}{10} \Rightarrow v = 30\text{ cm}$

(i)  $x = v + 2f \Rightarrow 30 + 2 \times 12 = 54\text{ cm}$

(ii)  $x = v - 2f = 30 - 2 \times 12 = 6\text{ cm}$



- Ex.** A convex lens of focal length  $f$  is producing real image which is  $\frac{1}{n}$  times of the size of the object. Find out position of the object.

**Sol.** Image is real so  $m = \frac{v}{u} = -\frac{1}{n} \Rightarrow v = -\frac{u}{n}$

from lens formula  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\frac{u}{n}} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{-(n+1)}{u} = \frac{1}{f} \Rightarrow u = -f(1+n)$

- Ex.** (a) If  $f = +0.5\text{m}$ , what is the power of the lens ?  
 (b) The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. Its focal length is 12 cm. What is the refractive index of glass ?  
 (c) A convex lens has 20 cm focal length in air. What is the focal length in water ? (Refractive index of air-water = 1.33, refractive index for air glass is 1.5)

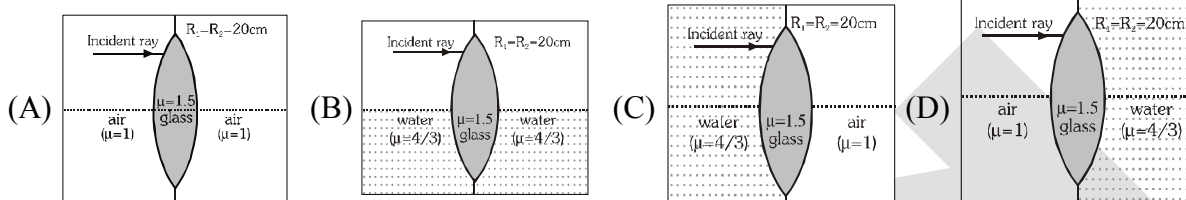
**Sol.** (a)  $P = \frac{1}{f(\text{m})} = \frac{1}{0.5} = +2\text{D}$

(b)  $\frac{1}{f} = (\mu - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{12} = (\mu - 1) \left( \frac{1}{10} - \frac{1}{-15} \right) \Rightarrow \mu = 1.5$

$$(c) \frac{1}{f} = (\mu - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \text{ so } \frac{1}{f} \propto (\mu - 1)$$

$$\frac{f_w}{f_a} = \frac{(\mu_l - 1)}{(\mu_w \mu_l - 1)} \Rightarrow f_w = \frac{(1.5 - 1)}{\left( \frac{1.5}{1.33} - 1 \right)} \times 20 = 78.2 \text{ cm}$$

**Ex. Column I (optical system)**



**Column II (focal length)**

(P) 80 cm

(Q) 40 cm

(R) 30 cm

(S) 20 cm

**Sol. Ans. (A) -S (B) -P (C) -R**

$$\text{For (A)} : \frac{1}{f} = (\mu - 1) \left( \frac{1}{p_1} + \frac{1}{p_2} \right) = (1.5 - 1) \left( \frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

$$\text{For (B)} : \frac{1}{f} = \left( \frac{1.5}{4/3} - 1 \right) \left( \frac{1}{20} + \frac{1}{20} \right) = \frac{1}{80} \Rightarrow f = 80 \text{ cm}$$

$$\text{For (C)} : \frac{1.5}{v_1} - \frac{4/3}{\infty} = \left( 1.5 - \frac{4}{3} \right) \left( \frac{1}{20} \right) \text{ \& } \frac{1}{f} - \frac{1.5}{v_1} = (1 - 1.5) \left( \frac{1}{-20} \right) \Rightarrow f = 30 \text{ cm}$$

$$\text{For (D)} : \frac{1.5}{v_1} - \frac{1}{\infty} = (1.5 - 1) \left( \frac{1}{20} \right) \text{ \& } \frac{4/3}{f} - \frac{1.5}{v_1} = \left( \frac{4}{3} - 1.5 \right) \left( \frac{1}{-20} \right) \Rightarrow f = 40 \text{ cm}$$

**DISPLACEMENT METHOD**

It is used for determination of focal length of convex lens in laboratory.

A thin convex lens of focal length  $f$  is placed between an object and a screen fixed at a distance  $D$  apart.

If  $D > 4f$  there are two position of lens at which a sharp image of the object is formed on the screen

$$\text{By lens formula } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{D-u} - \frac{1}{-u} = \frac{1}{f}$$

$$\Rightarrow u^2 - Du + Df = 0 \Rightarrow u = \frac{D \pm \sqrt{D(D-4f)}}{2}$$

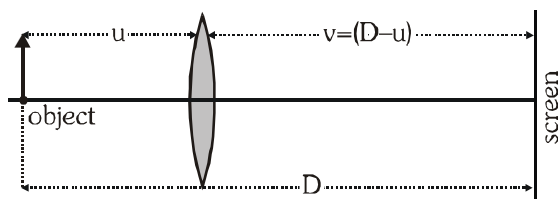
there are three possibilities

(i) for  $D < 4f$   $u$  will be imaginary hence physically no position of lens is possible

(ii) for  $D = 4f$   $u = \frac{D}{2} = 2f$  so only one position of lens is possible

$$\text{and since } v = D - u = 4f - 2f = u = 2f$$

(iii) for  $D > 4f$   $u_1 = \frac{D - \sqrt{D(D-4f)}}{2}$  and  $u_2 = \frac{D + \sqrt{D(D-4f)}}{2}$



So there are two positions of lens for which real image will be formed on the screen.  
(for two distances  $u_1$  and  $u_2$  of the object from lens)

If the distance between two positions of lens is  $x$

$$\text{then } x = u_2 - u_1 = \frac{D + \sqrt{D(D-4f)}}{2} - \frac{D - \sqrt{D(D-4f)}}{2}$$

$$= \sqrt{D(D-4f)} \Rightarrow x^2 = D^2 - 4Df \Rightarrow f = \frac{D^2 - x^2}{4D}$$

Distance of image corresponds to two positions of the lens :

$$v_1 = D - u_1 = D - \frac{1}{2}[D - \sqrt{D(D-4f)}] = \frac{1}{2}[D + \sqrt{D(D-4f)}] = u_2 \Rightarrow v_1 = u_2$$

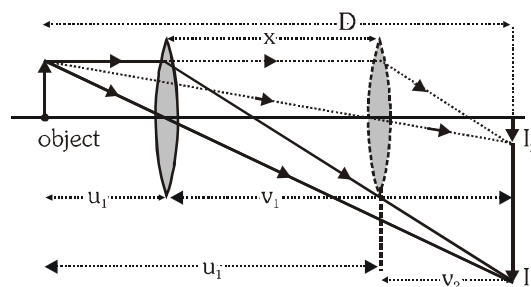
$$v_2 = D - u_2 = D - \frac{1}{2}[D + \sqrt{D(D-4f)}] = \frac{1}{2}[D - \sqrt{D(D-4f)}] = u_1 \Rightarrow v_2 = u_1$$

for two positions of the lens distances of object and image are interchangeable.

$$\text{Now } x = u_2 - u_1 \text{ and } D = v_1 + u_1 = u_2 + u_1 \quad [\because v_1 = u_2]$$

$$\text{so } u_1 (= v_2) = \frac{D-x}{2} \text{ and } v_1 = \frac{D+x}{2} = u_2; m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{D+x}{D-x} \text{ and } m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{D-x}{D+x}$$

$$\text{Now } m_1 \times m_2 = \frac{D+x}{D-x} \times \frac{D-x}{D+x} \Rightarrow \frac{I_1 I_2}{O^2} = 1 \Rightarrow O = \sqrt{I_1 I_2}$$



**Ex.** A convex lens is placed between an object and a screen which are at a fixed distance apart for one position of the lens. The magnification of the image obtained on the screen is  $m_1$ . When the lens is moved by a distance  $d$  the magnification of the image obtained on the same screen is  $m_2$ . Find the focal length of the lens.

**Sol.** If  $D$  is the distance between the object and the screen,  $d$  the separation of the two position of lens throwing two images on the screen then

$$m_1 = \frac{(D+d)}{(D-d)} \text{ and } m_2 = \frac{(D-d)}{(D+d)} \therefore m_1 - m_2 = \frac{4Dd}{D^2 - d^2} \text{ but } \frac{D^2 - d^2}{4D} = f \text{ so } m_1 - m_2 = \frac{d}{f} \Rightarrow f = \frac{d}{m_1 - m_2}$$

**Ex.** In a displacement method using lens, we obtain two images for separation of the lens  $d$ . One image is magnified as much as the other is diminished. If  $m$  is the magnifications of one image, find the focal length of the lens.

**Sol.** From above question  $f = \frac{d}{m_1 - m_2}$  here if  $m_1$  is taken as  $m$ ,  $m_2 = \frac{1}{m}$ , so  $f$  becomes  $\frac{md}{(m^2 - 1)}$

**Ex.** In the displacement method the distance between the object and the screen is 70 cm and the focal length of the lens is 16 cm, find the separations of the magnified and diminished image position of the lens.

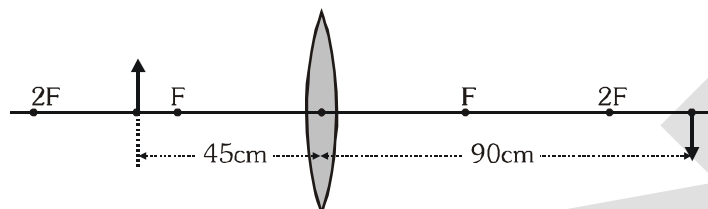
**Sol.**  $d = \sqrt{D^2 - 4fd} = \sqrt{(70)^2 - 4 \times 16 \times 70} = \sqrt{420} = 20.5 \text{ cm}$

**Ex.** An object 25 cm high is placed in front of a convex lens of focal length 30 cm. If the height of image formed is 50 cm, find the distance between the object and the image (real and virtual) ?

**Sol.** As object is in front of the lens, it is real and as  $h_1 = 25$  cm,  $f = 30$  cm,  $h_2 = -50$  cm ;

$$m = \frac{h_2}{h_1} = \frac{-50}{25} = -2$$

$$m = \frac{f}{f+u} \Rightarrow -2 = \frac{30}{30+u} \Rightarrow u = -45 \text{ cm} \Rightarrow m = \frac{v}{u} \Rightarrow -2 = \frac{v}{-45} \Rightarrow v = 90 \text{ cm}$$



As in this situation object and image are on opposite sides of lens, the distance between object and image  $d_1 = u + v = 45 + 90 = 135$  cm. If the image is erect (i.e., virtual)

$$m = \frac{f}{f+u} \Rightarrow 2 = \frac{30}{30+u} \Rightarrow u = -15 \text{ cm} \Rightarrow m = -\frac{v}{u} \Rightarrow 2 = \frac{-v}{-15} \Rightarrow v = 30 \text{ cm}$$

As in the situation both image and object are in front of the lens, the distance between object and image

$$d_2 = v - u = 30 - 15 = 15 \text{ cm.}$$

### COMBINATION OF LENSES AND MIRRORS

When several lenses or mirrors are used, the image formation is considered one after another in steps, The image formed by the lens facing the object serves as an object for the next lens or mirror, the image formed by the second lens acts as an object for the third, and so on, The total magnification in such situations will be given by

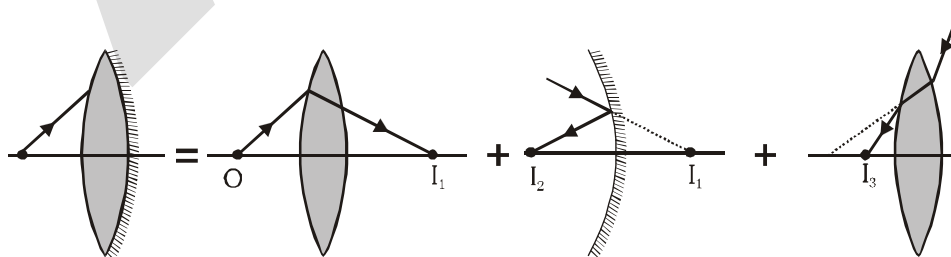
$$m = \frac{I}{O} = \frac{I_1}{O} \times \frac{I_2}{I_1} \times \dots \Rightarrow m = m_1 \times m_2 \times \dots$$

Power of Lens [in air]  $P_L = \frac{1}{f_L}$     Converging lens  $P_L = +ve$     Diverging lens  $P_L = -ve$

Power For mirror  $P_M = -\frac{1}{f_m}$     Convex mirror  $P_M = -ve$     Concave mirror  $P_M = +ve$

### SILVERING OF LENS

Calculate equivalent focal length of a equiconvex lens silvered at one side.



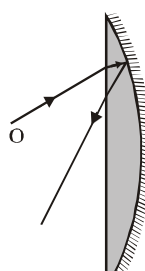
$$P = P_L + P_M + P_L = 2P_L + P_M$$

$$\frac{1}{F} = \frac{1}{f_L} + \frac{1}{f_m} + \frac{1}{f_L} = \frac{2}{f_L} + \frac{1}{f_m} = \frac{2(\mu-1) \times 2}{R} + \frac{2}{R} = \frac{4\mu-4+2}{R} \Rightarrow F = \frac{R}{4\mu-2}$$

**Ex.** Calculate equivalent focal length of plano convex lens for following case :-

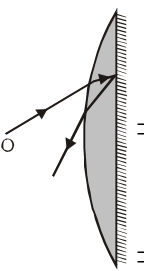
(i) When curved surface is silvered. (ii) When plane surface is silvered.

**Sol.** (i)  $P = 2P_L + P_M$  (ii)  $P = 2P_L + P_M$



$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_M} \Rightarrow \frac{1}{F} = \frac{2(\mu - 1)}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{F} = \frac{2\mu}{R} \Rightarrow F = \frac{R}{2\mu}$$



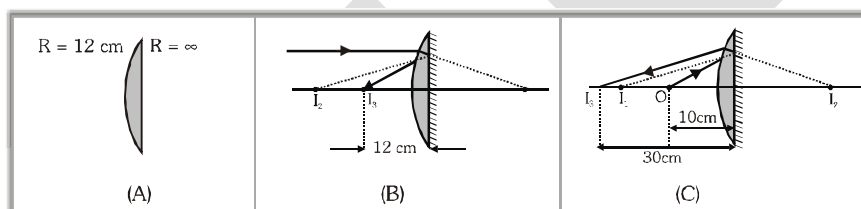
$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_M} \Rightarrow \frac{1}{F} = \frac{2(\mu - 1)}{R} + \frac{1}{\infty}$$

$$\Rightarrow F = \frac{R}{2(\mu - 1)}$$

**Ex.** The radius of curvature of the convex face of a plano-convex lens is 12 cm and its refractive index is 1.5. (a) Find the focal length of this lens. The plane surface of the lens is now silvered. (b) At what distance from the lens will parallel rays incident on the convex face converge? (c) Sketch the ray diagram to locate the image, when a point object is placed on the axis 20 cm from the lens. (d) Calculate the image distance when the object is placed as in (c).

**Sol.** (a) As for a lens, by lens-maker's formula  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$  Here  $\mu = 1.5$ ;  $R_1 = 12$  cm and  $R_2 = \infty$

So  $\frac{1}{f} = (1.5 - 1) \left[ \frac{1}{12} - \frac{1}{\infty} \right]$  i.e.  $f = 24$  cm i.e., the lens as convergent with focal length 24 cm.



(b) As light after passing through the lens will be incident on the mirror which will reflect it back through

the lens again, so  $P = P_L + P_M + P_L = 2P_L + P_M$  But  $P_L = \frac{1}{f_L} = \frac{1}{24}$  and  $P_M = -\frac{1}{\infty} = 0$  [as  $f_M = \frac{R}{2} = \infty$ ]

So  $P = 2 \times \frac{1}{24} + 0 = \frac{1}{12}$  D. The system is equivalent to a concave mirror of focal length  $F$ ,

$$P = -\frac{1}{F}$$

i.e.,  $F = -\frac{1}{P} = -0.12$  m = -12 cm i.e., the rays will behave as a concave mirror of focal length 12 cm.

So as for parallel incident rays  $u = -\infty$ , from mirror formula  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  we have  $\frac{1}{v} + \frac{1}{-\infty} = \frac{1}{-12}$   
 $\Rightarrow v = -12$  cm i.e., parallel incident rays will focus at a distance of 12 cm in front of the lens as shown in Figure (c) and (d) When object is at 20 cm in front of the given silvered lens which behaves

as a concave mirror of focal length 12 cm, from mirror formula  $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$  we have  $\frac{1}{v} + \frac{1}{-20} = \frac{1}{-12}$

$\Rightarrow v = -30$  cm i.e., the silvered lens will form image at a distance of 30 cm in front of it as shown in fig. (C)

**Ex.** A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image. Is the image real or virtual?

**Sol.** As radius of curvature of silvered surface is 22 cm, so

$$f_M = \frac{R}{2} = \frac{-22}{2} = -11 \text{ cm} = -0.11 \text{ m} \text{ and hence,}$$

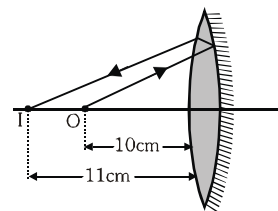
$$P_M = -\frac{1}{f_M} = -\frac{1}{-0.11} = \frac{1}{0.11} \text{ D}$$

Further as the focal length of lens is 20 cm, i.e., 0.20 m its power will be given by :

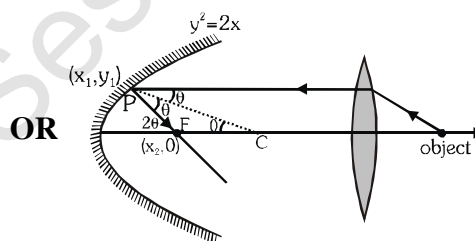
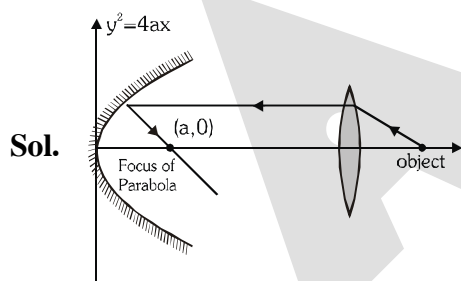
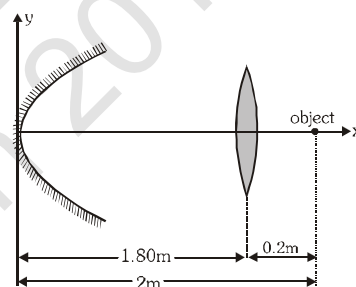
$$P_L = \frac{1}{f_L} = \frac{1}{0.20} \text{ D. Now as in image formation, light after passing through the lens will be reflected back}$$

$$\text{by the curved mirror through the lens again } P = P_L + P_M + P_L = 2P_L + P_M \text{ i.e. } P = \frac{2}{0.20} + \frac{1}{0.11} = \frac{210}{11} \text{ D.}$$

So the focal length of equivalent mirror  $F = -\frac{1}{P} = -\frac{11}{210} \text{ m} = -\frac{110}{21} \text{ cm}$  i.e., the silvered lens behave as a concave mirror of focal length  $(110/21) \text{ cm}$ . So for object at a distance 10 cm in front of it,  $\frac{1}{v} + \frac{1}{-10} = -\frac{21}{110}$  i.e.,  $v = -11 \text{ cm}$  i.e., image will be 11 cm in front of the silvered lens and will be real as shown in Figure.



**Ex.** A point object is kept at a distance of 2m from a parabolic reflecting surface  $y^2 = 2x$ . An equiconvex lens is kept at a distance of 1.80 m from the parabolic surface. The focal length of the lens is 20 cm. Find the position from origin of the image in cm, after reflection from the surface.



$$\text{Comparing with } y^2 = 4ax \Rightarrow a = 0.5$$

$$\Rightarrow \text{final position of image} = 0.5 \text{ m} = 50 \text{ cm}$$

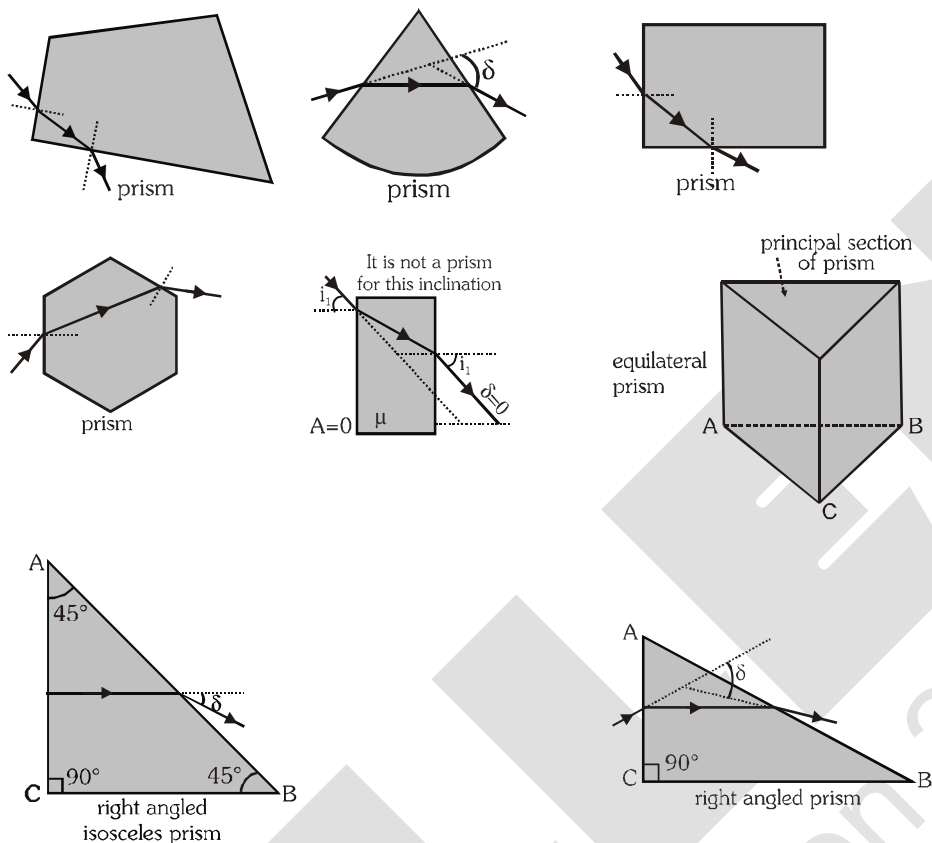
$$\text{PC is a normal so } \tan(\pi - \theta) = \frac{-1}{(dy/dx)_{x_1, y_1}} = -y_1$$

$$\text{But } \tan 2\theta = \frac{y_1 - 0}{x_2 - x_1} \text{ \& } \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\Rightarrow \frac{y_1}{x_2 - x_1} = \frac{2(y_1)}{1 - y_1^2} x_2 = \frac{1}{2} \text{ m}$$

## PRISM

A prism is a homogeneous, transparent medium (such as glass) enclosed by two plane surfaces inclined at an angle. These surfaces are called the 'refracting surfaces' and the angle between them is called the 'refracting angle' or the 'angle of prism'. The section cut by a plane perpendicular to the refracting surfaces is called the 'principal section' of the prism.



### DEVIATION

PQ = incident ray

QR = Refracted ray

RS = emergent ray

A = Prism angle

$i_1$  = incident angle on face AB

$i_2$  = emergent angle on face AC

$r_1$  = refracted angle on face AB

$r_2$  = incident angle on face AC

Angle of deviation on face AB.

Angle of deviation on face AC

Total angle of deviation

$$\delta = \delta_1 + \delta_2 \Rightarrow \delta = (i_1 - r_1) + (i_2 - r_2) = i_1 + i_2 - (r_1 + r_2) \dots (i)$$

$$\text{In } \triangle QOR \quad r_1 + r_2 + \theta = 180^\circ \dots (ii)$$

$$\text{In } \triangle AQR \quad A + \theta = 180^\circ \dots (iii)$$

$$\text{from (ii) and (iii)} \quad r_1 + r_2 = A \dots (iv)$$

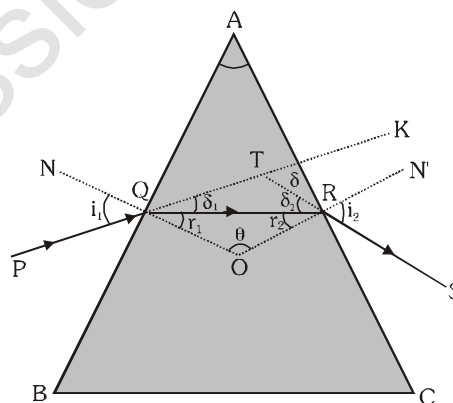
$$\text{from (i) and (iv)} \quad \text{Total angle of deviation } \delta = i_1 + i_2 - A$$

from Snell's law at surface AB

$$\mu_1 \sin i_1 = \mu_2 \sin r_1$$

and at surface AC

$$\mu_2 \sin r_2 = \mu_1 \sin i_2$$



## CONDITION OF MINIMUM DEVIATION

### For minimum deviation

In this condition  $i_1 = i_2 = i \Rightarrow r_1 = r_2 = r$  and since  $r_1 + r_2 = A \therefore r + r = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$

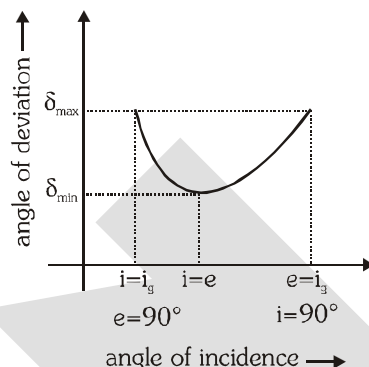
Minimum deviation  $\delta_{\min} = 2i - A$ ;  $i = \frac{A + \delta_{\min}}{2}$ ,  $r = \frac{A}{2}$

if prism is placed in air  $\mu_1 = 1$ ;  $1 \times \sin i = \mu \sin r$

$$\sin \left[ \frac{A + \delta_{\min}}{2} \right] = \mu \sin \frac{A}{2} \Rightarrow \mu = \frac{\sin \left[ \frac{A + \delta_{\min}}{2} \right]}{\sin \frac{A}{2}}$$

if angle of prism is small  $A < 10^\circ$  then  $\sin \theta \approx \theta$

$$\mu = \frac{\frac{A + \delta_{\min}}{2}}{\frac{A}{2}} = \frac{A + \delta_{\min}}{A} \Rightarrow A + \delta_{\min} = \mu A \Rightarrow \delta_{\min} = (\mu - 1)A$$



## CONDITION FOR MAXIMUM DEVIATION/GRAZING EMERGENCE

- **Angle of incidence ( $i_g$ ) for grazing emergence**

For  $i_g$ ,  $e = 90^\circ$

Applying Snell's law at face AC

$$\mu \sin r_2 = 1 \times 1 \Rightarrow \sin r_2 = \frac{1}{\mu}; r_2 = \sin^{-1} \left( \frac{1}{\mu} \right) = \theta_c$$

$$r_1 + r_2 = A \Rightarrow r_1 = A - \theta_c$$

Again, Applying Snell's law at face AB

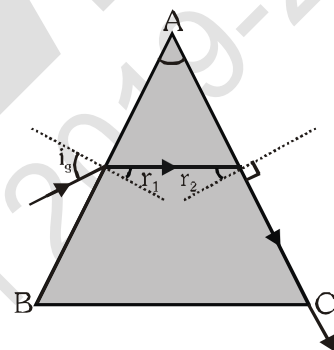
$$1 \times \sin i_g = \mu \sin r_1; 1 \times \sin i_g = \mu \sin (A - \theta_c)$$

$$\sin i_g = \mu [\sin A \cos \theta_c - \cos A \sin \theta_c]$$

$$i_g = \sin^{-1} \left[ \sqrt{\mu^2 - 1} \sin A - \cos A \right] \quad \left[ \begin{array}{l} \text{as } \sin \theta_c = \frac{1}{\mu}, \cos \theta_c = \frac{\sqrt{\mu^2 - 1}}{\mu} \end{array} \right]$$

If  $i$  increases beyond  $i_g$ ,  $r_1$  increases thus  $r_2$  decreases and becomes less than  $\theta_c$  and ray emerges.

Thus  $i \geq i_g \Rightarrow$  ray emerges, otherwise TIR.  $\delta_{\max} = i_g + 90^\circ - A$



## NO EMERGENCE CONDITION

Let maximum incident angle on the face AB  $i_{\max} = 90^\circ$

$$1 \times \sin 90^\circ = \mu \sin r_1; \sin r_1 = \frac{1}{\mu} = \sin \theta_c; r_1 = \theta_c \dots (i)$$

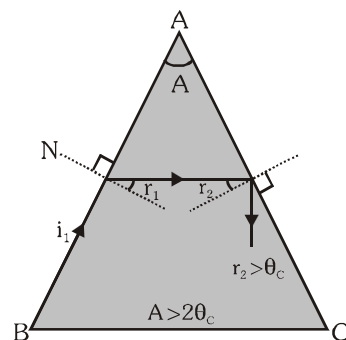
if TIR occur at face AC then

$$r_2 > \theta_c \dots (ii)$$

$$r_1 + r_2 = A \dots (iii)$$

$$\text{from (i) and (ii) } r_1 + r_2 > \theta_c + \theta_c \Rightarrow r_1 + r_2 > 2\theta_c \dots (iv)$$

$$\text{from (iii) and (iv) } A > 2\theta_c \Rightarrow \frac{A}{2} > \theta_c \Rightarrow \sin \frac{A}{2} > \sin \theta_c \Rightarrow \sin \frac{A}{2} > \frac{1}{\mu} \Rightarrow \frac{1}{\sin \frac{A}{2}} < \mu$$



**Ex.** A ray of light passes through an equilateral prism such that angle of incidence is equal of emergence and the later is equal to  $3/4^{\text{th}}$  of the angle of prism. Calculate the angle of deviation. Refractive index of prism is 1.5.

**Sol.**  $A = 60^\circ$ ,  $\mu = 1.5$ ;  $i_1 = i_2 = \frac{3}{4}A = 45^\circ$ ,  $\delta = ?$

$$\therefore A + \delta = i_1 + i_2$$

$$\therefore 60^\circ + \delta = 45^\circ + 45^\circ \Rightarrow \delta = 90^\circ - 60^\circ = 30^\circ$$

**Ex.** A prism of refractive index 1.53 is placed in water of refractive index 1.33. If the angle of prism is  $60^\circ$ , calculate the angle of minimum deviation in water. ( $\sin 35.1^\circ = 0.575$ )

**Sol.** Here,  ${}^a\mu_g = 1.33$ ,  ${}^a\mu_w = 1.53$ ,  $A = 60^\circ$ ,  $\delta_m = ?$   ${}^w\mu_g = \frac{{}^a\mu_g}{{}^a\mu_w} = \frac{1.53}{1.33} = 1.15$   $\therefore {}^w\mu_g = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$

$$\therefore \frac{\sin(A + \delta_m)}{2} = {}^w\mu_g \times \sin \frac{A}{2} = 1.15 \sin \frac{60^\circ}{2} = 0.575 \Rightarrow \frac{A + \delta_m}{2} = \sin^{-1}(0.575) = 35.1^\circ$$

$$\therefore \delta_m = 35.1 \times 2 - 60 = 10.2^\circ$$

### KEY POINTS

- Angle of prism or refracting angle of prism means the angle between the faces on which light is incident and from which it emerges.
- If the faces of a prism on which light is incident and from which it emerges are parallel then the angle of prism will be zero and as incident ray will emerge parallel to itself, deviation will also be zero, i.e., the prism will act as a transparent plate.
- If  $\mu$  of the material of the prism is equal to that of surroundings, no refraction at its faces will take place and light will pass through it undeviated, i.e.,  $\delta = 0$ .

### DISPERSION OF LIGHT

When white light is incident on a prism then it is splitted into seven colours. This phenomenon is known as dispersion. Prism introduces different refractive index with different wavelength

$$\text{As } \delta_{\min} = (\mu - 1)A \therefore \lambda_R > \lambda_V$$

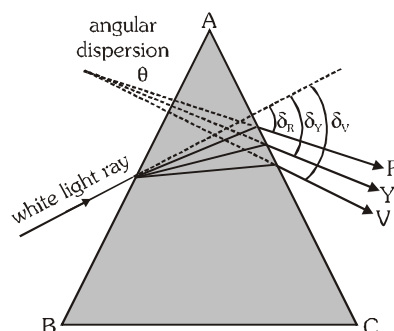
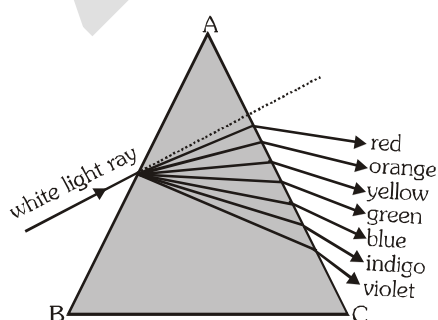
$$\text{So } \mu_V > \mu_R \Rightarrow \delta_{\text{m(violet)}} > \delta_{\text{m(red)}}$$

### ANGULAR DISPERSION

It is the difference of angle of deviation for violet colour and red colour

$$\text{Angular dispersion } \theta = \delta_V - \delta_R = (\mu_V - 1)A - (\mu_R - 1)A = (\mu_V - \mu_R)A$$

It depends on prism material and on the angle of prism  $\theta = (\mu_V - \mu_R)A$



### DISPERSIVE POWER ( $\omega$ )

It is ratio of angular dispersion ( $\theta$ ) to mean colour deviation ( $\delta_y$ )

$$\text{Dispersive power } \omega = \frac{\theta}{\delta_y} \Rightarrow \omega = \frac{(\mu_V - \mu_R)A}{(\mu_y - 1)A} = \frac{\mu_V - \mu_R}{\mu_y - 1} \Rightarrow \omega = \frac{\mu_V - \mu_R}{\mu_y - 1}$$

$$\text{Refractive index of mean colour } \mu_y = \frac{\mu_V + \mu_R}{2}$$

Dispersive power depends only on the material of the prism.

### COMBINATION OF PRISM

#### Deviation without dispersion ( $\theta = 0^\circ$ )

Two or more than two thin prism are combined in such a way that deviation occurs i.e. emergent light ray makes angle with incident light ray but dispersion does not occur i.e., light is not splitted into seven colours.

$$\text{Total dispersion } \theta = \theta_1 + \theta_2 = (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A'$$

$$\text{For no dispersion } \theta = 0; (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A' = 0$$

$$\text{Therefore, } A' = -\frac{(\mu_V - \mu_R)A}{\mu_V - \mu_R}$$

—ve sign indicates that prism angles are in opposite direction.

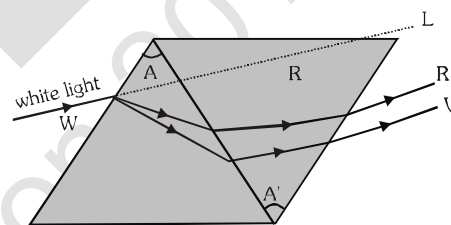
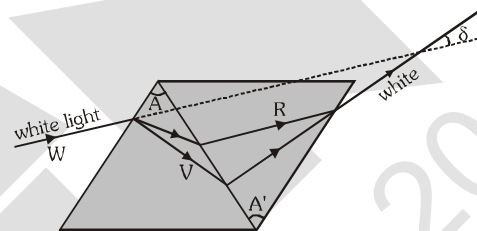
#### Dispersion without deviation ( $\delta = 0^\circ$ )

Two or more than two prisms combine in such a way that dispersion occurs i.e., light is splitted into seven colours but deviation do not occur i.e., emergent light ray becomes parallel to incident light ray.

$$\text{Total deviation } \delta = \delta_1 + \delta_2$$

$$\Rightarrow \delta = 0; (\mu - 1)A + (\mu' - 1)A' = 0 \Rightarrow A' = -\frac{(\mu - 1)A}{\mu' - 1}$$

—ve sign indicates that prism angles are in opposite direction.



### KEY POINTS

- Dispersive power like refractive index has no units and dimensions and depends on the material of the prism and is always positive.
- As for a given prism dispersive power is constant, i.e., dispersion of different wavelengths will be different and will be maximum for violet and minimum for red (as deviation is maximum for violet and minimum for red).
- As for a given prism  $\theta \propto \delta$  a single prism produces both deviation and dispersion simultaneously, i.e., a single prism cannot give deviation without dispersion or dispersion without deviation.

**Ex.** White light is passed through a prism of angle  $5^\circ$ . If the refractive indices for red and blue colours are 1.641

and 1.659 respectively, calculate the angle of dispersion between them.

**Sol.** As for small angle of prism  $\delta = (\mu - 1)A$ ,

$$\delta_B = (1.659 - 1) \times 5^\circ = 3.295^\circ \text{ and } \delta_R = (1.641 - 1) \times 5^\circ = 3.205^\circ$$

$$\text{so } \theta = \delta_B - \delta_R = 3.295^\circ - 3.205^\circ = 0.090^\circ$$

**Ex.** Prism angle of a prism is  $10^\circ$ . Their refractive index for red and violet color is 1.51 and 1.52 respectively. Then find the dispersive power.

**Sol.** Dispersive power of prism  $\omega = \left( \frac{\mu_v - \mu_r}{\mu_y - 1} \right)$  but  $\mu_y = \frac{\mu_v + \mu_r}{2} = \frac{1.52 + 1.51}{2} = 1.515$

$$\text{Therefore } \omega = \frac{1.52 - 1.51}{1.515 - 1} = \frac{0.01}{1.515} = 0.019$$

**Ex.** The refractive indices of flint glass for red and violet colours are 1.644 and 1.664. Calculate its dispersive power.

**Sol.** Here,  $\mu_r = 1.644$ ,  $\mu_v = 1.664$ ,  $\omega = ?$

$$\text{Now } \mu_y = \frac{\mu_v + \mu_r}{2} = \frac{1.664 + 1.644}{2} = 1.654 \quad \therefore \omega = \frac{\mu_v - \mu_r}{\mu_y - 1} = \frac{1.664 - 1.644}{1.654 - 1} = 0.0305$$

**Ex.** In a certain spectrum produced by a glass prism of dispersive power 0.031, it was found that  $\mu_r = 1.645$  and  $\mu_v = 1.665$ . What is the refractive index for yellow colour?

**Sol.** Here,  $\omega = 0.031$ ,  $\mu_r = 1.645$ ,  $\mu_v = 1.665$ ,  $\mu_y = ?$

$$\therefore \omega = \frac{\mu_v - \mu_r}{\mu_y - 1} \quad \therefore \mu_y - 1 = \frac{\mu_v - \mu_r}{\omega} = \frac{1.665 - 1.645}{0.031} = \frac{0.020}{0.31} = 0.645 \quad \therefore \mu_y = 0.645 + 1 = 1.645$$

**Ex.** A combination of two prisms, one of flint and other of crown glass produces dispersion without deviation. The angle of flint glass prism is  $15^\circ$ . Calculate the angle of crown glass prism and angular dispersion of red and violet. ( $\mu$  for crown glass = 1.52,  $\mu$  for flint glass = 1.65,  $\omega$  for crown glass 0.20,  $\omega$  for flint glass = 0.03).

**Sol.** Here,  $A = 15^\circ$ ,  $A' = ?$ ,  $\omega = 0.03$ ,  $\omega' = 0.02$ ,  $\mu = 1.65$ ,  $\mu' = 1.52$ ,  
For no deviation,  $\delta + \delta' = 0$

$$(\mu - 1)A + (\mu' - 1)A' = 0 \Rightarrow (1.65 - 1)15^\circ + (1.52 - 1)A' = 0 \Rightarrow A' = \frac{-0.65 \times 15}{0.52} = -18.75^\circ$$

Negative sign indicates that two prisms must be joined in opposition. Net angular dispersion

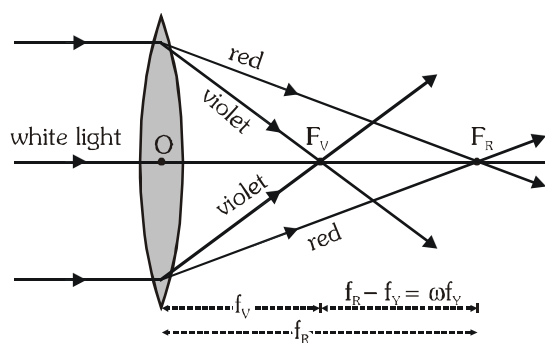
$$(\mu_v - \mu_r)A + (\mu'_v - \mu'_r)A' = \omega(\mu - 1)A + \omega'(\mu' - 1)A' = 0.03(1.65 - 1)15^\circ + 0.02(1.52 - 1)(-18.75^\circ) \\ = 0.2925 - 0.195 = 0.0975^\circ$$

## CHROMATIC ABERRATION

The image of an object in white light formed by a lens is usually colored and blurred. This defect of image is called chromatic aberration and arises due to the fact that focal length of a lens is different for different colors.

For a single lens  $\frac{1}{f} = (\mu - 1) \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$  and as  $\mu$  of lens is

maximum for violet while minimum for red, violet is focused nearest to the lens while red farthest from it. It is defect of lens.



### Longitudinal or Axial Chromatic Aberration

When a white object O is situated on the axis of a lens, then images of different colors are formed at different points along the axis. The formation of images of different colors at different positions is called 'axial' or longitudinal chromatic aberration. The axial distance between the red and the violet images  $I_R - I_V$  is known as longitudinal aberration. When white light is incident on lens, image is obtained at different point on the axis because focal length of lens depend on wavelength.  $f \propto \lambda \Rightarrow f_R > f_V$

$$f_R - f_V = \omega f_y \Rightarrow \text{Axial or longitudinal chromatic aberration}$$

If the object is at infinity, then the longitudinal chromatic aberration is equal to the difference in focal-lengths ( $f_R - f_V$ ) for the red and the violet rays.

### LATERAL CHROMATIC ABERRATION

As the focal-length of the lens varies from color

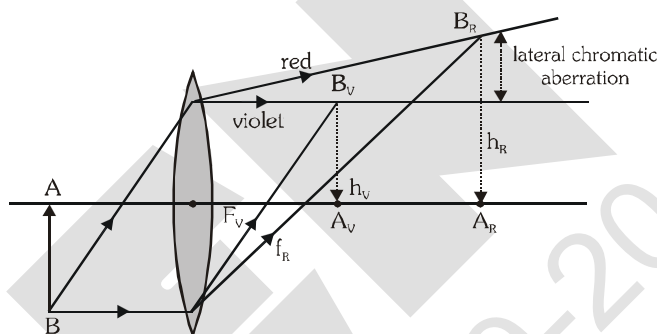
to color, the magnification  $m = \left[ \frac{f}{u+f} \right]$

produced by the lens also varies from color to color.

Therefore, for a finite-size white object AB, the images of different colors formed by the lens are of different sizes.

The formation of images of different colors in different sizes is called lateral chromatic aberration.

The difference in the height of the red image  $B_R A_R$  and the violet image  $B_V A_V$  is known as lateral chromatic aberration.  $LCA = h_R - h_V$



### ACHROMATISM

If two or more lens combined together in such a way that this combination produce image at a same point then this combination is known as achromatic combination of lenses.

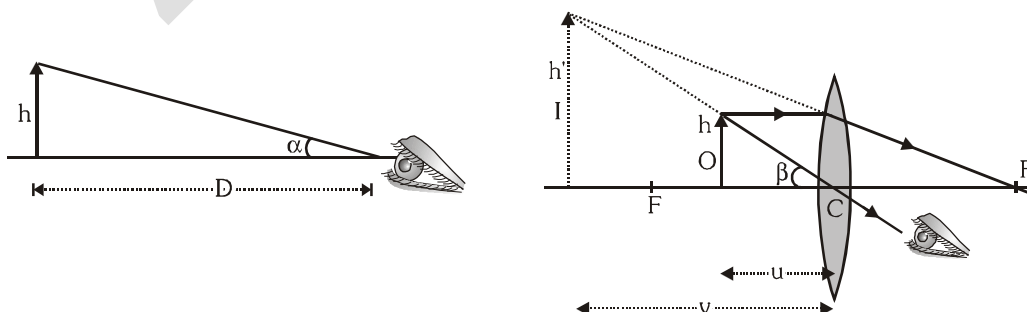
$$\frac{\omega}{f_y} + \frac{\omega'}{f'_y} = 0 \Rightarrow \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \Rightarrow \frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2}$$

For combination of lens.  $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$  (Apply sign convention in numerical)

### OPTICAL INSTRUMENTS

#### Simple microscope

When object is placed between focus and optical centre a virtual, magnified and erect image is formed



$$\text{Magnifying power (MP)} = \frac{\text{visual angle with instrument } (\beta)}{\text{maximum visual angle for unaided eye } (\alpha)} \Rightarrow \text{MP} = \frac{\frac{h}{-u}}{\frac{h}{-D}} = \frac{D}{u}$$

(i) When the image is formed at infinity :

$$\text{by lens equation } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\infty} - \frac{1}{-u} = \frac{1}{f} \Rightarrow u = f \quad \text{So } \text{MP} = \frac{D}{u} = \frac{D}{f}$$

(ii) If the image is at minimum distance of clear vision  $D$  :

$$\frac{1}{-D} - \frac{1}{-u} = \frac{1}{f} \Rightarrow \frac{1}{u} = \frac{1}{D} + \frac{1}{f} \quad [v = -D \text{ and } u = -ve]$$

$$\text{Multiplying by } D \text{ both the sides } \frac{D}{u} = 1 + \frac{D}{f} \Rightarrow \text{MP} = \frac{D}{u} = 1 + \frac{D}{f}$$

**Ex.** A man with normal near point 25 cm reads a book with small print using a magnifying glass, a thin convex lens of focal length 5 cm.

(a) What is the closest and farthest distance at which he can read the book when viewing through the magnifying glass ?

(b) What is the maximum and minimum MP possible using the above simple microscope ?

**Sol.** (a) As for normal eye far and near point are  $\infty$  and 25 cm respectively, so for magnifier  $v_{\text{max}} = \infty$  and

$$v_{\text{min}} = -25 \text{ cm. However, for a lens as } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow u = \frac{f}{(f/v) - 1}$$

$$\text{So } u \text{ will be minimum when } v = \text{minimum} = -25 \text{ cm i.e., } (u)_{\text{min}} = \frac{5}{-(5/25) - 1} = -\frac{25}{6} = -4.17 \text{ cm}$$

$$\text{Ans } u \text{ will be maximum when } v = \text{maximum} = \infty \text{ i.e., } u_{\text{max}} = \frac{5}{\left(\frac{5}{\infty} - 1\right)} = -5 \text{ cm}$$

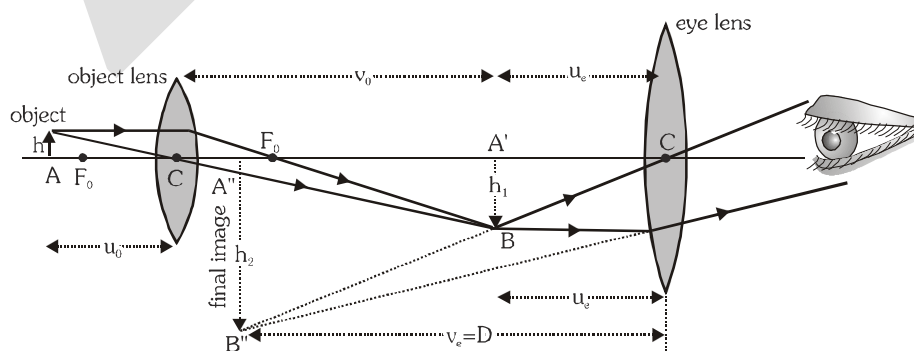
So the closest and farthest distance of the book from the magnifier (or eye) for clear viewing are 4.17 cm and 5 cm respectively.

(b) As in case of simple magnifier  $\text{MP} = (D/u)$ . So MP will be minimum when  $u = \text{max} = 5 \text{ cm}$

$$\Rightarrow (\text{MP})_{\text{min}} = \frac{-25}{-5} = 5 \left[ = \frac{D}{f} \right] \text{ and MP will be maximum when } u = \text{min} = (25/6) \text{ cm}$$

$$\Rightarrow (\text{MP})_{\text{max}} = \frac{-25}{-(25/6)} = 6 \left[ = 1 + \frac{D}{f} \right]$$

## COMPOUND MICROSCOPE



Compound microscope is used to get more magnified image. Object is placed in front of objective lens and image is seen through eye piece. The aperture of objective lens is less as compared to eye piece because object is very near so collection of more light is not required. Generally object is placed between  $F - 2F$  due to this a real inverted and magnified image is formed between  $2F - \infty$ . It is known as intermediate image  $A'B'$ . The intermediate image acts as an object for eye piece. Now the distance between both the lenses is adjusted in such a way that intermediate image falls between the optical centre of eye piece and its focus. In this condition, the final image is virtual, inverted and magnified.

$$\text{Total magnifying power} = \text{Linear magnification} \times \text{angular magnification} \quad MP = m_o m_e = \frac{v_o}{u_o} \frac{D}{u_e}$$

(i) When final image is formed at minimum, distance of distinct vision.

$$MP = \frac{v_o}{u_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[ 1 + \frac{D}{f_e} \right] = \frac{h_2}{h_1} \left( 1 + \frac{D}{f_e} \right)$$

$$\text{Length of the tube} = v_o + |u_e|$$

(ii) When final image is formed at infinity  $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{\infty} + \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e$

$$MP = \frac{v_o}{u_o} \left[ \frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[ \frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[ \frac{D}{f_e} \right] = \frac{h_2}{h_1} \left[ \frac{D}{f_e} \right]. \text{ Length of the tube } L = v_o + f_e$$

Sign convention for solving numerical  $u_o = -ve, v_o = +ve, f_o = +ve,$   
 $u_e = -ve, v_e = -ve, f_e = +ve, m_o = -ve, m_e = +ve, M = -ve$

**Ex.** A thin convex lens of focal length 5 cm is used as a simple microscope by a person with normal near point (25 cm). What is the magnifying power of the microscope?

**Sol.** Here,  $f = 5$  cm;  $D = 25$  cm,  $M = ?$   $MP = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$

**Ex.** A compound microscope consists of an objective lens of focal length 2.0 cm and an eye piece of focal length 6.25 cm, separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at (a) the least distance of distinct vision (25 cm) (b) infinity?

**Sol.** Here,  $f_o = 2.0$  cm;  $f_e = 6.25$  cm,  $u_o = ?$

$$(a) v_e = -25 \text{ cm} \therefore \frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \therefore \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{6.25} = \frac{-1-4}{25} = \frac{-5}{25} \Rightarrow u_e = -5 \text{ cm}$$

As distance between objective and eye piece = 15 cm;  $v_o = 15 - 5 = 10$  cm

$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \therefore \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2} = \frac{1-5}{10} \Rightarrow u_o = \frac{-10}{4} = -2.5 \text{ cm}$$

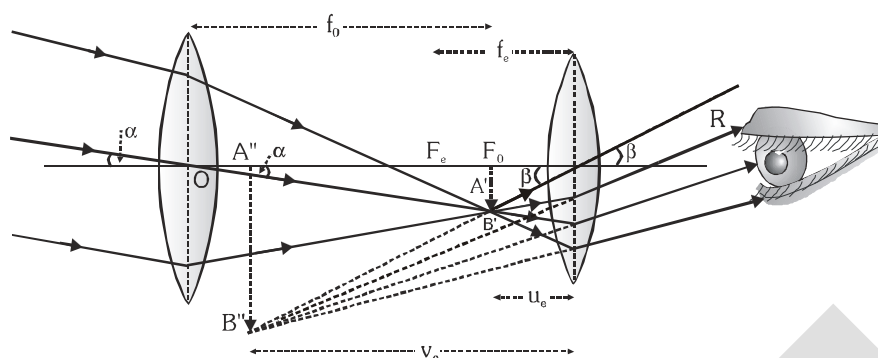
$$\text{Magnifying power} = \frac{v_o}{|u_o|} \left[ 1 + \frac{D}{f_e} \right] = \frac{10}{2.5} \left[ 1 + \frac{25}{6.25} \right] = 20$$

(b)  $\therefore v_e = \infty, u_e = f_e = 6.25$  cm  $\therefore v_o = 15 - 6.25 = 8.75$  cm.

$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \Rightarrow \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{8.75} - \frac{1}{2} = \frac{2-8.75}{17.5} \Rightarrow u_o = \frac{-17.5}{6.75} = -2.59 \text{ cm}$$

$$\text{Magnifying power} = \frac{v_o}{|u_o|} \times \left[ 1 + \frac{D}{f_e} \right] = \frac{v_o}{|u_o|} \times \frac{D}{|u_e|} = \frac{8.75}{2.59} \times \frac{25}{6.25} = 13.51$$

## ASTRONOMICAL TELESCOPE



A telescope is used to see distant object, objective lens forms the image  $A'B'$  at its focus. This image  $A'B'$  acts as an object for eyepiece and it forms final image  $A''B''$ .

$$MP = \frac{\text{visual angle with instrument } (\beta)}{\text{visual angle for unaided eye } (\alpha)} \Rightarrow MP = \frac{\frac{h'}{f_0}}{\frac{h}{-u_e}} = -\frac{f_0}{u_e} [A'B' = h']$$

- (i) If the final image is at infinity  $v_e = -\infty$ ,  $u_e = -ve$

$$\frac{1}{-\infty} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e. \text{ So } MP = -\frac{f_0}{f_e} \text{ and length of the tube } L = f_0 + f_e$$

- (ii) If the final image is at  $D$ :  $v_e = -D$   $u_e = -ve$

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} = \frac{1}{f_e} \left[ 1 + \frac{f_e}{D} \right] \text{ So } MP = -\frac{f_0}{u_e} = -\frac{f_0}{f_e} \left[ 1 + \frac{f_e}{D} \right]$$

Length of the tube is  $L = f_0 + u_e$

| S. No. | Compound - Microscope                                                                                                          | S. No. | Astronomical - Telescope                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | It is used to increase visual angle of near tiny object.                                                                       | 1.     | It is used to increase visual angle of distant large objects.                                                                         |
| 2.     | In it field and eye lens both are convergent, of short focal length and aperture.                                              | 2.     | In it objective lens is of large focal length and aperture while eye lens of short focal length and aperture and both are convergent. |
| 3.     | Final image is inverted, virtual and enlarged and at a distance $D$ to $\infty$ from the eye.                                  | 3.     | Final image is inverted, virtual and enlarged at a distance $D$ to $\infty$ from the eye.                                             |
| 4.     | MP does not change appreciably if objective and eye lens are interchanged as $[MP \sim (LD / f_0 f_e)]$                        | 4.     | MP becomes $(1/m^2)$ times of its initial value if objective and eye-lenses are interchanged as $MP \sim [f_0 / f_e]$                 |
| 5.     | MP is increased by decreasing the focal length of both the lenses.                                                             | 5.     | MP is increased by increasing the focal length of objective lens and by decreasing the focal length of eyepiece                       |
| 6.     | RP is increased by decreasing the wavelength of light used.<br>$\left( \because RP = \frac{2\mu \sin \theta}{\lambda} \right)$ | 6.     | RP is increased by increasing the aperture of objective.<br>$\left( \because RP = \frac{D}{1.22\lambda} \right)$                      |

**Ex.** A small telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6.0 cm. What is the magnifying power of the telescope? What is the separation between the objective and the eyepiece? When final image is formed at infinity.

**Sol.** Here,  $f_o = 144$  cm;  $f_e = 6.0$  cm,  $MP = ?$ ,  $L = ?$

$$MP = \frac{-f_o}{f_e} = \frac{-144}{6.0} = -24 \text{ and } L = f_o + f_e = 144 + 6.0 = 150.0 \text{ cm}$$

**Ex.** Diameter of the moon is  $3.5 \times 10^3$  km and its distance from earth is  $3.8 \times 10^5$  km. It is seen by a telescope whose objective and eyepiece have focal lengths 4m and 10cm respectively. What will the angular diameter of the image of the moon.

**Sol.**  $MP = -\frac{f_o}{f_e} = -\frac{400}{10} = -40$ . Angle subtended by the moon at the objective  $= \frac{3.5 \times 10^3}{3.8 \times 10^5} = 0.009$  radian.

$$\text{Thus angular diameter of the image} = MP \times \text{visual angle} = 40 \times 0.009 = 0.36 \text{ radian} = \frac{0.36 \times 180}{3.14} \approx 21^\circ$$

**Ex.** A telescope consisting of an objective of focal length 60 cm and a single-lens eyepiece of focal length 5 cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of  $2^\circ$  at the objective, then find the angular width of the image.

**Sol.**  $MP = \frac{f_o}{f_e} = \frac{\beta}{\alpha} \Rightarrow \beta = \alpha \frac{f_o}{f_e} = 2^\circ \times \frac{60}{5} = 24^\circ$

**Ex.** The focal lengths of the objective and the eye piece of an astronomical telescope are 60 cm and 5 cm respectively. Calculate the magnifying power and the length of the telescope when the final image is formed at (i) infinity, (ii) least distance of distinct vision (25 cm)

**Sol.** (i) When the final image is at infinity, then :

$$MP = -\frac{f_o}{f_e} = -\frac{60}{5} = -12 \text{ and length of the telescope is } L = f_o + f_e = 60 + 5 = 65 \text{ cm}$$

(ii) For least distance of distinct vision, the magnifying power is :

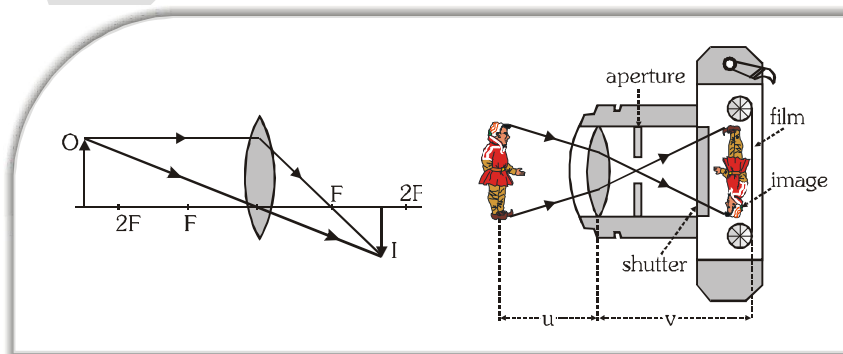
$$MP = -\frac{f_o}{f_e} \left( 1 + \frac{f_e}{D} \right) = -\frac{60}{5} \left( 1 + \frac{5}{25} \right) = -\frac{12 \times 6}{5} = -14.4$$

$$\text{Now } \frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e} \Rightarrow \frac{1}{5} = -\frac{1}{25} - \frac{1}{u_e} \Rightarrow \frac{-1}{u_e} = \frac{1}{25} + \frac{1}{5} \Rightarrow u_e = -4.17 \text{ cm} \Rightarrow |u_e| = 4.17 \text{ cm}$$

$$\text{The length of telescope in this position is } L = f_o + |u_e| = 60 + 4.17 = 64.17 \text{ cm}$$

## LENS – CAMERA

There is a convex lens whose aperture and distance from the film can be adjusted. Object is real and placed between  $\infty$  and  $2F$ , so the image is real, inverted diminished and between  $F$  and  $2F$ .



If  $I$  is the intensity of light,  $S$  is the light transmitting area of lens and  $t$  is the exposure time,

then for proper exposure,  $I \times S \times t = \text{constant}$

light transmitting area of a lens is proportional to the square of its aperture  $D$ ;  $I \times D^2 \times t = \text{constant}$

If aperture is kept fixed, for proper exposure,  $I \times t = \text{constant}$ , i.e.,  $I_1 t_1 = I_2 t_2$

If intensity is kept fixed, for proper exposure,  $D^2 \times t = \text{constant}$

$$\text{Time of exposure} \propto \frac{1}{(\text{aperture})^2} \dots (i)$$

The ratio of focal length to the aperture of lens is called  $f$ -number of the camera,

$$f\text{-number} = \frac{\text{focal length}}{\text{aperture}} \Rightarrow \text{Aperture} \propto \frac{1}{f\text{-number}} \dots (ii)$$

From equation (i) and (ii)  $\Rightarrow \text{Time of exposure} \propto (f\text{-number})^2$

**Ex.** With diaphragm of the camera lens set at  $\frac{f}{2}$ , the correct exposure time is  $\frac{1}{100}$ , then with diaphragm set at  $\frac{f}{4}$ . Calculate the correct exposure time.

**Sol.** As exposure time  $\propto \frac{1}{(\text{aperture})^2} \Rightarrow t_1 \propto \frac{1}{[f/2]^2}$  and  $t_2 \propto \frac{1}{[f/4]^2}$

$$\text{here } t_1 = \frac{1}{100} \text{ s then } \frac{t_2}{t_1} = \frac{16}{4} = 4 \Rightarrow t_2 = 4t_1 = \frac{4}{100} \text{ s}$$

**Ex.** A good photographic print is obtained by an exposure of two seconds at a distance of 20 cm from the lamp. Calculate the time of exposure required to get an equally good result at a distance of 40 cm.

**Sol.** We know that the intensity of light varies inversely as the  $(\text{distance})^2$ . When distance is doubled, the intensity becomes one-fourth. So, the time of exposure should be four times. Hence, time of exposure  $= 2 \times 4 = 8 \text{ s}$

**Ex.** Photograph of the ground are taken from an aircraft, flying at an altitude of 2000 m, by a camera with a lens of focal length 50 cm. The size of the film in the camera is  $18 \text{ cm} \times 18 \text{ cm}$ . What area of the ground can be photographed by this camera at any one time.

**Sol.** As here  $u = -2000 \text{ m}$ ,  $f = 0.50 \text{ m}$ , so from lens formula  $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ ,

$$\text{we have } \frac{1}{v} - \frac{1}{(-2000)} = \frac{1}{0.5} \Rightarrow \frac{1}{v} = \frac{1}{0.5} - \frac{1}{2000} \approx \frac{1}{0.5} \left[ \text{as } \frac{1}{0.5} \gg \frac{1}{2000} \right] \Rightarrow v = 0.5 \text{ m} = 50 \text{ cm} = f$$

$$\text{Now as in case of a lens, } m = \frac{v}{u} = \frac{0.5}{-2000} = -\frac{1}{4} \times 10^{-3} \text{ So } I_1 = (ma)(mb) = m^2 A [\because A = ab]$$

$$A = \frac{I_1}{m^2} = \frac{18 \text{ cm} \times 18 \text{ cm}}{\left[\left(\frac{1}{4}\right) \times 10^{-3}\right]^2} = (720 \text{ m} \times 720 \text{ m})$$

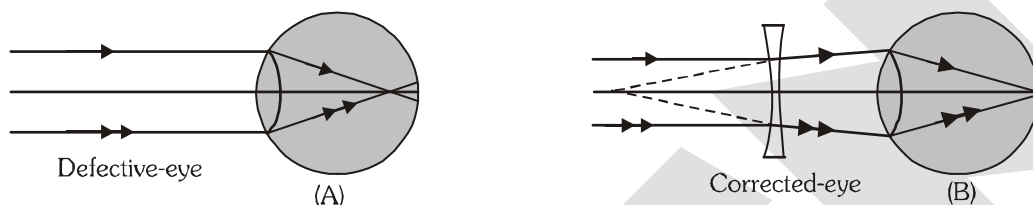
**Ex.** The proper exposure time for a photographic print is 20 s at a distance of 0.6 m from a 40 candle power lamp. How long will you expose the same print at a distance of 1.2 m from a 20 candle power lamp ?

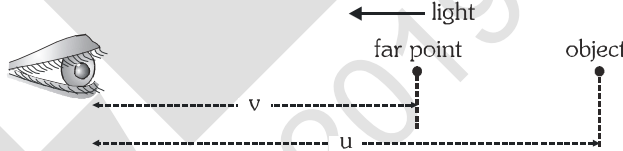
**Sol.** In case of camera, for proper exposure  $I_1 D_1^2 t_1 = I_2 D_2^2 t_2$

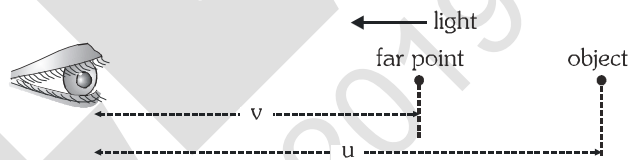
As here D is constant and  $I = (L/r^2)$ ;  $\frac{L_1}{r_1^2} \times t_1 = \frac{L_2}{r_2^2} \times t_2$  So  $\frac{40}{(0.6)^2} \times 20 = \frac{20}{(1.2)^2} t \Rightarrow t = 160 \text{ s}$

## DEFECTS OF EYES

**MYOPIA [or Short-sightedness or Near-sightedness]**



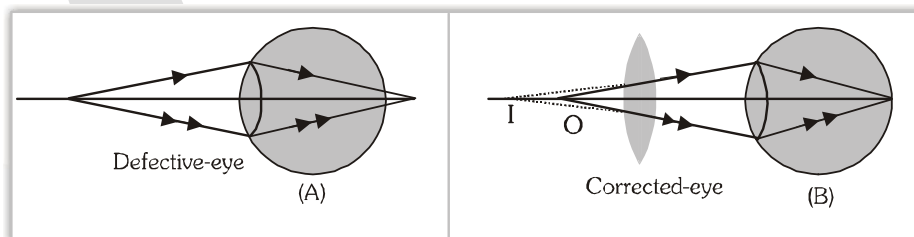
- (i) Distant object are not clearly visible, but near object are clearly visible because image is formed before the retina.
- (ii) To remove the defect concave lens is used.
- ◎ The maximum distance. Which a person can see without help of spectacles is known as far point.
- ◎ If the reference of object is not given then it is taken as infinity.
- ◎ In this case image of the object is formed at the far point of person.
- 



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of far point (in m)}} - \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

$$\frac{1}{100} - \frac{1}{\text{distance of object (in cm)}} = P$$

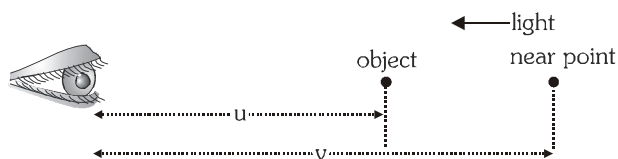
## LONG-SIGHTEDNESS OR HYPERMETROPIA



- (i) Near object are not clearly visible but far object are clearly visible.
- (ii) The image of near object is formed behind the retina.
- (iii) To remove this defect convex lens is used.

**Near Point :-**

The minimum distance which a person can see without help of spectacles.



- In this case image of the object is formed at the near point.
- If reference of object is not given it is taken as 25 cm.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of near point (in m)}} - \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

distance of near point = -ve, distance of object = -ve,  $P = +ve$

**PRESBYOPIA**

In this case both near and far object are not clearly visible. To remove this defect two separate spectacles one for myopia and other for hypermetropia are used or bifocal lenses are used.

**ASTIGMATISM**

In this defect eye can not see object in two orthogonal direction clearly. It can be removed by using cylindrical lens in particular direction.

**Ex.** A person can not see clearly an object kept at a distance beyond of 100 cm. Find the nature and the power of lens to be used for seeing clearly the object at infinity.

**Sol.** For lens  $u = -\infty$  and  $v = -100$  cm

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \frac{1}{v} = \frac{1}{f} \Rightarrow f = v = -100 \text{ cm (concave)} \therefore \text{Power of lens } P = \frac{1}{f} = -\frac{1}{1} = -1D$$

**Ex.** A far sighted person has a near point of 60 cm. What power lens should be used for eye glasses such that the person can read this book at a distance of 25 cm.

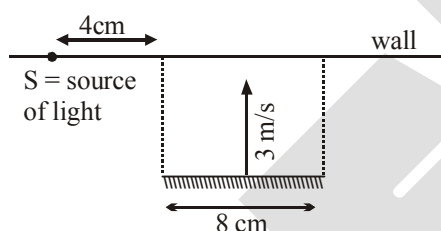
**Sol.** Here  $v = -60$  cm,  $u = -25$  cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = -\frac{1}{60} + \frac{1}{25} \Rightarrow f = \frac{300}{7} \text{ cm} \therefore \text{Power} = \frac{1}{f(\text{in m})} = \frac{1}{(3/7)} = +2.33D$$

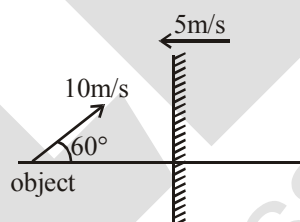
## EXERCISE (S-1)

### Plane mirror :

1. A fluorescent lamp of length 1 m is placed horizontally at a depth of 1.2 m below a ceiling. A plane mirror of length 0.6 m is placed below the lamp parallel to and symmetric to the lamp at a distance 2.4m from it. The length of the reflected patch of length on the ceiling \_\_\_\_\_.
2. A plane mirror of length 8 cm is moving with speed 3 m/s towards a wall in situation as shown in figure. Size of spot formed on the wall is  $32/x$  cm. Find the value of x.

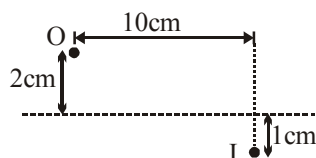


3. The angle between the velocity vector of object and image is  $\frac{\pi}{\alpha}$ . Velocity are shown in the figure. Then the value of  $\alpha$  is.



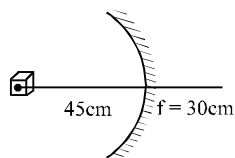
### Spherical mirror :

4. The principal axis of a spherical mirror is shown by dotted line. O is the point object whose real image is I. Find the distance of the pole and centre of curvature of the mirror from object measured along principal axis by drawing ray diagram.

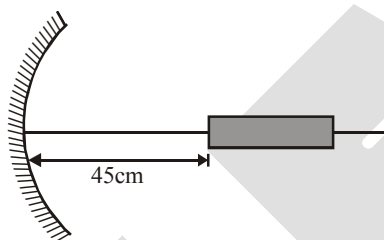


5. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 m while it is 0.30 m in front of the plane mirror. Find the radius of curvature (in cm) of the convex mirror.

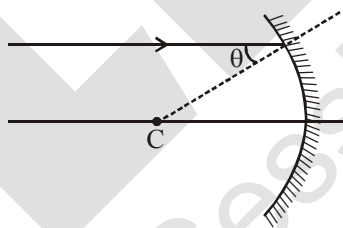
6. A thin rod of length  $d/3$  is placed along the principal axis of a concave mirror of focal length  $= d$  such that its image, which is real and elongated, just touches the rod. Find the length of the image?
7. A cube of side length 1 mm is placed on the axis of a concave mirror at a distance of 45 cm from the pole as shown in the figure. One edge of the cube is parallel to the axis. The focal length of the mirror is 30 cm. Find approximate volume of the image.



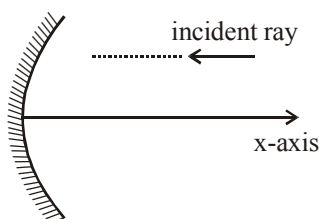
8. An object of length 30 cm is placed on principal axis of a concave mirror of focal length 30 cm. Its one end is at a distance of 45 cm as shown. If length of image is  $10x$  (in cm) find the value of  $x$ .



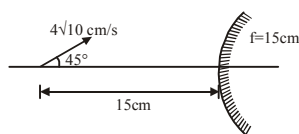
9. A ray incident parallel to principal axis on a concave mirror at an angle  $\theta$  after two reflection, reflected ray becomes parallel to incident ray. If  $\theta = \frac{\pi}{N}$  (in radian) then find the value of  $N$ .



10. A parabolic reflecting surface given by  $x = \frac{y^2}{2}$ , is placed at origin, as shown. An incident ray moving along  $y = 1$  is incident on it. The ray gets reflected by the surface twice. The deviation suffered by the ray is  $\frac{\pi}{n}$  radians. Find  $n$ .

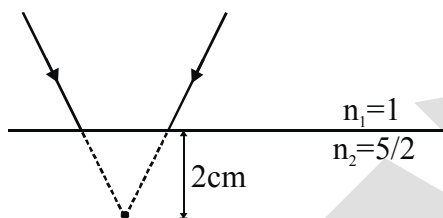


11. In the figure shown, the speed cm/s of image with respect to mirror is :



**Refraction on plane surface :**

12. In the given figure rays incident on an interface would converge 2 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet somewhere else. Find the distance of meeting point of refracted rays below the interface (in cm). (Assuming the rays to be making small angles with the normal to the interface)

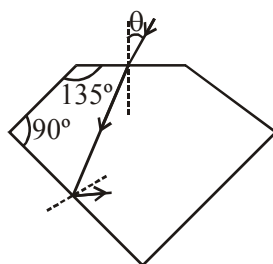


13. A point object is placed 33 cm from a convex mirror of curvature radius = 40 cm. A glass plate of thickness 6 cm and index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object?
14. An opaque cylindrical tank with an open top has a diameter of 3.00 m and is completely filled with water. When the setting sun reaches an angle of  $37^\circ$  above the horizon, sunlight ceases to illuminate any part of the bottom of the tank. How deep is tank in meter?
15. A cylindrical bucket of depth 60 cm is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as

seen by the observer is given as  $\alpha$  cm. Fill  $\left(\frac{\alpha}{5}\right)$  in OMR sheet.

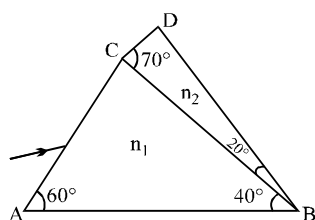
**Total internal reflection :**

16. A room contains air in which the speed of sound is 340 m/s. The walls of the room are made of concrete, in which the speed of sound is 1700 m/s. (a) Find the critical angle for total internal reflection of sound at the concrete–air boundary. (b) In which medium must the sound be traveling to undergo total internal reflection? (Assume sound waves follow snell's law)
17. A ray of light enters a diamond ( $n = 2$ ) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle  $\theta$  possible ?



18. A prism of refractive index  $n_1$  & another prism of refractive index  $n_2$  are stuck together without a gap as shown in the figure. The angles of the prisms are as shown.  $n_1$  &  $n_2$  depend on  $\lambda$ , the wavelength

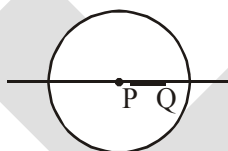
of light according to  $n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2}$  &  $n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$  where  $\lambda$  is in nm. [JEE 1998]



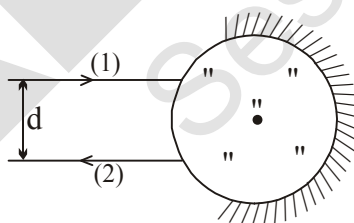
- Calculate the wavelength  $\lambda_0$  for which rays incident at any angle on the interface BC pass through without bending at that interface.
- For light of wavelength  $\lambda_0$ , find the angle of incidence  $i$  on the face AC such that the deviation produced by the combination of prisms is minimum.

### Refraction at curved surface :

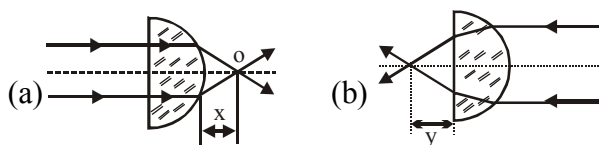
19. A small object of length 1 mm lies along the principal axis of a spherical glass of radius  $R = 10$  cm and refractive index is  $3/2$ . The object is seen from air along the principal axis from left. The distance of object from the centre is 5 cm. Find the size of the image. Is real, inverted ?



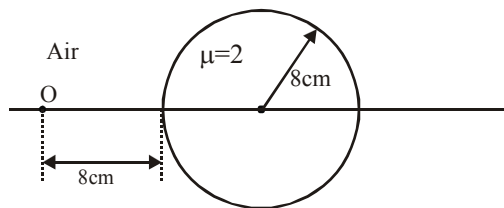
20. A transparent sphere of radius  $R = 2.0$  m has a mirrored surface on its right half as shown in figure. A light ray travelling in air is incident on the left side of the sphere. The incident light ray (1) & exiting light ray (2) are parallel & separated by distance  $d = 2.0$  m. Then find the refractive index of the material. (take :  $\sin 15^\circ = 0.25$ )



21. A narrow beam of light passing through the hemisphere of material with refractive index  $n$ , intersects at point O. Where does the beam converge (i.e.  $y$  in cm) if beam were to travel in the opposite direction as shown in figure (b)? The value of  $x$  is given to be 10 cm. Radius of the hemisphere is also 10 cm :-

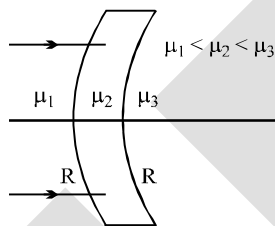


22. On the axis of a transparent sphere of refractive index ( $n = 2$ ) & radius 8 cm, an object is kept at the distance 8 cm from the surface of the sphere. Find the minimum distance of the image after all possible refraction from the surface of the sphere in cm.

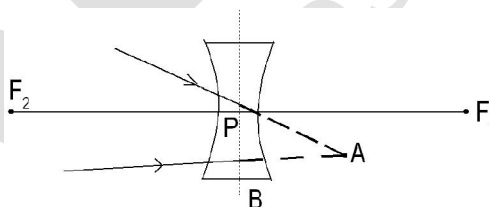


**Lens :**

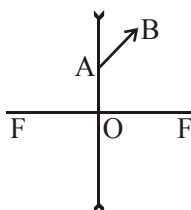
23. A meniscus lens is made of a material of refractive index  $\mu_2$ . Both its surfaces have radii of curvature  $R$ . It has two different media of refractive indices  $\mu_1$  and  $\mu_3$  respectively, on its two sides (see figure). Calculate its focal length for  $\mu_1 < \mu_2 < \mu_3$ , when light is incident on it as shown. [IIT-JEE 2003]



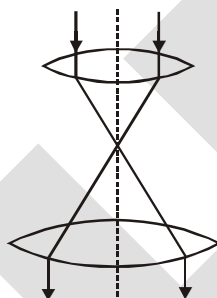
24. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed, the point where the rays meet will move 5 cm closer to the mounting that holds. Find the focal length (in cm) of the lens without sign.
25. The rays of a converging beam meet at a point A. A diverging lens is placed in their path in the plane B. Plot the position of the point where the rays meet after passing through the lens. The position of the principal foci FF is known.



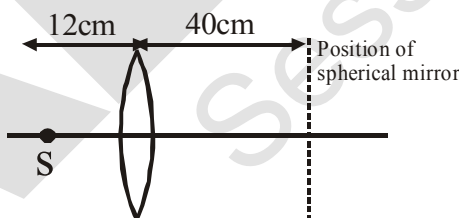
26. Figure shows ray AB that has passed through a divergent lens. Construct the path of the ray up to the lens if the position of its foci F is known.



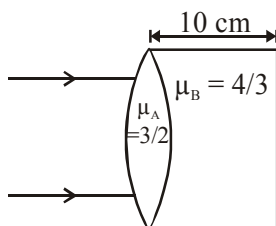
27. A plano convex lens ( $\mu = 1.5$ ) has a maximum thickness of 1 mm. If diameter of its aperture is 4 cm. Find (i) Radius of curvature of curved surface; (ii) its focal length in air.
28. A point source of light is kept at a distance of 150 cm from a converging lens, on its optical axis. The focal length of the lens is 100 cm and its diameter is 3 cm. A screen is placed on the other side of the lens, perpendicular to the axis of lens, at a distance 200 cm from it. Then find the area of the illuminated part of the screen? (Assume all rays incident on lens as paraxial)
29. There are two thin symmetrical lenses, one is converging, with refractive index  $n_1 = 1.70$ , and the other is diverging with refractive index  $n_2 = 1.50$ . Both lenses have the same curvature radius of their surface equal to  $R = 10$  cm. The lenses are put close together and submerged into water. The focal length of lens system is found to be  $\frac{100}{x}$  cm in water. What is the value of  $x$ . (R.I. of water =  $4/3$ )
30. Consider a 'beam expander' which consists of two converging lenses of focal lengths 40 cm and 100 cm having a common optical axis. A laser beam of diameter 2 mm is incident on the 40 cm focal length lens. Then what is the diameter (in mm) of the final beam (see figure)?



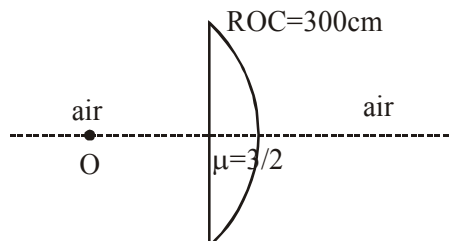
31. A converging lens and a spherical mirror are placed with their principal axis coinciding. Their separation equals 40 cm. A point source  $S$  is placed on the principal axis at a distance of 12 cm from the lens as shown in the figure. It is found that the final beam comes out parallel to the principal axis. Focal length of the lens equals 15 cm. Find the focal length of the mirror.



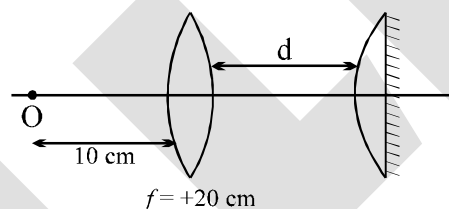
32. A parallel paraxial beam of light is incident on the arrangement as shown  $\mu_A = 3/2$ ,  $\mu_B = 4/3$ , the two spherical surfaces are very close and each has radius of curvature 10 cm. Find the point where the rays are focussed. (w.r.t. point of entry)



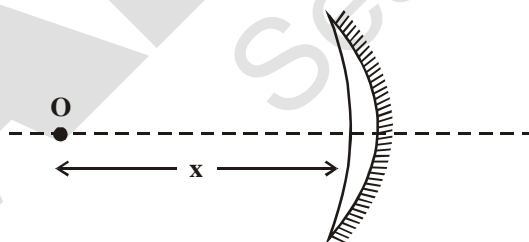
33. Plane surface of a thin planoconvex lens reflects 50% of light, while the curved surface is completely transparent, if final image of 'O' after refraction through thin lens coincides with the image formed due to partial reflection from plane surface. If distance between O and lens is  $x$  (in m) then find the value of  $x/4$ .



34. A lens is placed at origin, with  $x$ -axis as its principal axis. A ray of light is incident on it from the  $-ve$  side of  $x$ -axis along the line  $y = \frac{x}{400} + 0.1$ , where  $x, y$  are in cm. Focal length of lens is 30 cm. Find the equation of the ray after passing through the lens.
35. A convex lens of focal length 20 cm and another plano convex lens of focal length 40 cm are placed co-axially (see fig.). The plano convex lens is silvered on plane surface. What should be the  $5d$  (in m) so that final image of the object 'O' is formed on O itself.



36. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm and 20 cm as shown. The convex side is silvered. The distance  $x$  (in cm) on the principal axis where an object is placed so that its image is created on the object itself, is given as  $4\beta$ . Find the value of  $\beta$ .

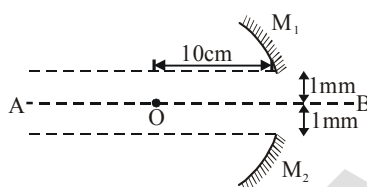


### Dispersion :

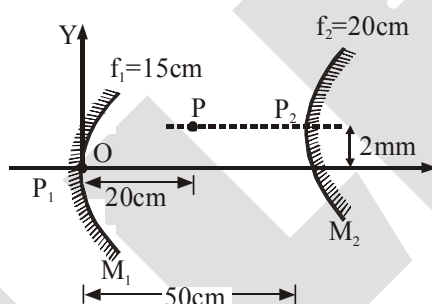
37. A flint glass convex lens of focal length 16 cm is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are  $\mu_v = 1.525$ ,  $\mu_r = 1.515$  and for flint glass  $\mu_v = 1.655$  and  $\mu_r = 1.645$ .
- (a) Find the nature and focal length of the crown glass lens which will form an achromatic combination with the flint glass lens.
- (b) What is the focal length of the combination?

## EXERCISE (S-2)

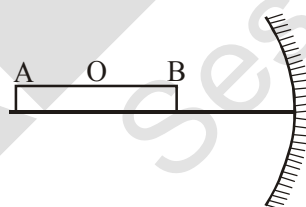
1. A concave mirror of focal length 20 cm is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 mm from the previous principal axis AB. If an object is placed at a distance of 10 cm from the mirror on the line AB then, find the distance between the images formed by the two parts?



2. Find the co-ordinates of image of point object P formed after two successive reflection in situation as shown in figure considering first reflection at concave mirror and then at convex.

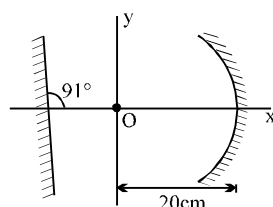


3. A rod of length  $\ell$  is moving with constant speed 1 m/s towards the pole of a concave mirror of focal length  $\ell$ . Find the rate of change in length of the image of the rod (in m/s) at the moment centre of the rod coincides with the centre of curvature of the mirror.

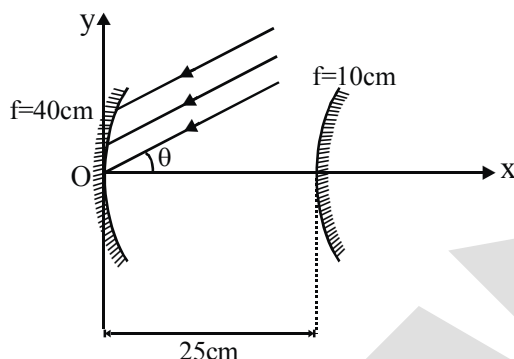


4. A point object is placed at the centre of curvature of a concave mirror (taken as origin). A plane mirror is also placed at a distance of 10 cm from the object as shown. Consider two reflection first at plane

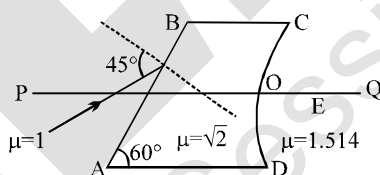
mirror and then at concave mirror ( $x_0, y_0$ ). Find  $\frac{\pi x_0}{y_0}$  the coordinates of the image thus formed are.



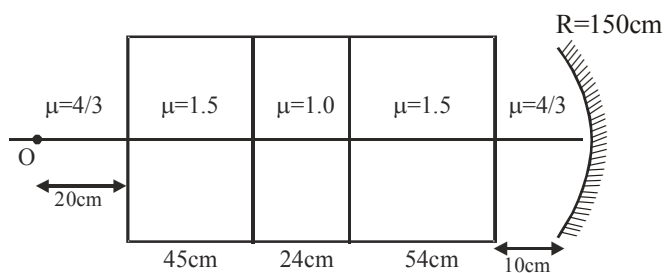
5. Consider a concave mirror kept at origin with focal length 40 cm. Parallel rays which subtend an angle  $\theta = \frac{1}{40}$  radian are incident on it. A convex mirror is kept at a distance 25 cm from the first mirror as shown. Find the y-coordinate of the image (in cm) formed by the system of mirrors after two reflections.



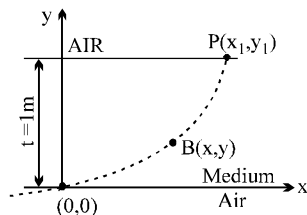
6. The x-y plane is the boundary between two transparent media. Medium-1 with  $z > 0$  has refractive index  $\sqrt{2}$  and medium - 2 with  $z < 0$  has a refractive index  $\sqrt{3}$ . A ray of light in medium -1 given by the vector  $A = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$  is incident on the plane of separation. Find the unit vector in the direction of refracted ray in medium -2. [1999, 10M]
7. Figure shows an irregular block of material of refractive index  $\sqrt{2}$ . A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4 m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE upto two places of decimal. [IIT-JEE 2004]



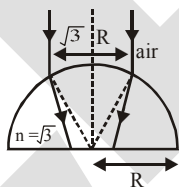
8. A composite slab consisting of different media is placed in front of a concave mirror of radius of curvature 150 cm. The whole arrangement is placed in water. An object O is placed at a distance 20cm from the slab. The R.I. of different media are given in the diagram. Find the position of the final image formed by the system.



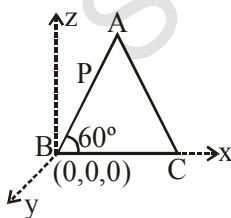
9. A ray of light travelling in air is incident at grazing angle (incident angle =  $90^\circ$ ) on a long rectangular slab of a transparent medium of thickness  $t = 1.0$  (see figure). The point of incidence is the origin A (O, O). The medium has a variable index of refraction  $n(y)$  given by:  $n(y) = [ky^{3/2} + 1]^{1/2}$ , where  $k = 1.0 \text{ m}^{-3/2}$ . The refractive index of air is 1.0.



- Obtain a relation between the slope ( $dy/dx$ ) of the trajectory of the ray at a point B(x, y) in the medium and the incident angle ( $i$ ) at that point. [JEE-1995]
  - Find the value of  $n \sin i$ .
  - Obtain an equation for the trajectory  $y(x)$  of the ray in the medium.
  - Determine the coordinates ( $x_1, y_1$ ) of the point P, where the ray intersects the upper surface of the slab-air boundary.
  - Indicate the path of the ray subsequently.
10. A light beam of diameter  $\sqrt{3}R$  is incident symmetrically on a glass hemisphere of radius  $R$  and of refractive index  $n = \sqrt{3}$ . Find the radius of the beam at the base of hemisphere.

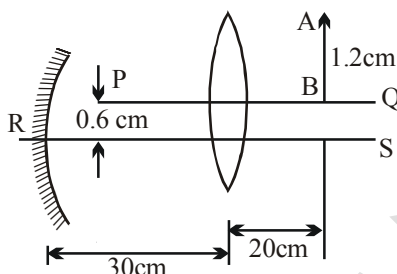


11. An equilateral prism ABC is placed in air with its base side C lying horizontally along X-axis as shown in the figure. A ray given by  $\sqrt{3}z + x = 10$  is incident at a point P on face AB of prism

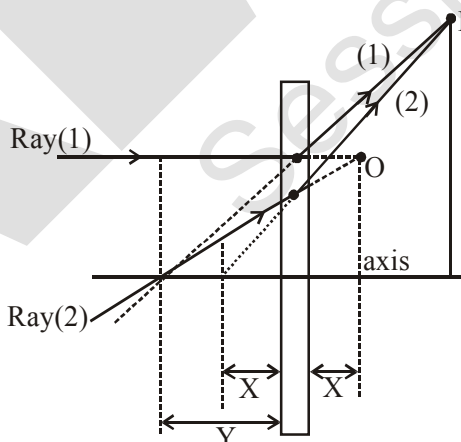
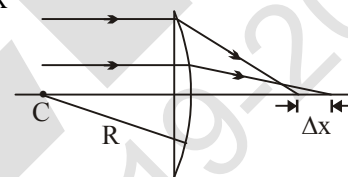


- Find the value of  $\mu$  for which the ray grazes the faces AC.
  - Find direction of the finally refracted ray if  $\mu = 3/2$ .
  - Find the equation of ray coming out of prism if bottom BC is silvered?
12. An object is approaching a thin convex lens of focal length 0.3 m with a speed of 0.01 m/s. Find the magnitudes of the rates of change of position and lateral magnification of image when the object is at a distance of 0.4 m from the lens. [IIT-JEE 2004]

13. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes  $PQ$  and  $RS$  parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object  $AB$  of height 1.2 cm is placed on the optic axis  $PQ$  of the lens at a distance of 20 cm from the lens. If  $A'B'$  is the image after refraction from the lens and reflection from the mirror, find the distance of  $A'B'$  from the pole of the mirror and obtain its magnification. Also locate positions of  $A'$  and  $B'$  with respect to the optic axis  $RS$ . [IIT-JEE 2000]

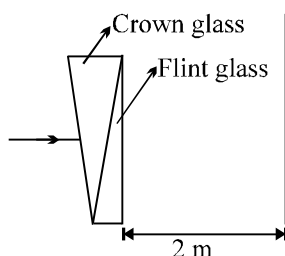


14. Two rays travelling parallel to the principal axis strike a large plano-convex lens having a refractive index of 1.60. If the convex face is spherical, a ray near the edge does not pass through the focal point (spherical aberration occurs). If this face has a radius of curvature of magnitude 20.0 cm and the two rays are  $h_1 = 0.500$  cm and  $h_2 = 12.0$  cm from the principal axis, find the difference in the positions where they cross the principal axis.
15. The rectangular box shown is the place of lens. By looking at the ray diagram, answer the following questions :



- (i) If  $X$  is 5 cm then what is the focal length of the lens ?
- (ii) If the point  $O$  is 1 cm above the axis then what is the position of the image ? Consider the optical center of the lens to be the origin.

16. The refractive indices of the crown glass for violet and red lights are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. A prism of angle  $6^\circ$  is made of crown glass. A beam of white light is incident at a small angle on this prism. The other thin flint glass prism is combined with the crown glass prism such that the net mean deviation is  $1.5^\circ$  anticlockwise.



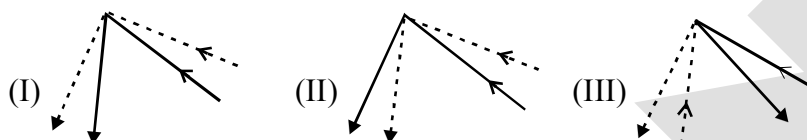
- (i) Determine the angle of the flint glass prism.  
(ii) A screen is placed normal to the emerging beam at a distance of 2m from the prism combination. Find the distance between red and violet spot on the screen. Which is the topmost colour on screen.
17. A large temple has a depression in one wall. On the floor plan it appears as a indentation having spherical shape of radius 2.50 m. A worshiper stands on the center line of the depression, 2.00 m out from its deepest point, and whispers a prayer. Where is the sound concentrated after reflection from the back wall of the depression?

## EXERCISE (O-1)

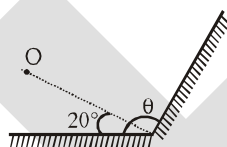
### SINGLE CORRECT TYPE QUESTIONS

#### Plane mirror :

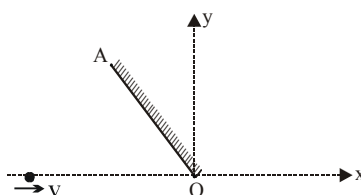
1. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same mirror. Which option is correct :-



- (A) Only I (B) Only II (C) Only III (D) All
2. If two mirrors are kept at  $60^\circ$  to each other, then the number of images formed by them is :-  
[AIEEE- 2002]
- (A) 5 (B) 6 (C) 7 (D) 8
3. If  $\theta = 110^\circ$  then total number of images formed by the mirror system will be :-

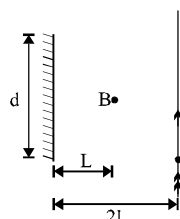


- (A) 2 (B) 3 (C) 4 (D) 5
4. A point object approaches a plane mirror with a speed of  $10 \text{ ms}^{-1}$ , while the image recedes away from the mirror with a speed of  $6 \text{ ms}^{-1}$  (as seen by stationary observer). The direction and magnitude of the velocity of mirror is :-
- (A) towards the object,  $8 \text{ ms}^{-1}$  (B) towards the image,  $6 \text{ ms}^{-1}$   
(C) away from the object,  $8 \text{ ms}^{-1}$  (D) away from the object,  $2 \text{ ms}^{-1}$
5. A boy of TOA3 batch is 1.8m tall and can see his image in a plane mirror fixed on a wall. His eyes are 1.6m from the floor level. The minimum length of the mirror to see his full image is :-
- (A) 0.9 m (B) 0.85 m (C) 0.8 m (D) Can't be determined
6. A plane mirror OA of length 2 m is kept along the line  $y = -x$  as shown in the figure. An insect having velocity of  $4\hat{i} \text{ cm/s}$  is moving along the x-axis from far away. The time span for which the insect can see its image will be :-



- (A) 50 sec (B) 25 sec (C)  $25\sqrt{2}$  sec (D)  $50\sqrt{2}$  sec

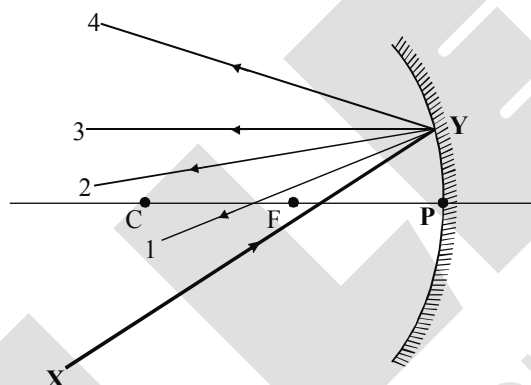
7. A point source of light  $B$  is placed at a distance  $L$  in front of the centre of a mirror of width  $d$  hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance  $2L$  from it as shown. The greatest distance over which he can see the image of the light source in the mirror is :-  
[IIT-JEE '2000 (Scr)]



- (A)  $d/2$                       (B)  $d$                       (C)  $2d$                       (D)  $3d$

### Spherical mirror :

8. Figure shows a small concave mirror with  $CP$  as its principal axis. A ray  $XY$  is incident on the mirror. Which of the four rays can be the reflected ray.

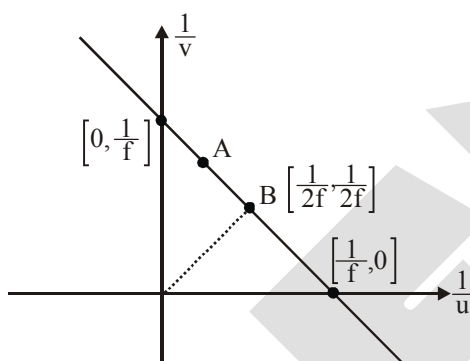


- (A) 1                      (B) 2                      (C) 3                      (D) 4

9. The distance of an object from a spherical mirror is equal to the focal length of the mirror. Then the image:  
(A) must be at infinity                      (B) may be at infinity  
(C) may be at the focus                      (D) none
10. An object is placed in front of a spherical mirror whose 2 times magnified image is formed on screen. Then choose **CORRECT** option :-  
(A) Mirror is concave  $m = +2$                       (B) Mirror is concave  $m = -2$   
(C) Mirror is convex  $m = +2$                       (D) Mirror is convex  $m = -2$
11. The table below lists object and image positions for four objects placed in front of mirrors, using Cartesian sign convention with pole of the mirror as origin and direction of incident rays as positive. In the following cases, the case in which images is formed by a convex spherical mirror, is :-

| Object position | Image position |
|-----------------|----------------|
| (A) $-25.0$ cm  | $-16.7$ cm     |
| (B) $-5.0$ cm   | $10.0$ cm      |
| (C) $-20.0$ cm  | $5.71$ cm      |
| (D) $-40.0$ cm  | $80.0$ cm      |

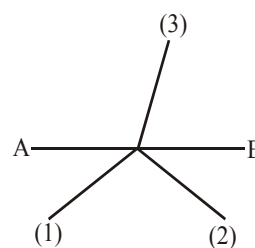
12. A particle approaches from very large distance towards concave mirror along the principal axis. By the time the particle reaches the mirror the distance between the particle and its image :-  
 (A) first decreases then increases  
 (B) first increases then decreases  
 (C) first increases then decreases and then again increases  
 (D) first decreases then increases and then again decreases
13. Figure shows  $\frac{1}{v}$  vs  $\frac{1}{u}$  curve for convex mirror. Nature of image at point A is :-



- (A) Real, erect, magnified  
 (B) virtual, inverted, magnified  
 (C) virtual, inverted, diminished  
 (D) virtual, erect, diminished
14. A boy of height 1 m stands in front of a convex mirror. His distance from the mirror is equal to its focal length. The height of his image is :- (Assume paraxial ray approximation holds)  
 (A) 0.25 m (B) 0.33 m (C) 0.5 m (D) 0.67 m

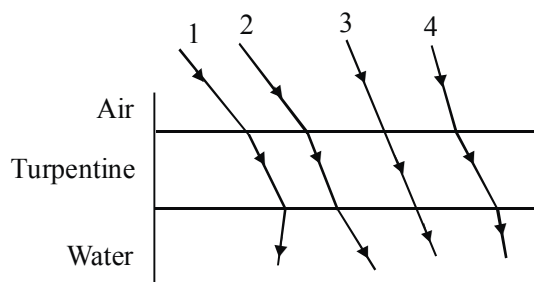
### Refraction at plane surface :

15. AB is a boundary separating two media of different refractive indices. A ray is incident on the boundary is partially reflected and partially transmitted. Choose the **CORRECT** statement.  
 (A) 3 is incident ray and 1 is refracted ray  
 (B) 2 is incident ray and 1 is partially reflected ray  
 (C) 1 is incident ray and 3 is refracted ray  
 (D) 3 is incident ray and 2 is partially reflected ray

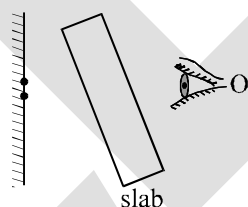


16. **Statement-1:** You see a geostationary satellite above the horizon. You desire to communicate with the satellite by sending a beam of laser light. You should aim your laser slightly higher than the line of sight of the satellite.  
**Statement-2:** Light bends away from the normal while moving from denser to rarer medium.  
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

17. The optical density of turpentine is higher than that of water while its mass density is lower. Figure shows a layer of turpentine floating over water in a container. For which one of the four rays incident on turpentine in figure, the path shown is **CORRECT** ?

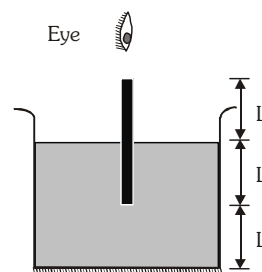


- (A) 1 (B) 2 (C) 3 (D) 4
18. When a ray of light of frequency  $6 \times 10^{14}$  Hz travels from water of refractive index  $4/3$  to the glass of refractive index  $8/5$ , its :-  
 (A) frequency decreases to  $5/6$  of its initial value  
 (B) speed decreases to  $5/6$  of its initial value  
 (C) wavelength decreases to  $6/5$  of its initial value  
 (D) speed increases to  $6/5$  of its initial value
19. The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown. As seen by observer, the spots appear.

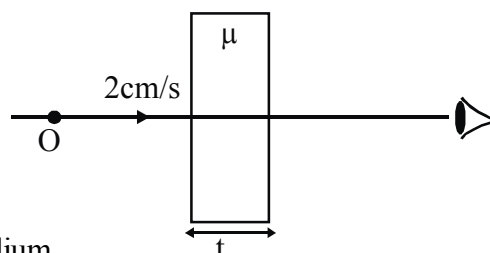


- (A) shifted upward (B) shifted downward  
 (C) spaced farther apart (D) spaced closer together
20. What is the length of the image of the rod in mirror, according to the observer in air? (refractive index of the liquid is  $\mu$ )

- (A)  $\mu L + L$  (B)  $L + \frac{L}{\mu}$   
 (C)  $L\mu + \frac{L}{\mu}$  (D) None of these

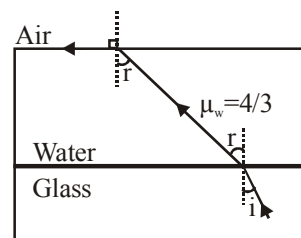


21. A glass slab of width ' $t$ ', refractive index ' $\mu$ ' is placed as shown in the figure. If the point object, moves with a speed  $2 \text{ cm/s}$  towards the slab the speed observed will be :-  
 (A)  $2 \text{ cm/s}$   
 (B) less than  $2 \text{ cm/s}$   
 (C) greater than  $2 \text{ cm/s}$   
 (D) dependent on the refractive index of surrounding medium



22. A ray of light is incident at the glass–water interface at an angle  $i$ , it emerges finally parallel to the surface of water, then the value of  $\mu_g$  would be :-  
[IIT-JEE 2003]

- (A)  $(4/3) \sin i$   
(B)  $1/\sin i$   
(C)  $4/3$   
(D) 1



**Total internal reflection :**

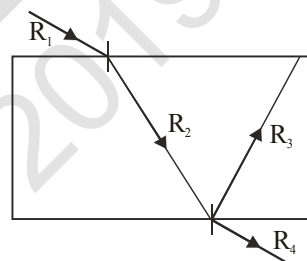
23. **Statement-1:** A point source of light is placed inside water. A light detector present out side, in air can detect light only in a conical region, with the apex at the source and circumscribing the circle of illuminance.

**Statement-2:** Ray incident from denser to rarer medium undergoes total internal reflection when their angle of incidence become more than the critical angle, this situation create a circular region from which light escapes and is called circle of illuminance.

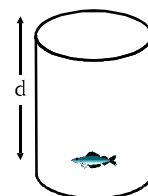
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(C) Statement-1 is true, statement-2 is false.  
(D) Statement-1 is false, statement-2 is true.

24. A ray  $R_1$  is incident on the plane surface of the glass slab (kept in air) of refractive index  $\sqrt{2}$  at an angle of incidence equal to the critical angle for this air glass system. The refracted ray  $R_2$  undergoes partial reflection and refraction at the other surface. The angle between reflected ray  $R_3$  and the refracted ray  $R_4$  at that surface is :

- (A)  $45^\circ$  (B)  $135^\circ$   
(C)  $105^\circ$  (D)  $75^\circ$



25. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 cm, and the critical angle for liquid is  $37^\circ$ . At what maximum depth below the



surface of the liquid,  $d$ , must the fish be floating?  $\left( \sin 37^\circ = \frac{3}{5} \right)$

- (A) 16 cm (B) 20 cm (C) 11.25 cm (D) 25 cm

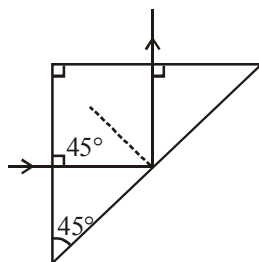
26. **Statement-1:** When light falls on a sphere made of diamond total internal reflection takes place which makes it shine more than a similar sphere made of common glass.

**Statement-2:** Refractive index for diamond is more than refractive index of cheap glass.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(C) Statement-1 is true, statement-2 is false.  
(D) Statement-1 is false, statement-2 is true.

27. A light ray is incident perpendicular to one face of a  $90^\circ$  prism and is totally internally reflected at the glass-air interface. If the angle of reflection is  $45^\circ$ , we conclude that the refractive index  $n$  :

[AIEEE-2004]



- (A)  $n < \frac{1}{\sqrt{2}}$       (B)  $n > \sqrt{2}$       (C)  $n > \frac{1}{\sqrt{2}}$       (D)  $n < \sqrt{2}$

**Prism :**

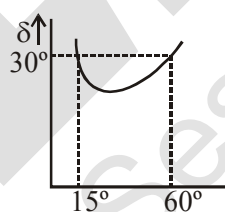
28. A ray of light is incident at  $60^\circ$  on a prism of refracting angle  $30^\circ$ . The emerging ray is at an angle  $30^\circ$  with the incident ray. The value of refractive index of the prism is:-

- (A)  $\frac{\sqrt{3}}{4}$       (B)  $\frac{\sqrt{3}}{2}$       (C)  $\sqrt{3}$       (D)  $\frac{2}{\sqrt{3}}$

29. The refracting angle of the prism is  $60^\circ$  and minimum deviation of  $30^\circ$ , then the angle of incidence is:-

- (A)  $30^\circ$       (B)  $45^\circ$       (C)  $25^\circ$       (D)  $60^\circ$

30. Figure shows graph of deviation  $\delta$  versus angle of incidence for a light ray striking a prism. Angle of prism is :-



- (A)  $30^\circ$       (B)  $45^\circ$       (C)  $60^\circ$       (D)  $75^\circ$

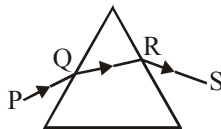
31. There is a prism with refractive index equal to  $\sqrt{2}$  and the refracting angle equal to  $30^\circ$ . One of the refracting surface of the prism is polished. A beam of monochromatic light will retrace its path if its angle of incidence over the first refracting surface of the prism is :-

- (A)  $0^\circ$       (B)  $30^\circ$       (C)  $45^\circ$       (D)  $60^\circ$

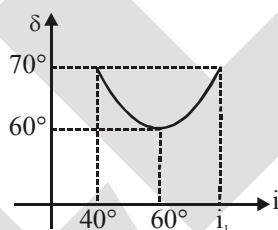
32. The refractive index for the material of a  $60^\circ$  prism is 1.50. Then the angle of incidence for minimum deviation is nearly. ( $\sin 42^\circ \approx \frac{2}{3}$  and  $\sin 49^\circ \approx \frac{3}{4}$ )

- (A)  $30^\circ$       (B)  $49^\circ$       (C)  $38^\circ$       (D)  $28^\circ$

33. A ray of light is incident on an equilateral glass prism placed on a horizontal table. For minimum deviation which of the following is true ?



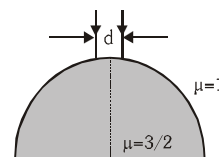
- (A) PQ is horizontal (B) QR is horizontal  
(C) RS is horizontal (D) Either PQ or RS is horizontal
34. An isosceles glass prism having refractive index  $\mu$  has one of its faces coated with silver. A ray of light is incident normally on the other face (which is equal to the silvered face). The ray of light is reflected twice on the same sized faces and then emerges through the base of the prism perpendicularly. The angles of prism are :-  
(A)  $40^\circ, 70^\circ, 70^\circ$  (B)  $50^\circ, 65^\circ, 65^\circ$  (C)  $36^\circ, 72^\circ, 72^\circ$  (D) data insufficient
35. A given ray of light suffers minimum deviation in an equilateral prism P. If refractive index increases slightly then the ray will now suffer :-  
(A) greater deviation (B) no deviation  
(C) same deviation as before (D) Lesser deviation
36. The curve of angle of incidence versus angle of deviation shown has been plotted for prism. The value of refractive index of the prism used is :



- (A)  $\sqrt{3}$  (B)  $\sqrt{2}$  (C)  $\frac{\sqrt{3}}{\sqrt{2}}$  (D)  $\frac{2}{\sqrt{3}}$
37. A beam of monochromatic light is incident at  $i = 50^\circ$  on one face of an equilateral prism, the angle of emergence is  $40^\circ$ , then the angle of minimum deviation is :  
(A)  $30^\circ$  (B)  $< 30^\circ$  (C)  $\leq 30^\circ$  (D)  $\geq 30^\circ$
38. The angle of a prism is  $6^\circ$  and its refractive index for green light is 1.5. If a green ray passes through it, the deviation will be :-  
(A)  $30^\circ$  (B)  $15^\circ$  (C)  $9^\circ$  (D)  $3^\circ$

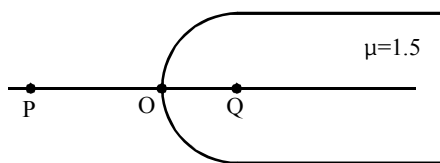
### Refraction at curved surface :

39. A beam of diameter 'd' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be :-



- (A)  $\frac{3}{4}d$  (B) d (C)  $\frac{d}{3}$  (D)  $\frac{2}{3}d$

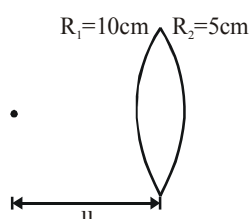
40. One end of a glass rod of refractive index  $n = 1.5$  is a spherical surface of radius of curvature  $R$ . The centre of the spherical surface lies inside the glass. A point object placed in air on the axis of the rod at the point  $P$  has its real image inside glass at the point  $Q$  (see fig.). A line joining the points  $P$  and  $Q$  cuts the surface at  $O$  such that  $OP = 2OQ$ . The distance  $PO$  is :-



- (A)  $8R$  (B)  $7R$  (C)  $2R$  (D) None of these
41. An air bubble is inside water. The refractive index of water is  $4/3$ . At what distance from the air bubble should a point object be placed so as to form a real image at the same distance from the bubble:
- (A)  $2R$  (B)  $3R$   
(C)  $4R$  (D) The air bubble cannot form a real image
42. A point object is placed at the centre of a glass sphere of radius  $6\text{ cm}$  and refractive index  $1.5$ . The distance of the virtual image from the surface of the sphere is :- [IIT-JEE 2004 (Scr)]
- (A)  $2\text{ cm}$  (B)  $4\text{ cm}$  (C)  $6\text{ cm}$  (D)  $12\text{ cm}$

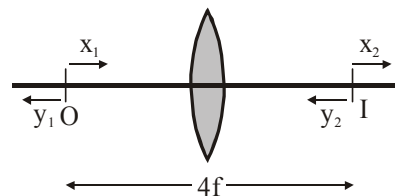
### Lens :

43. A concave lens of glass, refractive index  $1.5$ , has both surfaces of same radius of curvature  $R$ . On immersion in a medium of refractive index  $1.75$ , it will behave as a :-
- (A) convergent lens of focal length  $3.5R$   
(B) convergent lens of focal length  $3.0R$   
(C) divergent lens of focal length  $3.5R$   
(D) divergent lens of focal length  $3.0R$
44. A convergent (biconvex) lens is placed inside a jar filled with a liquid. The lens has focal length  $20\text{ cm}$ , when in air and its material has a refractive index of  $1.5$ . If the liquid has a refractive index of  $1.6$ , the focal length of the lens while in the jar, is :-
- (A)  $-110\text{ cm}$  (B)  $-130\text{ cm}$  (C)  $-160\text{ cm}$  (D)  $-180\text{ cm}$
45. A thin convex lens is made of a material of refractive index  $1.6$ . An object is kept at a distance of  $u$  from the lens on the principal axis as shown in the figure. The radius of curvature of the surfaces are  $10\text{ cm}$  and  $5\text{ cm}$ . Now, the lens is reversed such that the face having radius of curvature  $5\text{ cm}$  lies close to the object. The difference in image position as obtained for both the cases is equal to :-



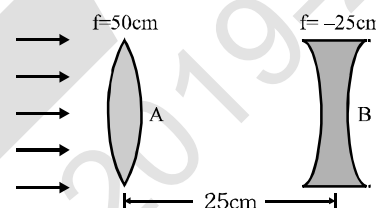
- (A)  $0.4u$  (B)  $0.6u$  (C)  $0.8u$  (D)  $0$

46. In a converging lens of focal length  $f$  and the distance between real object and its real image is  $4f$ . If the object moves  $x_1$  distance towards lens its image moves  $x_2$  distance away from the lens and when object moves  $y_1$  distance away from the lens its image moves  $y_2$  distance towards the lens, then choose the correct option:-

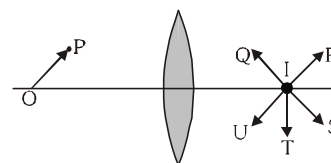


- (A)  $x_1 > x_2$  and  $y_1 > y_2$  (B)  $x_1 < x_2$  and  $y_1 < y_2$   
 (C)  $x_1 < x_2$  and  $y_1 > y_2$  (D)  $x_1 > x_2$  and  $y_2 > y_1$
47. A point object is placed on the principal axis of a converging lens and its image ( $I_1$ ) is formed on its principal axis. If the lens is rotated by a small angle  $\theta$  about its optical centre such that its principal axis also rotates by the same amount then the image ( $I_2$ ) of the same object is formed at point P. Choose the correct option.

- (A) Point P lies on the new principal axis.  
 (B) Point P lies on the old principal axis.  
 (C) Point P is anywhere between the two principal axes  
 (D) None of these
48. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is :



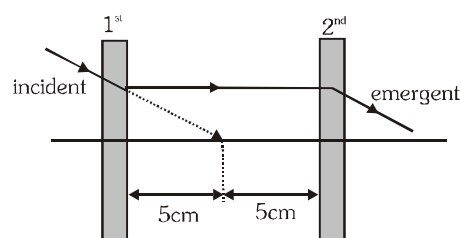
- (A) Initially parallel and then diverging  
 (B) Always diverging  
 (C) Initially converging and finally parallel  
 (D) Always parallel
49. A point object O moves from the principal axis of a converging lens in a direction OP. I the image of O, will move initially in the direction :



- (A) IQ (B) IR  
 (C) IS (D) IU
50. When the object is at distances  $u_1$  and  $u_2$  the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :

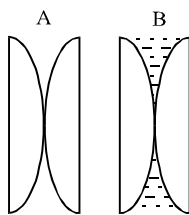
- (A)  $\frac{1}{2}\sqrt{u_1 u_2}$  (B)  $\frac{u_1 + u_2}{2}$  (C)  $\sqrt{u_1 u_2}$  (D)  $\sqrt{(u_1 + u_2)}$

51. Look at the ray diagram shown, what will be the focal length of the 1<sup>st</sup> and the 2<sup>nd</sup> lens, if the incident light ray passes without any deviation ?

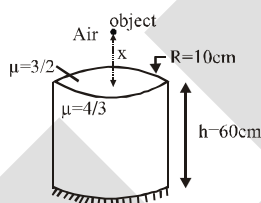


- (A) -5 cm and +10 cm  
 (B) +5cm and +10cm  
 (C) -5cm and +5cm  
 (D) +5cm and - 5cm

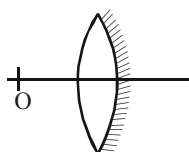
52. Figure A shows two identical plano-convex lenses in contact as shown. The combination has focal length 24 cm. Figure B shows the same with a liquid introduced between them. If refractive index of glass of the lenses is 1.50 and that of the liquid is 1.60, the focal length of the system in figure B will be :-



- (A) -120 cm      (B) 120 cm      (C) -24 cm      (D) 24 cm
53. In the arrangement shown given that a biconvex lens of radius of curvature equal to 10 cm and concave mirror has focal length equal to 20cm. Then find the distance 'x' such that the final image formed by the system coincides with the object :-



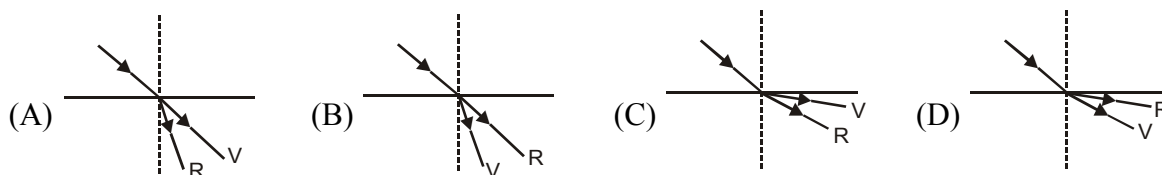
- (A) 15 cm      (B) 22.5 cm      (C) 20 cm      (D) 17.5 cm
54. A convex lens of focal length 30 cm forms an image of height 2 cm for an object situated at infinity. If a concave lens of focal length 20 cm is placed coaxially at a distance of 26 cm in front of convex lens then size image would be [IIT-JEE 2003 (Scr)]
- (A) 2.5 cm      (B) 5.0      (C) 1.25      (D) None
55. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now, this lens has been used to form the image of an object. At what distance from this lens, an object be placed in order to have a real image of the size of the object? [AIEEE-2004]
- (A) 20 cm      (B) 30 cm      (C) 60 cm      (D) 80 cm
56. An equiconvex lens of refractive index  $\mu$  and radius of curvature  $R$  has its one surface silvered. A point source  $O$  is placed before the silvered lens so that its image is coincident with it, the distance of the object from the lens is :-



- (A)  $\frac{R}{(\mu-1)}$       (B)  $\frac{2R}{(\mu-1)}$       (C)  $\frac{R}{(2\mu-1)}$       (D)  $\frac{2R}{(2\mu-1)}$

# Dispersion :

57. A ray of light is coming from air to water. Which of the following figure shows correct dispersion of light ?



58. When white light passes through a glass prism, one gets spectrum on the other side of the prism. In the emergent beam, the ray which is deviating least is or deviation by a prism is lowest for :-

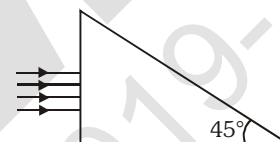
(A) Violet ray (B) Green ray (C) Red ray (D) Yellow ray

59. The angle of prism is  $5^\circ$  and its refractive indices for red and violet colours are 1.5 and 1.6 respectively. The angular dispersion produced by the prism is :-

(A)  $7.75^\circ$  (B)  $5^\circ$  (C)  $0.5^\circ$  (D)  $0.17^\circ$

60. A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will :-

(A) separate part of the red color from the green and blue colors  
(B) separate part of the blue color from the red and green colors  
(C) separate all the three colors from the other two colors  
(D) not separate even partially, any colors from the other two colors



61. Two thin prisms of flint glass, with refracting angles of  $6^\circ$  and  $8^\circ$  respectively, possess dispersive powers in the ratio :

(A) 4 : 3 (B) 3 : 4 (C) 1 : 1 (D) 9 : 16

62. Chromatic aberration will be absent if for two thin lenses in contact :-

(A)  $(\omega_1/F_1) + (\omega_2/F_2) = 0$  (B)  $(\omega_1/F_2) + (\omega_2/F_1) = 0$   
(C)  $(F_1/\omega_2) + (F_2/\omega_1) = 0$  (D)  $(\omega_1/\omega_2) + (F_1 + F_2) = 0$

63. The dispersive powers of the materials of the two lenses are in the ratio 4 : 3. If the achromatic combination of these two lenses in contact is a convex lens of focal length 60 cm then the focal lengths of the component lenses are :-

(A) - 20 cm and 25 cm (B) 20 cm and - 25 cm  
(C) - 15 cm and 40 cm (D) 15 cm and - 20 cm

64. An achromatic convergent doublet of two lenses in contact has a power of + 2D. The convex lens has a power + 5D. What is the ratio of the dispersive powers of the convergent and divergent lenses :-

(A) 2 : 5 (B) 3 : 5 (C) 5 : 2 (D) 5 : 3

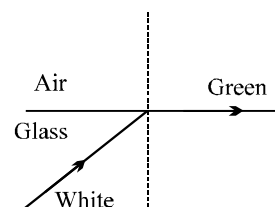
65. It is desired to make an achromatic combination of two lenses ( $L_1$  &  $L_2$ ) made of materials having dispersive powers  $\omega_1$  and  $\omega_2$  ( $< \omega_1$ ). If the combination of lenses is converging then :-

(A)  $L_1$  is converging  
(B)  $L_2$  is converging  
(C) Power of  $L_1$  is greater than the power of  $L_2$   
(D) None of these

66. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains

[IIT-JEE 2004 (Scr)]

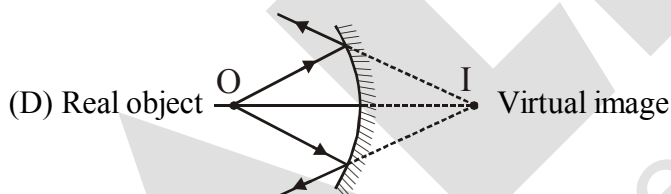
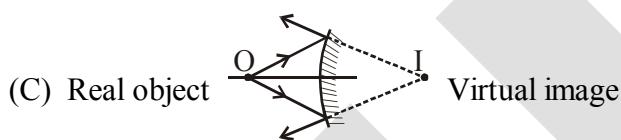
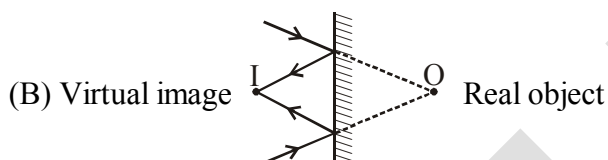
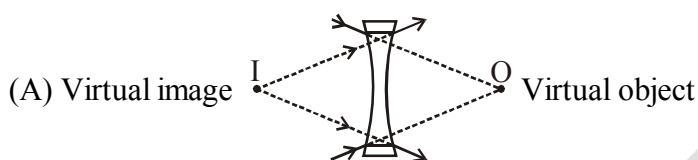
- (A) yellow, orange, red  
(B) violet, indigo, blue  
(C) all colours  
(D) all colours except green



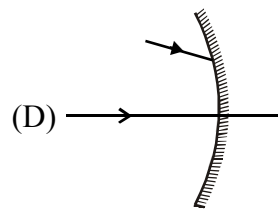
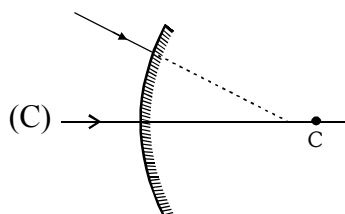
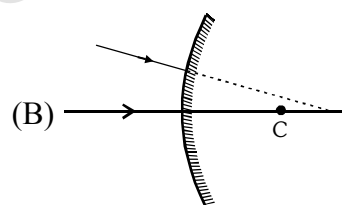
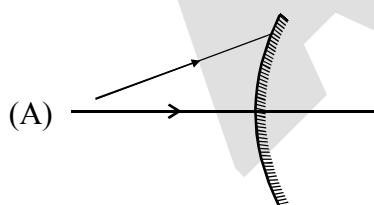
### MULTIPLE CORRECT TYPE QUESTIONS

#### Spherical mirror :

67. The nature of object and image given with each of the optical condition is shown. Choose the **correct** option(s) :-

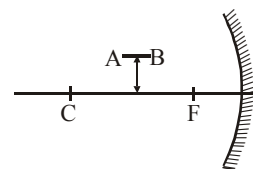


68. In which of the following diagrams the image formed is virtual and inverted ?



69. An object AB is placed parallel and close to the optical axis between focus F and centre of curvature C of a converging mirror of focal length  $f$  as shown in figure.

- (A) Image of A will be closer than that of B from the mirror.  
 (B) Image of AB will be parallel to the optical axis.  
 (C) Length of image is equal to AB.  
 (D) Length of image is more than AB.



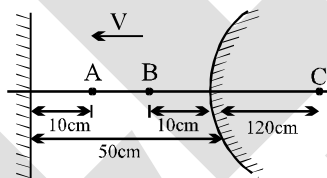
70. Mark the **CORRECT** statement :-

- (A) Virtual image of virtual object can be formed by a single optical element  
 (B) Virtual image of virtual object formed by a single optical element can be inverted  
 (C) Virtual image of real object can be formed by a single optical element  
 (D) Virtual image of real object formed by a single optical element can be inverted

71. A concave mirror forms an image of the sun at a distance of 12 cm from it.

- (A) the radius of curvature of this mirror is 6 cm.  
 (B) to use it as a shaving mirror, it can be held at a distance of 8-10 cm from the face  
 (C) if an object is kept at a distance of 24 cm from it, the image formed will be of the same size as the object  
 (D) all the above alternatives are correct.

72. In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. AB is object.

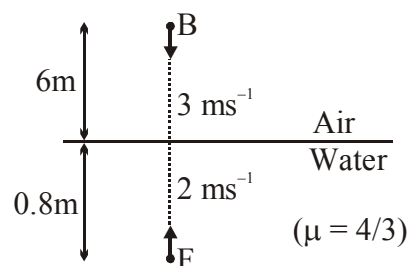


- (A) the second image is real, inverted of  $1/5^{\text{th}}$  magnification w.r.t AB  
 (B) the second image is virtual and erect with magnification  $1/5$  w.r.t AB  
 (C) the second image moves towards the convex mirror  
 (D) the second image moves away from the convex mirror.

### Refraction at plane surface :

73. A fish, F in the pond, is at a depth of 0.8 m from water surface and is moving vertically upwards with velocity  $2 \text{ ms}^{-1}$ . At the same instant, a bird B is at a height of 6 m from water surface and is moving downwards with velocity  $3 \text{ ms}^{-1}$ . At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?

- (A) Height of B, observed by F (from itself) is equal to 8.00 m.  
 (B) Depth of F, observed by B (from itself) is equal to 6.60 m.  
 (C) Velocity of B, observed by F (relative to itself) is equal to  $5.00 \text{ ms}^{-1}$ .  
 (D) Velocity of F, observed by B (relative to itself) is equal to  $4.50 \text{ ms}^{-1}$ .



**Total internal reflection :**

74. In the figure (a) light is incident at an angle  $\theta$  which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index  $n_3$  is placed on surface AB in fig. (b). Which of the following statements are correct :-

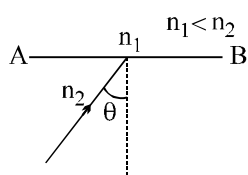


Fig. (a)

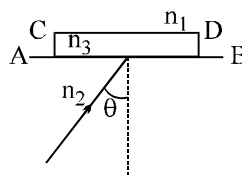
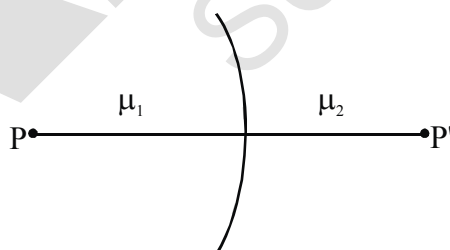


Fig. (b)

- (A) total internal reflection occurs at AB for  $n_3 < n_1$   
 (B) total internal reflection occurs at AB for  $n_3 > n_1$   
 (C) the ray will return back to the same medium for all values of  $n_3$   
 (D) total internal reflection occurs at CD for  $n_3 < n_1$
75. For the refraction of light through a prism :-  
 (A) For every angle of deviation there are two angles of incidence.  
 (B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.  
 (C) There are two angles of incidence for maximum deviation.  
 (D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if  $\mu_p > \mu_s$ .

**Refraction at curved surface :**

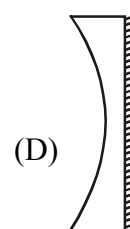
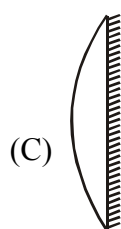
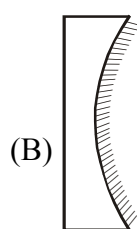
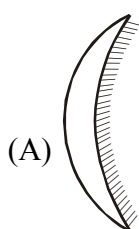
76. Two refracting media are separated by a spherical interface as shown in the figure.  $PP'$  is the principal axis,  $\mu_1$  and  $\mu_2$  are the refractive indices of medium of incidence and medium of refraction respectively, then :-



- (A) if  $\mu_2 > \mu_1$ , then there cannot be a real image of real object.  
 (B) if  $\mu_2 > \mu_1$ , then there cannot be a real image of virtual object.  
 (C) if  $\mu_1 > \mu_2$ , then there cannot be a virtual image of virtual object.  
 (D) if  $\mu_1 > \mu_2$ , then there cannot be a real image of real object.

**Lens :**

77. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are  $(-20 \text{ cm}, 1 \text{ cm})$  and  $(25 \text{ cm}, -2 \text{ cm})$  respectively :-  
 (A) the lens is located at  $x = 5 \text{ cm}$   
 (B) the lens is located at  $x = -5 \text{ cm}$   
 (C) the focal length of the lens is  $10 \text{ cm}$   
 (D) the focal length of the lens is  $15 \text{ cm}$
78. The principal axis of an optical device is along  $y = -1$ , image of a small body placed at  $(-30, 3)$  is formed at a point  $(60, -3)$ . Then the optical device is :-  
 (A) A convex lens of focal length  $20 \text{ cm}$   
 (B) A concave mirror of focal length  $60 \text{ cm}$   
 (C) A concave lens of focal length  $20 \text{ cm}$   
 (D) A convex mirror of focal length  $60 \text{ cm}$
79. In displacement method, the distance between object and screen is  $96 \text{ cm}$ . The ratio of length of two images formed by a convex lens placed between them is  $4.84$  :-  
 (A) Ratio of the length of object to the length of shorter image is  $11/5$ .  
 (B) Distance between the two positions of the lens is  $36 \text{ cm}$ .  
 (C) Focal length of the lens is  $20.625 \text{ cm}$ .  
 (D) Distance of the lens from the shorter image is  $30 \text{ cm}$ .
80. A lens is formed by a material having refracting index  $\frac{3}{2}$  & radii of curvature  $20 \text{ cm}$  &  $10 \text{ cm}$ . Then choose the correct option(s) :-  
 (A) If lens is concavo-convex & light is falling on the surface having radius of curvature  $20 \text{ cm}$ , then focal length is  $40 \text{ cm}$   
 (B) If lens is concavo-convex & light is falling on the surface having radius of curvature  $10 \text{ cm}$ , then focal length is  $-40 \text{ cm}$   
 (C) If lens is convexo-concave & light is falling on the surface having radius of curvature  $20 \text{ cm}$ , then focal length is  $40 \text{ cm}$   
 (D) If lens is convexo-concave & light is falling on the surface having radius of curvature  $10 \text{ cm}$ , then focal length is  $-40 \text{ cm}$
81. Which of the following silvered lenses kept in air may form real image of a real object.



82. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.
- (A) the image will look like a white donkey on the photograph.  
 (B) the image will look like a Zebra on the photograph.  
 (C) the image will be more intense compared to the case in which no such glass is used.  
 (D) the image will be less intense compared to the case in which no such glass is used.

### COMPREHENSION TYPE QUESTIONS

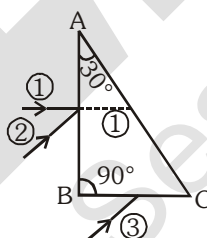
#### Paragraph for Question No. 83 to 85

A real object is placed perpendicular to the principal axis of a concave mirror at a position (1) such that the image formed is real with a magnification 2. The object is now shifted to another position (2) at a distance 15 cm from the position (1) and a real image is obtained with a magnification 8. Shifting the position of object from (2) to a third position (3) gives an image with magnification  $1/2$ .

83. Position of the image when the object is at position (3) is :-  
 (A) between centre of curvature and  $\infty$  (B) between focus and centre of curvature  
 (C) at focus (D) between pole and focus
84. Focal length of the mirror is :-  
 (A) -60 cm (B) -40 cm (C) -30 cm (D) -25 cm
85. Distance of position (1) of the object from pole and that of position (2) of object from pole are, respectively :-  
 (A) 60 cm, 75 cm (B) 75 cm, 60 cm (C) 60 cm, 45 cm (D) 45 cm, 30 cm

#### Paragraph for Question No. 86 to 88

ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC.



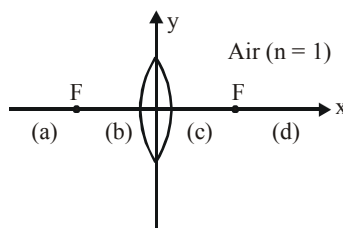
Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) to have the same frequency, answer the following questions.

86. Refractive index of the material of prism is :-  
 (A)  $\sqrt{2}$  (B) 2 (C) 1.5 (D) 2.2
87. Angle of incidence of ray (2) is :-  
 (A)  $0^\circ$  (B)  $45^\circ$  (C)  $60^\circ$  (D)  $90^\circ$
88. Deviation suffered by ray (3) is :-  
 (A)  $60^\circ$  (B)  $30^\circ$  (C)  $90^\circ$  (D)  $120^\circ$

**Paragraph for Question no. 89 and 90**

This question concerns a symmetrical lens shown, along with its two focal points. It is made of plastic with ( $n = 1.2$ ), and has a focal length  $f$ . Four different regions are shown :

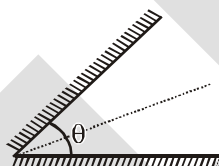
Here (a)  $-\infty < x < -f$ ; (b)  $-f < x < 0$ ; (c)  $0 < x < f$ ; (d)  $f < x < \infty$



89. If an object is placed somewhere in region (a), in which region does the image appear ?  
 (A) (a) (B) (b) (C) (c) (D) (d)
90. If incident rays are converging then in which region does the image appear ?  
 (A) (a) (B) (b) (C) (c) (D) (d)

**MATRIX MATCH TYPE QUESTIONS**

91. Angle between two mirrors ' $\theta$ ' and location of object is given in column-I and some possible number of images are given in column-II.



**Column-I**

- (A)  $\theta = 30^\circ$  and object is on bisector  
 (B)  $\theta = 36^\circ$  and object is on bisector  
 (C)  $\theta = 22.5^\circ$  and object is on bisector  
 (D)  $\theta = 50^\circ$  and object is on bisector

**Column-II**

- (P) 9  
 (Q) 11  
 (R) 8  
 (S) 15

92. An object O is kept perpendicular to the principal axis of a spherical mirror. In column-I, coordinate of, object,  $u$  with sign in cm, type of mirror and its focal length in cm is given :-

**Column-I**

- (A)  $u = -18$ , concave mirror, 12  
 (B)  $u = -12$ , concave mirror, 18  
 (C)  $u = -8$ , convex mirror, 10  
 (D)  $u = -10$ , convex mirror, 8

**Column-II**

- (P) Image is real  
 (Q) Image is virtual  
 (R) Image diminished  
 (S) Image is enlarged  
 (T) Image is inverted

93. **Column-I**

- (A) An object is placed at a distance equal to focal length from pole before convex mirror  
 (B) An object is placed at focus before a concave mirror  
 (C) An object is placed at the centre of curvature before a concave mirror.  
 (D) An object is placed at a distance equal to radius of curvature before a convex mirror.

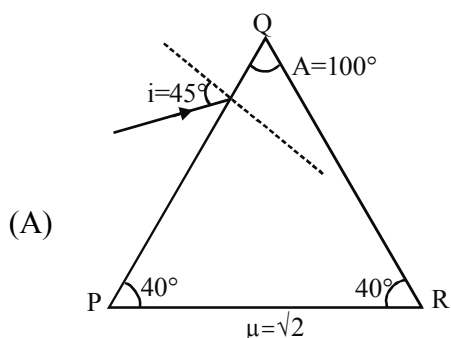
**Column-II**

- (P) Magnification is ( $\infty$ )  
 (Q) Magnification is (0.5)  
 (R) Magnification is ( $1/3$ )  
 (S) Magnification is ( $-1$ )

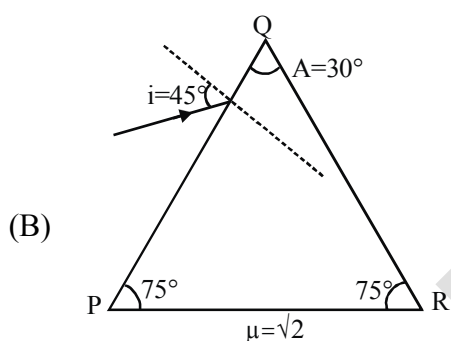
94. Light is incident at surface PQ of prism as shown in column-I then match the column-I with column-II (surrounding medium is air in all cases)

## Column-I

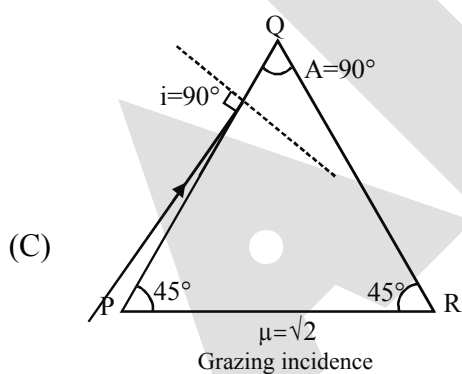
## Column-II



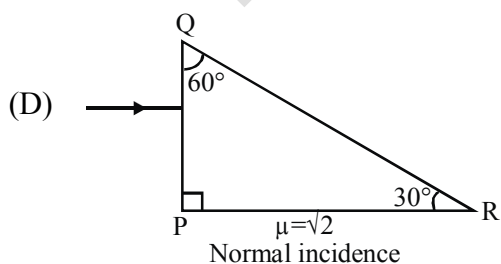
(P) Total internal reflection takes place at surface QR.



(Q) Light emerges normally from the surface QR



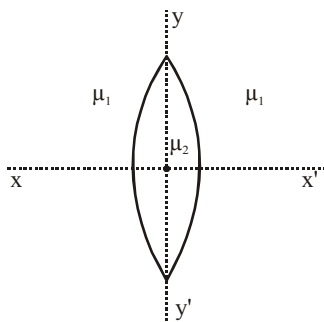
(R) Light emerges parallel to surface QR



(S) The light ray emerges from face PR perpendicularly

(T) When light ray passes through the prism it is parallel to the base PR.

95. An equi-convex lens of refractive index  $\mu_2$  and focal length  $f$  (in air) is kept in medium of refractive index  $\mu_1$ .



**Column-A**

- (P) If lens is cut in two equal parts by a plane  $yy'$ . ( $\mu_1 = 1$ )  
 (Q) If lens is cut in two equal parts by a plane  $xx'$ . ( $\mu_1 = 1$ )  
 (R) If  $\mu_1 = \mu_2$ .  
 (S) If  $\mu_1 > \mu_2$ .

**Column-B**

- (1) Lens will act as a glass slab  
 (2) Lens will be converging and focal length will change.  
 (3) Lens will be converging and focal length will remain same  
 (4) Lens will be diverging and focal length will change.

**Codes :**

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 4 | 1 | 3 |
| (B) | 2 | 3 | 1 | 4 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 1 | 2 | 3 | 4 |

96. Column-II shows the optical phenomenon that can be associated with optical components given in column-I. Note that column-I may have more than one matching options in column-II.

**Column-I**

- (A) Convex mirror  
 (B) Converging lens  
 (C) Thin prism  
 (D) Glass slab

**Column-II**

- (P) Dispersion  
 (Q) Deviation  
 (R) Real image of real object  
 (S) Virtual images of real object

## SUPPLEMENT FOR MAINS

**Optical Instruments :**

97. Myopia can be removed by using a lenses of :-  
(A) concave lens (B) convex lens  
(C) cylindrical lens (D) by surgical removal
98. Loss of the ability of eye to focus on near and far objects with advancing age is called :-  
(A) Presbyopia (B) Astigmatism (C) Hypermetropia (D) Myopia
99. A long sighted person has a minimum distance of distinct vision of 50 cm. He wants to reduce it to 25cm. He should use a :-  
(A) Concave lens of focal length 50 cm (B) Convex lens of focal length 25 cm  
(C) Convex lens of focal length 50 cm (D) Concave lens of focal length 25 cm
100. A person's near point is 50 cm and his far point is 3m. Powers of the lenses he required for (i) for reading (ii) for viewing distinct stars :-  
(A) - 2D and 0.33 D (B) 2D and - 0.33 D (C) - 2D and 3D (D) 2D and - 3 D
101. In a simple microscope, if the final image is located at infinity then its magnifying power :-  
(A)  $25/F$  (B)  $25/D$  (C)  $F/25$  (D)  $(1+25/F)$
102. In a compound microscope, the intermediate image, in normal use is :-  
(A) Virtual, erect and magnified (B) Real, erect and magnified  
(C) Real, inverted and magnified (D) Virtual, inverted and magnified
103. If tube length of astronomical telescope is 105 cm and magnifying power is 20 for normal setting. Calculate the focal length of objective :-  
(A) 100 cm (B) 10 cm (C) 20 cm (D) 25 cm
104. When length of a compound microscope tube increase, its magnifying power :-  
(A) decreases (B) increases  
(C) does not change (D) may increase or decrease
105. An astronomical telescope of angular magnification 10, when final image is at infinite has a length of 44 cm. The focal length of the objective is :-  
(A) 4 cm (B) 40 cm (C) 44 cm (D) 440 cm
106. When the length of an astronomical telescope tube increases its magnifying power :-  
(A) Decreases (B) Increases  
(C) Does not change (D) May increase or decrease
107. The objective of a small telescope has focal length of 120 cm and diameter 5 cm. The focal length of the eye-piece is 2 cm. The magnifying power of telescope for distant object is :-  
(A) 12 (B) 24 (C) 60 (D) 300

108. An astronomical telescope has an angular magnification of magnitude 5 for distance objects. The separation between the objective and the eye-piece is 36 cm and the final image is formed at infinity. The focal length  $f_o$  of the objective and  $f_e$  of the eye-piece are :-
- (A)  $f_o = 45$  cm and  $f_e = -9$  cm (B)  $f_o = 50$  cm and  $f_e = 10$  cm  
(C)  $f_o = 7.2$  cm and  $f_e = 5$  cm (D)  $f_o = 30$  cm and  $f_e = 6$  cm
109. The magnifying power of telescope is found to be 9 and separation between lenses is 20cm. what are focal lengths of component lenses :-
- (A) 18cm, 2cm (B) 4cm, 16cm (C) 10cm, 10cm (D) 12cm, 8cm

**Paragraph for Question 110 to 113**

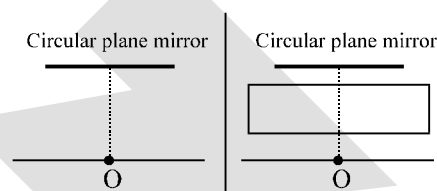
110. If an astronomical telescope has objective and eye-pieces of focal lengths 200 cm and 4 cm respectively, then the magnifying power of the telescope for the normal vision is :-
- (A) 42 (B) 50 (C) 58 (D) 204
111. In the above question the length of the telescope for normal vision, is :-
- (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm
112. In the above question magnifying power of the telescope for distinct vision is :-
- (A) 42 (B) 50 (C) 58 (D) 204
113. In the above question the length of the telescope for distinct vision is :-
- (A) 204 cm (B) 200 cm (C) 196 cm (D) 203.45 cm

## EXERCISE O-2

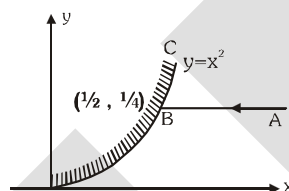
## SINGLE CORRECT TYPE QUESTIONS

1. In the diagram shown below, a point source O is placed vertically below the center of a circular plane mirror. The light rays starting from the source are reflected from the mirror such that a circular area A on the ground receives light. Now, a glass slab is placed between the mirror and the source O. What will the magnitude of the new area on the ground receiving light?

- (A) A  
(B) Greater than A  
(C) Less than A  
(D) Cannot tell, as the information given is insufficient

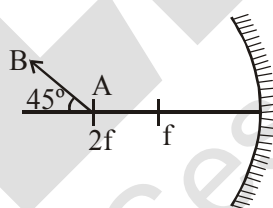


2. In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is :-



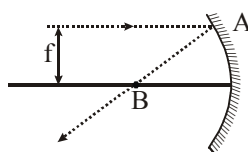
- (A)  $30^\circ$  (B)  $60^\circ$  (C)  $45^\circ$  (D) None of these

3. The shape of image formed of an object AB due to the concave mirror shown in the figure is best represented by (assume point A is at the centre of curvature of the mirror) :-



- (A) (B) (C) (D)

4. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray) :-

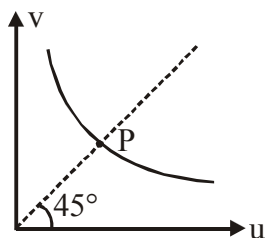


- (A)  $\frac{2}{\sqrt{3}}$  (B)  $\frac{\sqrt{3}}{2}$  (C)  $\frac{2}{3}$  (D)  $\frac{1}{2}$

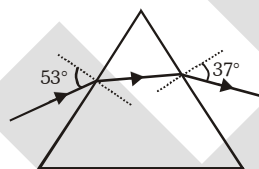
5. An infinitely long rod lies along the axis of a concave mirror of focal length  $f$ . The near end of the rod is at distance  $u > f$  from the mirror. Its image will have a length :-

(A)  $\frac{uf}{u-f}$       (B)  $\frac{uf}{u+f}$       (C)  $\frac{f^2}{u+f}$       (D)  $\frac{f^2}{u-f}$

6. The graph shows the variation of  $v$  with change in  $u$  for a mirror. Points plotted above the point P on the curve are for values of  $v$  :-

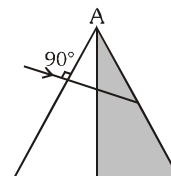


- (A) smaller than  $f$       (B) smaller than  $2f$       (C) larger than  $2f$       (D) larger than  $f$
7. A ray incident at an angle  $53^\circ$  on a prism emerges at an angle  $37^\circ$  as shown. If the angle of incidence is made  $50^\circ$ , which of the following is a possible value of the angle of emergence?

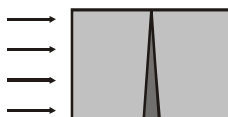


- (A)  $35^\circ$       (B)  $42^\circ$       (C)  $40^\circ$       (D)  $38^\circ$
8. A ray of light is incident normally on the first refracting face of the prism of refracting angle  $A$ . The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will :-

- (A) Emerge at an angle of emergence  $\sin^{-1}\left(\frac{1}{2}\sec A/2\right)$
- (B) Not emerge out of the prism
- (C) Emerge at an angle of emergence  $\sin^{-1}\left(\frac{1}{2}\sec A/4\right)$
- (D) None of these

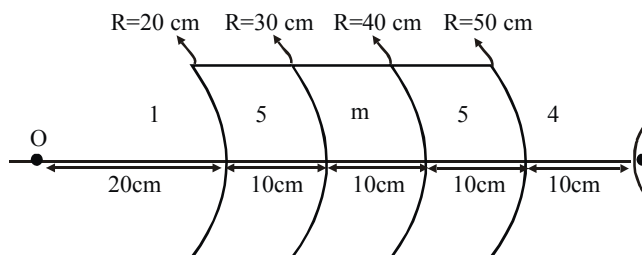


9. A thin isosceles prism with angle  $4^\circ$  and refractive index  $5/4$  is placed inside a transparent tube with liquid (refractive index = 1.5) as shown. The deviation of light due to prism will be :-

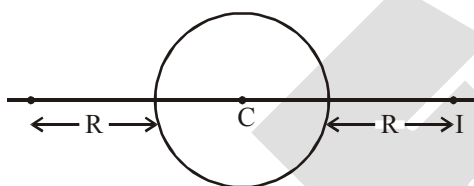


- (A)  $0.8^\circ$  upward      (B)  $0.8^\circ$  downward      (C)  $0.67^\circ$  upward      (D)  $0.67^\circ$  downward

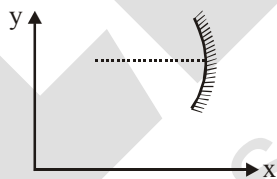
10. In the arrangement shown, separation between observer and object as seen by observer is :-



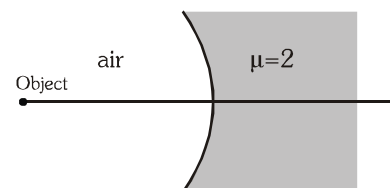
- (A) 60 cm      (B) more than 60 cm      (C) less than 60 cm      (D) depends on  $\mu$
11. In the given situation, a spherical transparent surface of radius  $R$  is placed. The refractive index of sphere is such that image for an object kept at distance  $R$  is formed at the same distance of the opposite side. What is refractive index of sphere?



- (A) 1.5      (B) 1.75      (C) 1.25      (D) 2
12. Figure shows a concave mirror with its principal axis parallel to x-axis and focus on the y-axis. The centre of curvature is at  $(-108 \text{ cm}, 54 \text{ cm})$ . The coordinates of the image of a point source A  $(81 \text{ cm}, 48 \text{ cm})$  will be :-

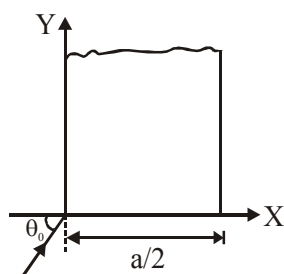


- (A)  $(144 \text{ cm}, 46 \text{ cm})$       (B)  $(144 \text{ cm}, 62 \text{ cm})$   
 (C)  $(36 \text{ cm}, 8 \text{ cm})$       (D) None of these
13. A light ray is incident on a transparent sphere of index  $=\sqrt{2}$ , at an angle of incidence  $= 45^\circ$ . What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air ?
- (A)  $270^\circ$       (B)  $240^\circ$       (C)  $120^\circ$       (D)  $180^\circ$
14. A concave spherical surface with radius  $R$  separates a medium with index of refraction 2 from air. If an object is moved toward the surface from far away along the central axis, its image :
- (A) Changes from virtual to real when it is  $R/2$  from the surface  
 (B) Changes from virtual to real when it is  $2R$  from the surface  
 (C) Changes from real to virtual when it is  $R/2$  from the surface  
 (D) Remains virtual

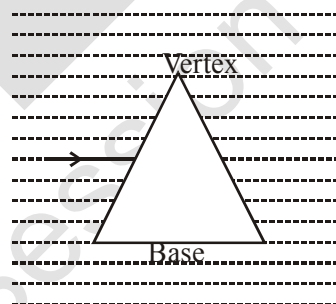
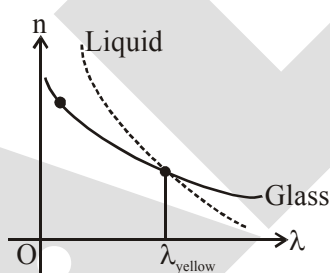


MULTIPLE CORRECT TYPE QUESTIONS

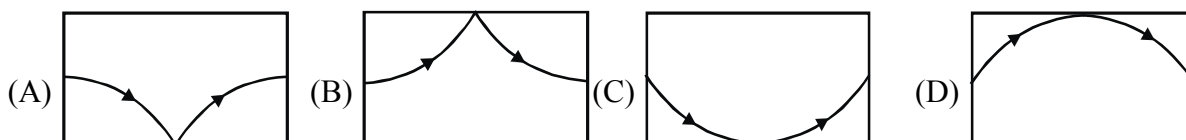
15. A glass slab is of thickness  $d = a/2$  is shown in figure. The refractive index of the slab varies according to relation  $\mu = \frac{a}{a-x}$  where  $a$  is positive constant. A ray of light enters the slab at origin at an angle  $\theta_0 \rightarrow \pi/2$ , then choose the **CORRECT** statement (s):-



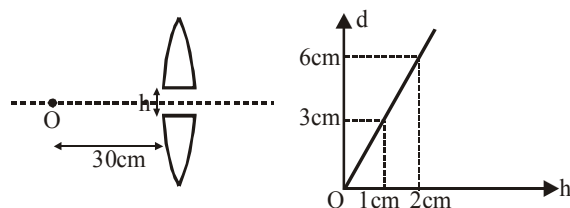
- (A) The path of the light ray inside medium is a circle given by  $(x-a)^2 + y^2 = a^2$   
 (B) The path of the light ray inside medium is a parabola given by  $(x-a)^2 = y$   
 (C) The deviation of ray from its initial direction is  $\delta = \pi/3$   
 (D) Ray emerges grazing at the surface.
16. A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index  $n$  as a function of wavelength  $\lambda$  for glass and liquid are as shown in the figure. When a ray of white light is incident on the prism parallel to the base :



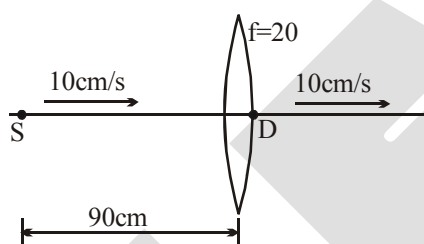
- (A) yellow ray travels without deviation  
 (B) blue ray is deviated towards the vertex  
 (C) red ray is deviated towards the base  
 (D) there is no dispersion
17. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following figures may represent the path of a ray of light inside the liquid?



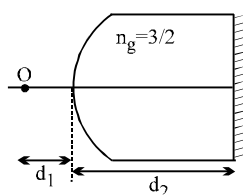
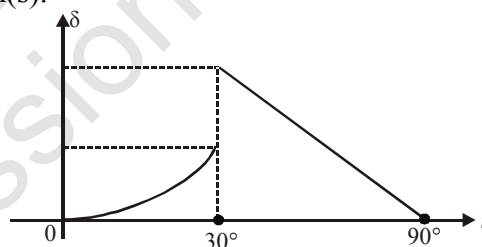
18. Figure shows a convex lens cut symmetrically into two equal halves and separated laterally by a distance  $h$ . A point object  $O$  placed symmetrically at a distance 30 cm, from the lens halves, forms two images separated by a distance  $d$ . A plot of  $d$  versus  $h$  is shown in figure. The focal length of the lens is :-



- (A) 22.5 cm      (B) 40 cm      (C) 45 cm      (D) 20 cm
19. The figure shows initial positions of a source of light  $S$  and a light detector  $D$ . Both  $S$  &  $D$  move with velocity 10 cm/s, as shown. The converging lens of focal length 20 cm is fixed. Time instant(s) when the detector detects maximum intensity of light is /are :-

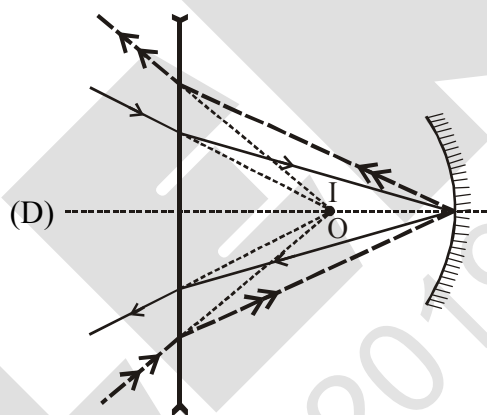
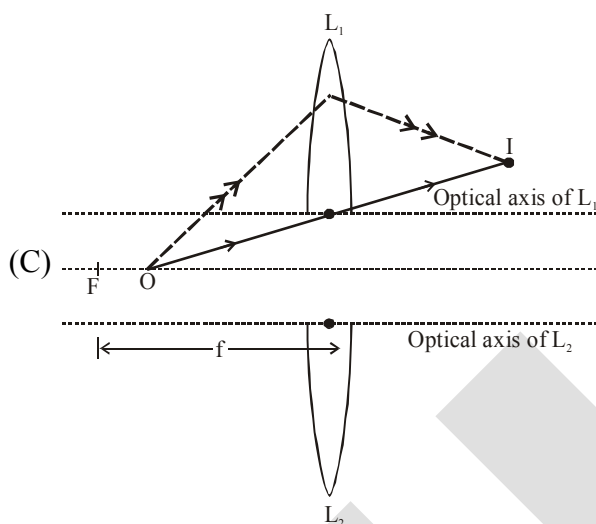
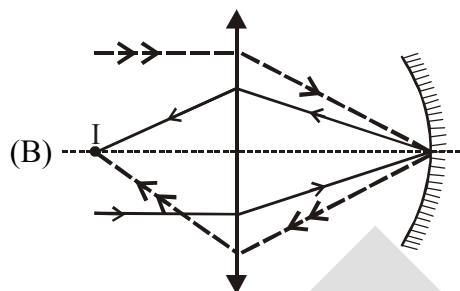


- (A) 3 sec      (B) 4.5 sec  
(C) 6 sec      (D) equal intensity at all time
20. Figure shows graph of angle of deviation  $\delta$  vs angle of incidence  $i$  for a light ray. Incident ray goes from medium 1 ( $\mu_1$ ) to medium 2 ( $\mu_2$ ). Mark the **correct** option(s).
- (A)  $\frac{\mu_1}{\mu_2} = \frac{1}{2}$   
(B) Critical angle is  $30^\circ$   
(C)  $\mu_1 > \mu_2$   
(D) Maximum deviation is  $120^\circ$
21. In the figure shown a point object  $O$  is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 cm.  $I_f$  is the final image formed after all the refractions and reflections.

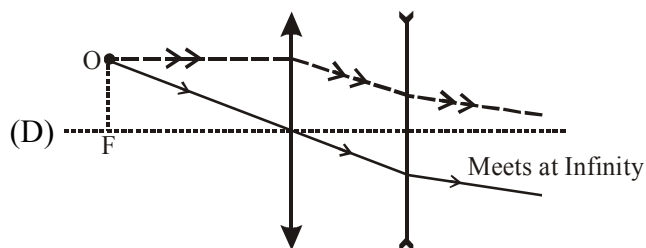
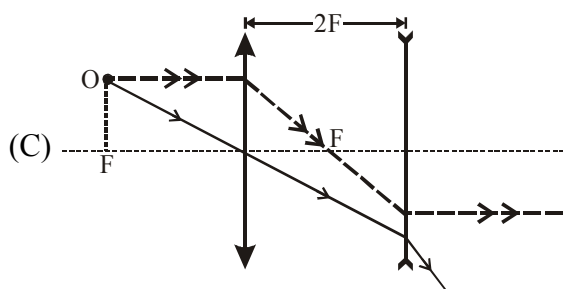
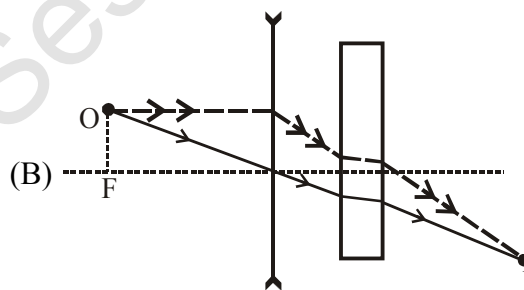
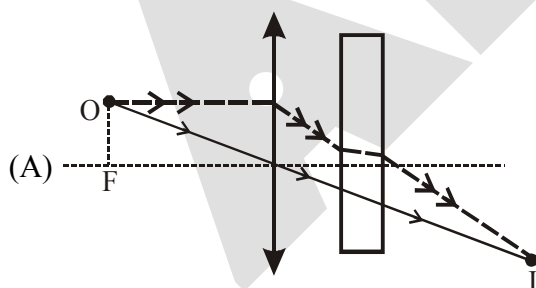


- (A) If  $d_1 = 120 \text{ cm}$ , then the ' $I_f$ ' is formed on ' $O$ ' for any value of  $d_2$ .  
(B) If  $d_1 = 240 \text{ cm}$ , then the ' $I_f$ ' is formed on ' $O$ ' only if  $d_2 = 360 \text{ cm}$ .  
(C) If  $d_1 = 240 \text{ cm}$ , then the ' $I_f$ ' is formed on ' $O$ ' for all values of  $d_2$ .  
(D) If  $d_1 = 240 \text{ cm}$ , then the ' $I_f$ ' cannot be formed on ' $O$ '.

$\begin{bmatrix} \updownarrow \end{bmatrix}$  denotes converging lens and  $\begin{bmatrix} \updownarrow \end{bmatrix}$  denotes diverging lens]



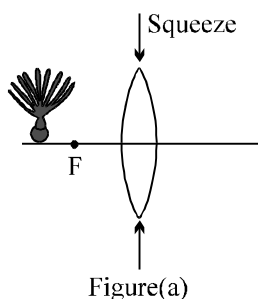
All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is  $F$ )



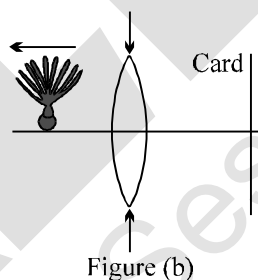
## COMPREHENSION TYPE QUESTIONS

## Paragraph for Question 24 to 26

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(a)], you can change the curvature of its front and rear sides.



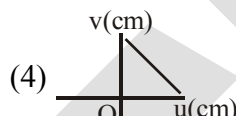
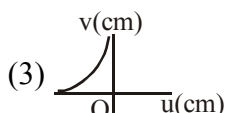
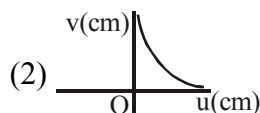
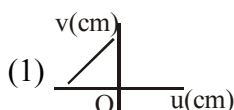
24. When you squeeze the lens, the image.  
 (A) moves towards the lens (B) moves away from the lens  
 (C) shifts up (D) remains as it is
25. When you squeeze the lens, the lateral height of image.  
 (A) increases (B) decreases (C) remains same (D) data insufficient
26. Suppose that a sharp image must be formed on a card which is at a certain distance behind the lens [figure(b)], while you move the turnip away from the lens, then you should



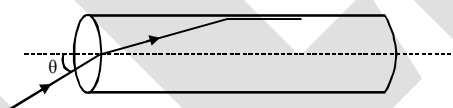
- (A) decrease the squeeze of the lens (B) increase the squeeze of the lens  
 (C) keep the card and lens as it is. (D) move the card away from the lens

# EXERCISE-(JM)

1. A student measures the focal length of a convex lens by putting an object pin at a distance 'u' from the lens and measuring the distance 'v' of the image pin. The graph between 'u' and 'v' plotted by the student should look like :- [AIEEE - 2008]



2. A transparent solid cylindrical rod has a refractive index of  $\frac{2}{\sqrt{3}}$ . It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure. [AIEEE - 2009]



The incident angle  $\theta$  for which the light ray grazes along the wall of the rod is :-

(1)  $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(2)  $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(3)  $\sin^{-1}\left(\frac{1}{2}\right)$

(4)  $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

3. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s. The speed of the image of the second car as seen in the mirror of the first one is:- [AIEEE- 2011]

(1) 10 m/s

(2) 15 m/s

(3)  $\frac{1}{10}$  m/s

(4)  $\frac{1}{15}$  m/s

4. When monochromatic red light is used instead of blue light in a convex lens, its focal length will :- [AIEEE- 2011]

(1) Does not depend on colour of light

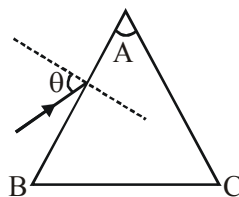
(2) Increase

(3) Decrease

(4) Remain same



11. Monochromatic light is incident on a glass prism of angle  $A$ . If the refractive index of the material of the prism is  $\mu$ , a ray, incident at an angle  $\theta$ , on the face  $AB$  would get transmitted through the face  $AC$  of the prism provided : [JEE-Main- 2015]



- (1)  $\theta > \cos^{-1} \left[ \mu \sin \left( A + \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$       (2)  $\theta < \cos^{-1} \left[ \mu \sin \left( A + \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$   
 (3)  $\theta > \sin^{-1} \left[ \mu \sin \left( A - \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$       (4)  $\theta < \sin^{-1} \left[ \mu \sin \left( A - \sin^{-1} \left( \frac{1}{\mu} \right) \right) \right]$

12. An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears : [JEE-Main- 2016]

- (1) 20 times nearer      (2) 10 times taller  
 (3) 10 times nearer      (4) 20 times taller

13. In an experiment for determination of refractive index of glass of a prism by  $i - \delta$ , plot, it was found that a ray incident at angle  $35^\circ$ , suffers a deviation of  $40^\circ$  and that it emerges at angle  $79^\circ$ . In that case which of the following is closest to the maximum possible value of the refractive index? [JEE-Main- 2016]

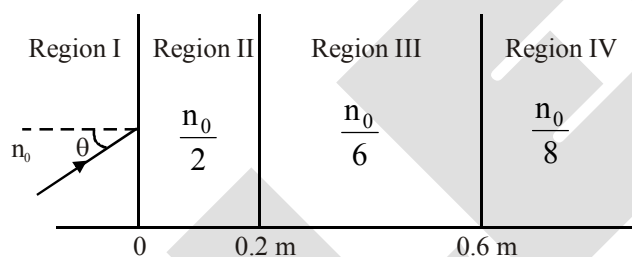
- (1) 1.8      (2) 1.5      (3) 1.6      (4) 1.7

14. A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15 cm from a converging lens of magnitude of focal length 20 cm. A beam of parallel light falls on the diverging lens. The final image formed is : [JEE-Main- 2017]

- (1) real and at a distance of 40 cm from the divergent lens  
 (2) real and at a distance of 6 cm from the convergent lens  
 (3) real and at a distance of 40 cm from convergent lens  
 (4) virtual and at a distance of 40 cm from convergent lens.

## EXERCISE-(JA)

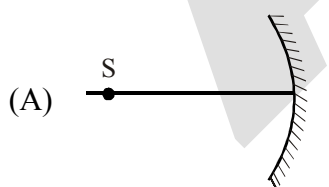
- Two beams of red and violet colours are made to pass separately through a prism (angle of the prism is  $60^\circ$ ). In the position of minimum deviation, the angle of refraction will be : **[IIT-JEE 2008]**  
 (A)  $30^\circ$  for both the colours (B) greater for the violet colour  
 (C) greater for the red colour (D) equal but not  $30^\circ$  for both the colours
- A light beam is traveling from Region I to Region IV (Refer Figure). The refractive index in Regions I, II, III and IV are  $n_0$ ,  $\frac{n_0}{2}$ ,  $\frac{n_0}{6}$  and  $\frac{n_0}{8}$ , respectively. The angle of incidence  $\theta$  for which the beam just misses entering Region IV is figure **[IIT-JEE 2008]**



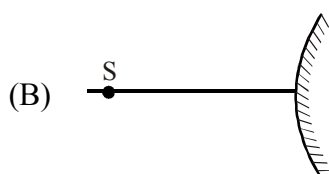
- (A)  $\sin^{-1} \left( \frac{3}{4} \right)$  (B)  $\sin^{-1} \left( \frac{1}{8} \right)$  (C)  $\sin^{-1} \left( \frac{1}{4} \right)$  (D)  $\sin^{-1} \left( \frac{1}{3} \right)$
- An optical component and an object S placed along its optic axis are given in **Column-I**. The distance between the object and the component can be varied. The properties of images are given in **Column-II**. Match all the properties of images from **Column-II** with the appropriate components given in **Column I**. Indicate your answer by darkening the appropriate bubbles of the  $4 \times 4$  matrix given in the ORS. **[IIT-JEE 2008]**

Column-I

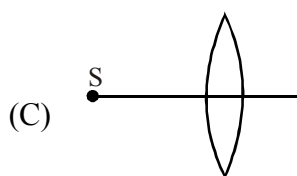
Column-II



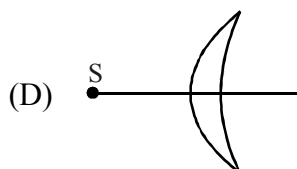
(p) Real image



(q) Virtual image



(r) Magnified image



(s) Image at infinite

4. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is  $4/3$ . A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant, when the ball is 12.8 m above the water surface, the fish sees the speed of ball as [ $g = 10 \text{ m/s}^2$ .]

[IIT-JEE 2009]

(A) 9 m/s (B) 12 m/s (C) 16 m/s (D) 21.33 m/s

5. A student performed the experiment of determination of focal length of a concave mirror by u-v method using an optical bench of length 1.5 m. The focal length of the mirror used is 24 cm. The maximum error in the location of the image can be 0.2 cm. The 5 sets of (u, v) values recorded by the student (in cm) are : (42, 56), (48, 48), (60, 40), (66, 33), (78, 39). The data set(s) that cannot come from experiment and is (are) incorrectly recorded, is (are) :

[IIT-JEE 2009]

(A) (42, 56) (B) (48, 48) (C) (66, 33) (D) (78, 39)

6. A biconvex lens of focal length 15 cm is in front of a plane mirror. The distance between the lens and the mirror is 10 cm. A small object is kept at a distance of 30 cm from the lens. The final image is

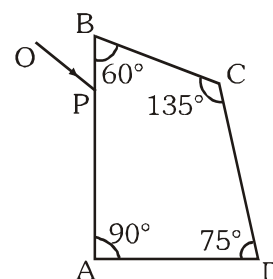
[IIT-JEE 2010]

(A) virtual and at a distance of 16 cm from the mirror  
(B) real and at a distance of 16 cm from the mirror  
(C) virtual and at a distance of 20 cm from the mirror  
(D) real and at a distance of 20 cm from the mirror

7. A ray OP of monochromatic light is incident on the face AB of prism ABCD near vertex B at an incident angle of  $60^\circ$  (see figure). If the refractive index of the material of the prism is  $\sqrt{3}$ , which of the following is (are) correct?

[IIT-JEE 2010]

(A) The ray gets totally internally reflected at face CD  
(B) The ray comes out through face AD  
(C) The angle between the incident ray and the emergent ray is  $90^\circ$   
(D) The angle between the incident ray and the emergent ray is  $120^\circ$



8. Two transparent media of refractive indices  $\mu_1$  and  $\mu_3$  have a solid lens shaped transparent material of refractive index  $\mu_2$  between them as shown in figures in **Column II**. A ray traversing these media is also shown in the figures. In **Column I** different relationships between  $\mu_1$ ,  $\mu_2$  and  $\mu_3$  are given. Match them to the ray diagrams shown in **Column II**. [IIT-JEE 2010]

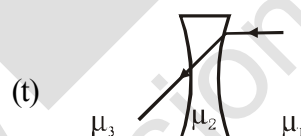
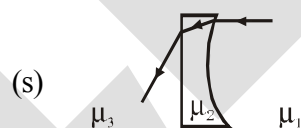
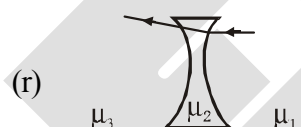
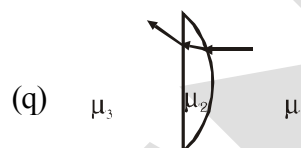
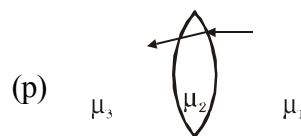
**Column I**

(A)  $\mu_1 < \mu_2$

(B)  $\mu_1 > \mu_2$

(C)  $\mu_2 = \mu_3$

(D)  $\mu_2 > \mu_3$

**Column II**

9. The focal length of a thin biconvex lens is 20 cm. When an object is moved from a distance of 25 cm in front of it to 50 cm, the magnification of its image changes from  $m_{25}$  to  $m_{50}$ . The ratio  $\frac{m_{25}}{m_{50}}$  is

[IIT-JEE 2010]

10. Image of an object approaching a convex mirror of radius of curvature 20 m along its optical axis is observed to move from  $\frac{25}{3}$  m to  $\frac{50}{7}$  m in 30 seconds. What is the speed of the object in km per hour?

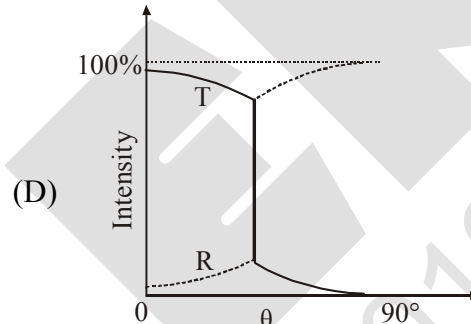
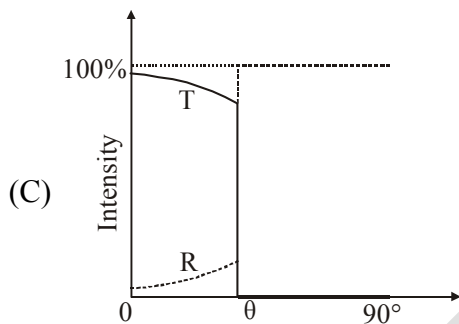
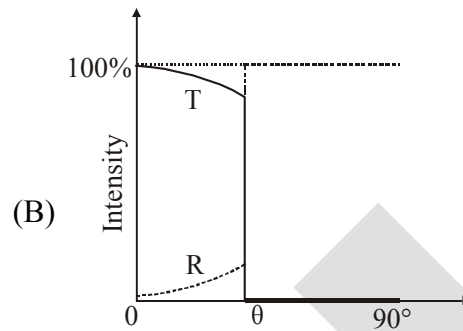
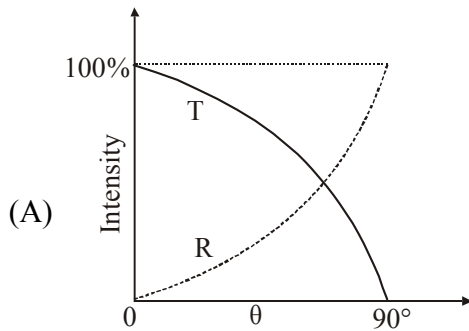
[IIT-JEE 2010]

11. A large glass slab ( $\mu = \frac{5}{3}$ ) of thickness 8 cm is placed over a point source of light on a plane surface.

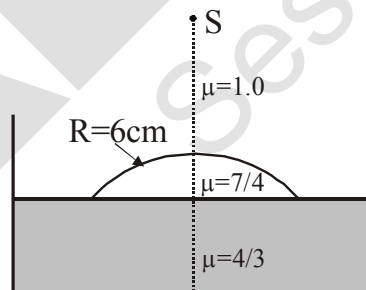
It is seen that light emerges out of the top surface of the slab from a circular area of radius R cm. What is the value of R?

[IIT-JEE 2010]

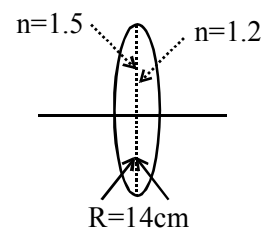
12. A light ray traveling in glass medium is incident on glass-air interface at an angle of incidence  $\theta$ . The reflected (R) and transmitted (T) intensities, both as function of  $\theta$ , are plotted. The correct sketch is [IIT-JEE 2011]



13. Water (with refractive index  $= \frac{4}{3}$ ) in a tank is 18 cm deep. Oil of refractive index  $\frac{7}{4}$  lies on water making a convex surface of radius of curvature ' $R = 6$  cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at ' $x$ ' cm above the bottom of the tank. Then ' $x$ ' is [IIT-JEE 2011]



14. A biconvex lens is formed with two thin plano-convex lenses as shown in the figure, Refractive index  $n$  of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature  $R = 14$  cm. For this biconvex lens, for an object distance of 40 cm, the image distance will be :- [IIT-JEE 2012]



- (A) -280.0 cm  
(B) 40.0 cm  
(C) 21.5 cm  
(D) 13.3 cm



17. A ray of light travelling in the direction  $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$  is incident on a plane mirror. After reflection, it

travels along the direction  $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$ . The angle of incidence is :- [JEE-Advance-2013]

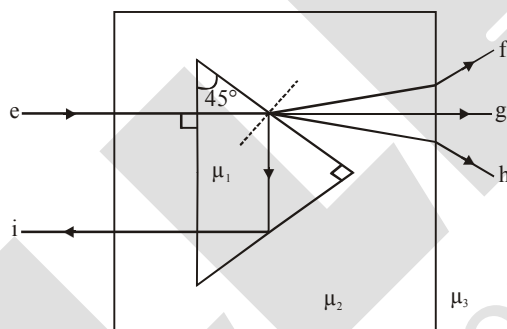
- (A)  $30^\circ$  (B)  $45^\circ$  (C)  $60^\circ$  (D)  $75^\circ$

18. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is  $\frac{2}{3}$  times the wavelength

in free space. The radius of the curved surface of the lens is :- [JEE-Advance-2013]

- (A) 1 m (B) 2 m (C) 3 m (D) 6 m

19. A right angled prism of refractive index  $\mu_1$  is placed in a rectangular block of refractive index  $\mu_2$ , which is surrounded by a medium of refractive index  $\mu_3$ , as shown in the figure. A ray of light 'e' enters the rectangular block at normal incidence. Depending upon the relationships between  $\mu_1$ ,  $\mu_2$ , and  $\mu_3$ , it takes one of the four possible paths 'ef', 'eg', 'eh' or 'ei'. [JEE-Advance-2013]



Match the paths in List-I with conditions of refractive indices in List-II and select the correct answer using the codes given below the lists :

**List-I**

- P.  $e \rightarrow f$   
Q.  $e \rightarrow g$   
R.  $e \rightarrow h$   
S.  $e \rightarrow i$

**List-II**

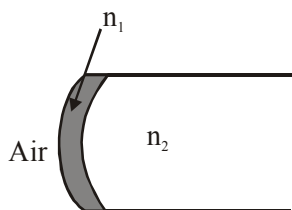
1.  $\mu_1 > \sqrt{2}\mu_2$
2.  $\mu_2 > \mu_1$  and  $\mu_2 > \mu_3$
3.  $\mu_1 = \mu_2$
4.  $\mu_2 < \mu_1 < \sqrt{2}\mu_2$  and  $\mu_2 > \mu_3$

**Codes :**

|     | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 3 | 1 | 4 |
| (B) | 1 | 2 | 4 | 3 |
| (C) | 4 | 1 | 2 | 3 |
| (D) | 2 | 3 | 4 | 1 |

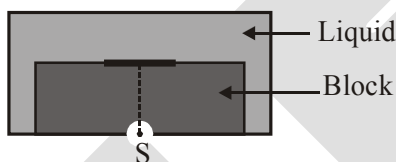
20. A transparent thin film of uniform thickness and refractive index  $n_1 = 1.4$  is coated on the convex spherical surface of radius  $R$  at one end of a long solid glass cylinder of refractive index  $n_2 = 1.5$ , as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance  $f_1$  from the film, while rays of light traversing from glass to air get focused at distance  $f_2$  from the film. Then

[JEE-Advance-2014]



- (A)  $|f_1| = 3R$       (B)  $|f_1| = 2.8R$       (C)  $|f_2| = 2R$       (D)  $|f_2| = 1.4R$
21. A point source  $S$  is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is :-

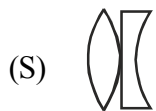
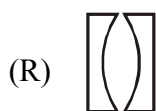
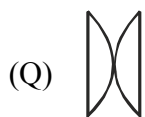
[JEE-Advance-2014]



- (A) 1.21      (B) 1.30      (C) 1.36      (D) 1.42
22. Four combinations of two thin lenses are given in List-I. The radius of curvature of all curved surfaces is  $r$  and the refractive index of all the lenses is 1.5. Match lens combinations in List-I with their focal length in List-II and select the correct answer using the code given below the lists.

[JEE-Advance-2014]

## List-I



## List-II

(1)  $2r$

(2)  $\frac{r}{2}$

(3)  $-r$

(4)  $r$

## Code :

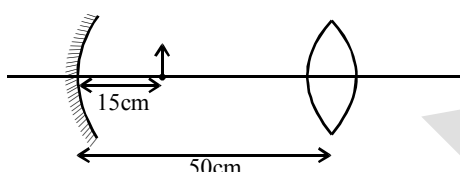
- (A) P-1, Q-2, R-3, S-4  
(C) P-4, Q-1, R-2, S-3

- (B) P-2, Q-4, R-3, S-1  
(D) P-2, Q-1, R-3, S-4

23. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification  $M_1$ . When the set-up is kept in a medium of refractive index  $7/6$  the magnification

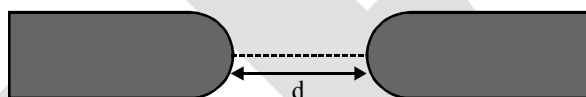
becomes  $M_2$ . The magnitude  $\left| \frac{M_2}{M_1} \right|$  is.

[JEE-Advance-2015]



24. Two identical glass rods  $S_1$  and  $S_2$  (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance  $d$  as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light  $P$  is placed inside rod  $S_1$  on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside  $S_2$ . The distance  $d$  is :

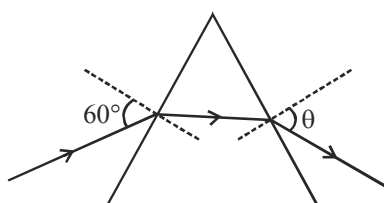
[JEE-Advance-2015]



- (A) 60 cm  
(B) 70 cm  
(C) 80 cm  
(D) 90 cm
25. A monochromatic beam of light is incident at  $60^\circ$  on one face of an equilateral prism of refractive index  $n$  and emerges from the opposite face making an angle  $\theta(n)$  with the normal (see the figure). For

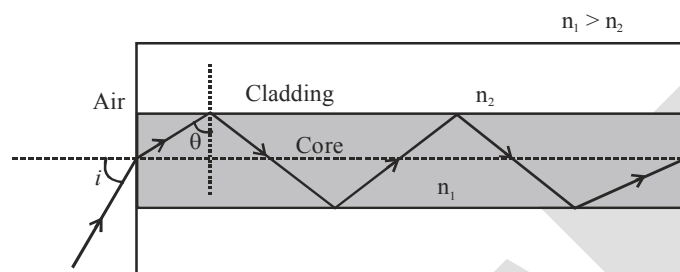
$n = \sqrt{3}$  the value of  $\theta$  is  $60^\circ$  and  $\frac{d\theta}{dn} = m$ . The value of  $m$  is

[JEE-Advance-2015]

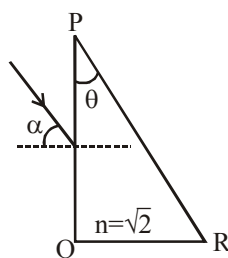


## Paragraph for Question No. 26 and 27

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index  $n_1$  surrounded by a medium of lower refractive index  $n_2$ . The light guidance in the structure takes place due to successive total internal reflections at the interface of the media  $n_1$  and  $n_2$  as shown in the figure. All rays with the angle of incidence  $i$  less than a particular value of  $i_m$  are confined in the medium of refractive index  $n_1$ . The numerical aperture (NA) of the structure is defined as  $\sin i_m$ . [JEE-Advance-2015]

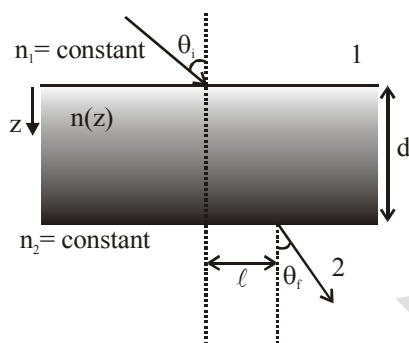


26. For two structures namely  $S_1$  with  $n_1 = \sqrt{45}/4$  and  $n_2 = 3/2$ , and  $S_2$  with  $n_1 = 8/5$  and  $n_2 = 7/5$  and taking the refractive index of water to be  $4/3$  and that of air to be 1, the correct option(s) is (are)
- (A) NA of  $S_1$  immersed in water is the same as that of  $S_2$  immersed in liquid of refractive index  $\frac{16}{3\sqrt{15}}$ .
- (B) NA of  $S_1$  immersed in liquid of refractive index  $\frac{6}{\sqrt{15}}$  is the same as that of  $S_2$  immersed in water.
- (C) NA of  $S_1$  placed in air is the same as that of  $S_2$  immersed in liquid of refractive index  $\frac{4}{\sqrt{15}}$ .
- (D) NA of  $S_1$  placed in air is the same as that of  $S_2$  placed in water.
27. If two structures of same cross-sectional area, but different numerical apertures  $NA_1$  and  $NA_2$  ( $NA_2 < NA_1$ ) are joined longitudinally, the numerical aperture of the combined structure is
- (A)  $\frac{NA_1 NA_2}{NA_1 + NA_2}$  (B)  $NA_1 + NA_2$  (C)  $NA_1$  (D)  $NA_2$
28. A parallel beam of light is incident from air at an angle  $\alpha$  on the side PQ of a right angled triangular prism of refractive index  $n = \sqrt{2}$ . Light undergoes total internal reflection in the prism at the face PR when  $\alpha$  has a minimum value of  $45^\circ$ . The angle  $\theta$  of the prism is : [JEE-Advance-2016]

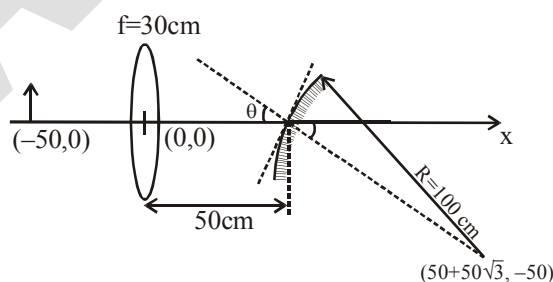


- (A)  $15^\circ$  (B)  $22.5^\circ$  (C)  $30^\circ$  (D)  $45^\circ$

29. A transparent slab of thickness  $d$  has a refractive index  $n(z)$  that increases with  $z$ . Here  $z$  is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices  $n_1$  and  $n_2$  ( $> n_1$ ), as shown in the figure. A ray of light is incident with angle  $\theta_i$  from medium 1 and emerges in medium 2 with refraction angle  $\theta_f$  with a lateral displacement  $\ell$ . Which of the following statement(s) is(are) true? [JEE-Advance-2016]



- (A)  $\ell$  is independent of  $n_2$  (B)  $\ell$  is dependent on  $n(z)$   
 (C)  $n_1 \sin \theta_i = (n_2 - n_1) \sin \theta_f$  (D)  $n_1 \sin \theta_i = n_2 \sin \theta_f$
30. A plano-convex lens is made of a material of refractive index  $n$ . When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true? [JEE-Advance-2016]
- (A) The refractive index of the lens is 2.5  
 (B) The radius of curvature of the convex surface is 45 cm  
 (C) The faint image is erect and real  
 (D) The focal length of the lens is 20 cm.
31. A small object is placed 50 cm to the left of thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle  $\theta = 30^\circ$  to the axis of the lens, as shown in the figure. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point  $(x, y)$  at which the image is formed are : [JEE-Advance-2016]



- (A)  $(25, 25\sqrt{3})$  (B)  $\left(\frac{125}{3}, \frac{25}{\sqrt{3}}\right)$  (C)  $(50 - 25\sqrt{3}, 25)$  (D)  $(0, 0)$

32. For an isosceles prism of angle  $A$  and refractive index  $\mu$ , it is found that the angle of minimum deviation  $\delta_m = A$ . Which of the following options is/are correct ? **[JEE-Advance-2017]**

(A) At minimum deviation, the incident angle  $i_1$  and the refracting angle  $r_1$  at the first refracting surface are related by  $r_1 = (i_1/2)$

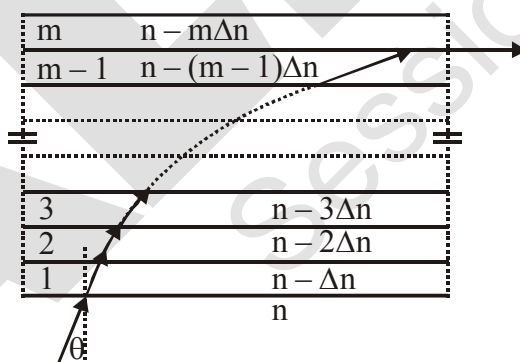
(B) For this prism, the refractive index  $\mu$  and the angle of prism  $A$  are related as  $A = \frac{1}{2} \cos^{-1} \left( \frac{\mu}{2} \right)$

(C) For this prism, the emergent ray at the second surface will be tangential to the surface when the

angle of incidence at the first surface is  $i_1 = \sin^{-1} \left[ \sin A \sqrt{4 \cos^2 \frac{A}{2} - 1} - \cos A \right]$

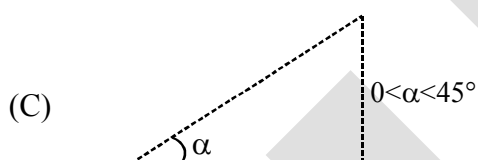
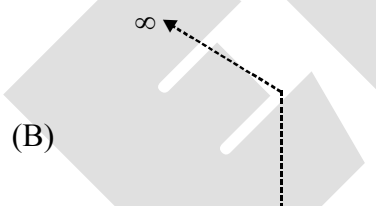
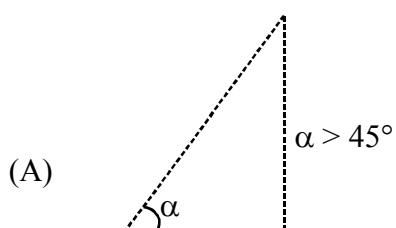
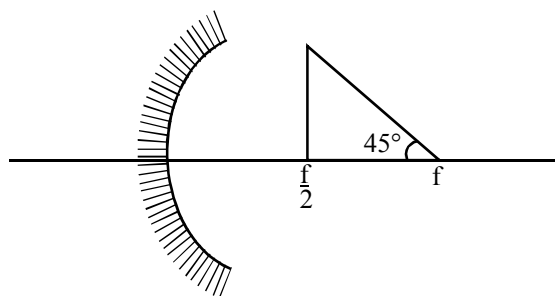
(D) For the angle of incidence  $i_1 = A$ , the ray inside the prism is parallel to the base of the prism.

33. A monochromatic light is travelling in a medium of refractive index  $n = 1.6$ . It enters a stack of glass layers from the bottom side at an angle  $\theta = 30^\circ$ . The interfaces of the glass layers are parallel to each other. The refractive indices of different glass layers are monotonically decreasing as  $n_m = n - m\Delta n$ , where  $n_m$  is the refractive index of the  $m^{\text{th}}$  slab and  $\Delta n = 0.1$  (see the figure). The ray is refracted out parallel to the interface between the  $(m-1)^{\text{th}}$  and  $m^{\text{th}}$  slabs from the right side of the stack. What is the value of  $m$  ? **[JEE-Advance-2017]**



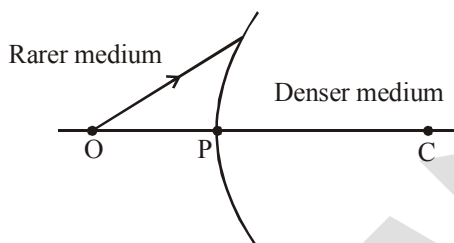
34. Sunlight of intensity  $1.3 \text{ kW m}^{-2}$  is incident normally on a thin convex lens of focal length 20 cm. Ignore the energy loss of light due to the lens and assume that the lens aperture size is much smaller than its focal length. The average intensity of light, in  $\text{kW m}^{-2}$ , at a distance 22 cm from the lens on the other side is \_\_\_\_\_. **[JEE-Advance-2018]**

35. A wire is bent in the shape of a right angled triangle and is placed in front of a concave mirror of focal length  $f$ , as shown in the figure. Which of the figures shown in the four options qualitatively represent(s) the shape of the image of the bent wire ? (These figures are not to scale.) ? [JEE-Advance-2018]



## CBSE PREVIOUS YEAR'S QUESTIONS

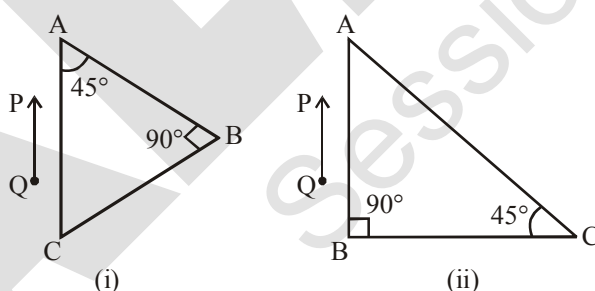
1. Draw a ray diagram of an astronomical telescope in the normal adjustment position. Write down the expression for its magnifying power. [2; CBSE-2004]
2. A spherical surface of radius of curvature  $R$ , separates a rarer and a denser medium as shown in the figure.



Complete the path of the incident ray of light, showing the formation of a real image. Hence derive the relation connecting object distance ' $u$ ', image distance ' $v$ ', radius of curvature  $R$  and the refractive indices  $n_1$  and  $n_2$  of the two media.

Briefly explain, how the focal length of a convex lens changes, with increase in wavelength of incident light. [5; CBSE-2004]

3. A right-angled crown glass prism with critical angle  $41^\circ$  is placed before an object PQ, in two positions as shown in the figures (i) and (ii). Trace the paths of the rays from P and Q the prisms in the two cases. [2; CBSE-2005]



4. (a) Draw a labelled ray diagram to show the formation of an image by a compound microscope. Write the expression for its magnifying power.  
(b) How does the resolving power of a compound microscope change, when (i) refractive index of the medium between the object and the objective lens increases; and (ii) wavelength of the radiation used is increased? [3; CBSE-2005]
5. What is the function of cladding in a typical optical fiber? [1; CBSE-2005]
6. Draw a labelled ray diagram to show the image formation in a refracting type astronomical telescope. Why should the diameter of the objective of a telescope be large? [2; CBSE-2006]

7. A beam of light converges to a point P. A lens is placed in the path of the convergent beam 12 cm from P. At what point does the beam converge if the lens is
- a convex lens of focal length 20 cm,
  - a concave lens of focal length 16 cm ?

Do the required calculations.

[3; CBSE-2006]

8. Define resolving power of a compound microscope. How does the resolving power of a compound microscope change when

- refractive index of the medium between the object and objective lens increases ?
- wavelength of the radiation used is increased

[2; CBSE-2007]

9. A double convex lens of glass of refractive index 1.6 has its both surfaces of equal radii of curvature of 30 cm each. An object of height 5 cm is placed at a distance of 12.5 cm from the lens. Calculate the size of the image formed.

[2; CBSE-2007]

10. A glass lens of refractive index 1.5 is placed in a trough of liquid. What must be the refractive index of the liquid in order to make the lens disappear?

[1; CBSE-2008]

11. A ray of light passing through an equilateral triangular glass prism from air undergoes minimum deviation when angle of incidence is  $3/4^{\text{th}}$  of the angle of prism. Calculate the speed of light in the prism.

[2; CBSE-2008]

12. (a) For a ray of light traveling from a denser medium of refractive index  $n_1$  to a rarer medium of

refractive index  $n_2$ , prove that  $\frac{n_2}{n_1} = \sin i_c$  where  $i_c$  is the critical angle of incidence for the media.

(b) Explain with the help of a diagram, how the above principle is used for transmission of video signals using optical fibres.

[5; CBSE-2008]

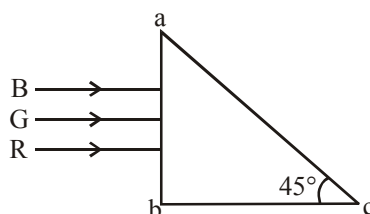
13. You are given following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope?

[1; CBSE-2009]

| Lenses | Power (P) | Aperture (A) |
|--------|-----------|--------------|
| L1     | 3D        | 8 cm         |
| L2     | 6D        | 1 cm         |
| L3     | 10D       | 1 cm         |

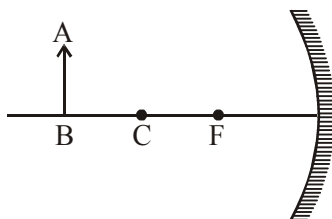
14. Three light rays red (R), green (G) and blue (B) are incident on a right angled prism 'abc' at face 'ab'. The refractive indices of the material of the prism for red, green and blue wavelength are 1.39, 1.44 and 1.47 respectively. Out of the three which colour ray will emerge out of face 'ac'? Justify your answer. Trace the path of these rays after passing through face 'ab'.

[3; CBSE-2009]

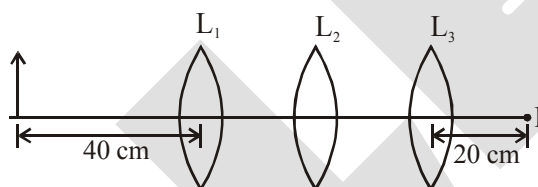


15. Trace the rays of light showing the formation of an image due to a point object placed on the axis of a spherical surface separating the two media of refractive indices  $n_1$  and  $n_2$ . Establish the relation between the distances of the object, the image and the radius of curvature from the central point the spherical surface. Hence derive the expression of lens maker's formula. [5; CBSE-2009]
16. Draw the labeled ray diagram for the formation of image by a compound microscope. Derive the expression for the total magnification of a compound microscope. Explain why both the objective and the eyepieces of a compound microscope must have short focal lengths. [5; CBSE-2009]
17. A converging lens is kept coaxially in contact with a diverging lens- both the lenses being of equal focal lengths. What is the focal length of the combination? [1; CBSE-2010]
18. (i) Draw a net labelled ray diagram of an astronomical telescope in normal adjustment. Explain briefly its working.  
(ii) An astronomical telescope uses two lenses of powers 10 D and 1 D. What is its magnifying power in normal adjustment? [3; CBSE-2010]
19. (i) Draw a neat labelled ray diagram of a compound microscope. Explain briefly its working,  
(ii) Why must both the objective and the eye piece of a compound microscope have short focal lengths? [3; CBSE-2010]
20. An illuminated object and a screen are placed 90 cm apart. Determine the focal length and nature of the lens required to produce a clear image on the screen, twice the size of the object, [3; CBSE-2010]
21. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in (i) a medium of refractive index 1.65, (ii) a medium of refractive index 1.33.  
(a) Will it behave as a converging or a diverging lens in the two cases ?  
(b) How will its focal length change in the two media ? [3; CBSE-2011]
22. Use the mirror equation to show that (a) an object placed between  $f$  and  $2f$  of a concave mirror produces a real image beyond  $2f$ . (b) a convex mirror always produces a virtual image independent of the location of the object, (c) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image. [3; CBSE-2011]
23. A compound microscope uses an objective lens of focal length 4 cm and eyepiece lens of focal length 10 cm. An object is placed at 6 cm from the objective lens. Calculate the magnifying power of the compound microscope. Also calculate the length of the microscope. [3; CBSE-2011]
24. A giant refracting telescope at an observatory has an objective lens of focal length 1 m. If an eyepiece lens of focal length 1.0 cm is used, find the angular magnification of the telescope. If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens ? The diameter of the moon is  $3.42 \times 10^6$  m and the radius of the lunar orbit is  $3.8 \times 10^8$  m. [3; CBSE-2011]
25. For the same value of angle of incidence, the angles of refraction in three media A, B and C are  $15^\circ$ ,  $25^\circ$  and  $35^\circ$  respectively. In which medium would the velocity of light be minimum? [1; CBSE-2012]

26. An object AB is kept in front of a concave mirror as shown in the figure.



- (i) Complete the ray diagram showing the image formation of the object.  
 (ii) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black? [2; CBSE-2012]
27. Draw a labeled ray diagram of a reflecting telescope. Mention its two advantages over the refracting telescope. [2; CBSE-2012]
28. You are given three lenses  $L_1$ ,  $L_2$  and  $L_3$  each of focal length 20 cm. A object is kept at 40 cm in front of  $L_1$ , as shown. The final real image is formed at the focus "I" of  $L_3$ . Find the separation between  $L_1$ ,  $L_2$  and  $L_3$ . [3; CBSE-2012]

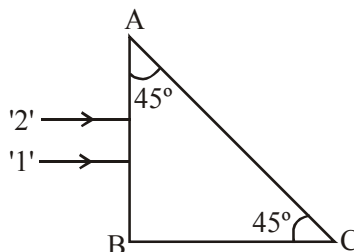


29. A convex lens of focal length  $f_1$  is kept in contact with a concave lens of focal length  $f_2$ . Find the focal length of the combination. [CBSE-2013]
30. Draw a labelled ray diagram of a refracting telescope. Define its magnifying power and write the expression for it. Write two important limitations of a refracting telescope over a reflecting type telescope. [CBSE-2013]
31. One day Chetan's mother developed a severe stomach ache all of a sudden. She was rushed to the doctor who suggested for an immediate endoscopy test and gave an estimate of expenditure for the same. Chetan immediately contacted his class teacher and shared the information with her. The class teacher arranged for the money and rushed to the hospital. On releasing that Chetan belonged to a below average income group family, even the doctor offered concession for the test fee. The test was conducted successfully. [CBSE-2013]

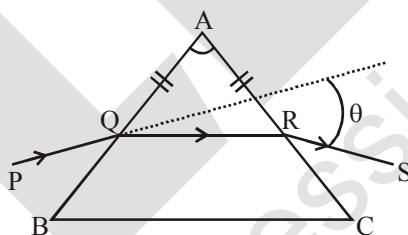
Answer the following questions based on the above information:

- (a) Which principle in optics is made use of in endoscopy?  
 (b) Briefly explain the values reflected in the action taken by the teacher.  
 (c) In what way do you appreciate the response of the doctor on the given situation?
32. A biconvex lens made of a transparent material of refractive index 1.5 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens? Give reason. [CBSE-2014]

33. Two monochromatic rays of light are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays '1' and '2' are respectively 1.3 and 1.5. Trace the path of these rays after entering through the prism. [CBSE-2014]



34. A Convex lens of focal length 20 cm is placed coaxially with a concave mirror of focal length 10 cm at a distance of 50 cm apart from each other. A beam of light coming parallel to the principal axis is incident on the convex lens. Find the position of the final image formed by this combination. Draw the ray diagram showing the formation of the image. [CBSE-2014]
35. When light travels from an optically denser medium to a rarer medium, why does the critical angle of incidence depend on the colour of light ? [1; CBSE-2015]
36. A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that  $AQ = AR$ . If the angle of prism  $A = 60^\circ$  and refractive index of material of prism is  $\sqrt{3}$ , calculate angle  $\theta$ . [CBSE-2016]

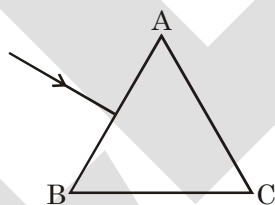


37. (i) A screen is placed at a distance of 100 cm from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by 20 cm. Calculate the focal length of the lens used.  
 (ii) A converging lens is kept coaxially in contact with a diverging lens-both the lenses being of equal focal length. What is the focal length of the combination ? [CBSE-2016]
38. How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced by red light ? Give reason. [CBSE-2017]
39. (a) Monochromatic light of wavelength 589 nm is incident from air on a water surface. If  $\mu$  for water is 1.33, find the wavelength, frequency and speed of the refracted light. [CBSE-2017]  
 (b) A double convex lens is made of a glass of refractive index 1.55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm.

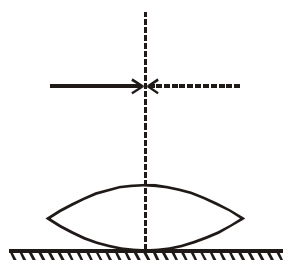
40. (a) Draw a ray diagram depicting the formation of the image by an astronomical telescope in normal adjustment. [CBSE-2017]  
 (b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? Give reason.

| Lenses | Power (D) | Aperture (cm) |
|--------|-----------|---------------|
| $L_1$  | 3         | 8             |
| $L_2$  | 6         | 1             |
| $L_3$  | 10        | 1             |

41. (a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.  
 (b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is  $\frac{3}{4}$ th of the angle of prism. Calculate the speed of light in the prism. [CBSE-2017]
42. (b) The figure shows a ray of light falling normally on the face AB of an equilateral glass prism having refractive index  $\frac{3}{2}$ , placed in water of refractive index  $\frac{4}{3}$ . Will this ray suffer total internal reflection on striking the face AC? Justify your answer. [CBSE-2018]



43. A symmetric biconvex lens of radius of curvature  $R$  and made of glass of refractive index 1.5, is placed on a layer of liquid placed on top of a plane mirror as shown in the figure. An optical needle with its tip on the principal axis of the lens is moved along the axis until its real, inverted image coincides with the needle itself. The distance of the needle from the lens is measured to be  $x$ . On removing the liquid layer and repeating the experiment, the distance is found to be  $y$ . Obtain the expression for the refractive index of the liquid in terms of  $x$  and  $y$ . [CBSE-2018]



44. (a) Draw a ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object.  
 (b) Obtain the mirror formula and write the expression for the linear magnification.  
 (c) Explain two advantages of a reflecting telescope over a refracting telescope. [CBSE-2018]

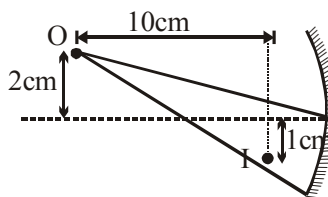
## ANSWER KEY

## EXERCISE (S-1)

1. Ans. 3m

2. Ans. 2

3. Ans. 2

4. Ans. 20 cm from object,  $20/3$  cm from object

5. Ans. 25

6. Ans.  $d/2$ 7. Ans.  $16 \text{ mm}^3$ 

8. Ans. 4

9. Ans. 4

10. Ans. 1

11. Ans. 5

12. Ans. 5

13. Ans. 42 cm

14. Ans. 4

15. Ans. 8

16. Ans. (a)  $\sin^{-1}\left(\frac{1}{5}\right)$  (b) air17. Ans.  $\theta < \sin^{-1}(2 \sin 15^\circ)$  18. Ans. (i)  $\lambda_0 = 600 \text{ nm}$ ,  $n = 1.5$  (ii)  $i = \sin^{-1}(0.75) = 48.59^\circ$ 19. Ans.  $\left(\frac{8}{3} \text{ mm}\right)$ 

20. Ans. 2

21. Ans. 5cm

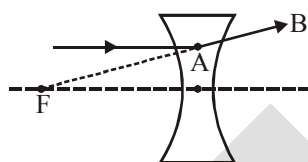
22. Ans. 8

23. Ans.  $f = v = \frac{\mu_3 R}{\mu_3 - \mu_1}$ 

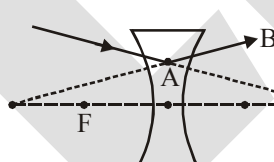
24. Ans. 30

25. Ans. Real, below principal axis, anywhere b/w P &  $\infty$ 

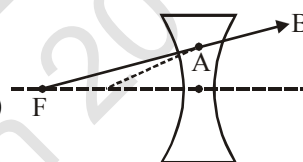
26. Ans. (1)



(2)



(3)



27. Ans. (i) 0.2 m, (ii) 0.4 m

28. Ans.  $(\pi/4) \text{ cm}^2$ 

29. Ans. 3

30. Ans. 5

31. Ans. (-20 cm)

32. Ans. 17.5 cm

33. Ans. 3

34. Ans.  $y = -\frac{x}{1200} + 0.1$ 

35. Ans. 1

36. Ans. 4

37. Ans. (a)  $f = -20 \text{ cm}$ , (b)  $+80 \text{ cm}$ , convergent achromatic lens]

## EXERCISE (S-2)

1. Ans. 2 mm

2. Ans. (30 cm, -14 mm)

3. Ans.  $\frac{32}{9}$ 

4. Ans. 0180

5. Ans. 2

6. Ans.  $\vec{r} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{2\sqrt{2}}{5}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$  (angle of incidence  $= 60^\circ$ ;  $r = 45^\circ$ )7. Ans.  $\frac{1.514 \times 0.4}{0.1} = 6.06 \text{ m}$  correct upto two places of decimal.

8. Ans. On the object itself

9. Ans. (i)  $\tan \theta = \frac{dy}{dx} = \cot i$ ; (ii) 1; (iii)  $y = k^2(x/4)^4$ ; (iv) 4.0, 1; (v) It will become parallel to x-axis

10. Ans.  $\frac{R}{\sqrt{3}}$

11. Ans. (a)  $\frac{2}{\sqrt{3}}$  (b) Normal to surface (c) Retrace the path,  $\sqrt{3}z + x = 10$

12. Ans. 0.09 m/s; Magnitude of the rate of change of lateral magnification is  $0.3 \text{ s}^{-1}$ .

13. Ans. 15 cm right of the mirror, magnification = 1.5; (-15 cm, -1.5 cm; -15 cm, 0.3 cm)

14. Ans.  $\frac{64}{3} \text{ cm}$

15. Ans. (i) -10 cm (ii) (10, 2)

16. Ans. (i)  $2^\circ$ , (ii)  $\frac{4\pi}{9} \text{ mm}$

17. Ans.  $\left(\frac{10}{3} \text{ m}\right)$

### EXERCISE (O-1)

- |                                                                                                                |                 |                 |                  |               |              |
|----------------------------------------------------------------------------------------------------------------|-----------------|-----------------|------------------|---------------|--------------|
| 1. Ans. (A)                                                                                                    | 2. Ans. (A)     | 3. Ans. (B)     | 4. Ans. (C)      | 5. Ans. (A)   | 6. Ans. (D)  |
| 7. Ans. (D)                                                                                                    | 8. Ans. (D)     | 9. Ans. (B)     | 10. Ans. (B)     | 11. Ans. (C)  | 12. Ans. (D) |
| 13. Ans. (C)                                                                                                   | 14. Ans. (C)    | 15. Ans. (C)    | 16. Ans. (D)     | 17. Ans. (B)  | 18. Ans. (B) |
| 19. Ans. (A)                                                                                                   | 20. Ans. (B)    | 21. Ans. (A)    | 22. Ans. (B)     | 23. Ans. (D)  | 24. Ans. (C) |
| 25. Ans. (B)                                                                                                   | 26. Ans. (D)    | 27. Ans. (B)    | 28. Ans. (C)     | 29. Ans. (B)  | 30. Ans. (B) |
| 31. Ans. (C)                                                                                                   | 32. Ans. (B)    | 33. Ans. (B)    | 34. Ans. (C)     | 35. Ans. (A)  | 36. Ans. (A) |
| 37. Ans. (B)                                                                                                   | 38. Ans. (D)    | 39. Ans. (D)    | 40. Ans. (A)     | 41. Ans. (D)  | 42. Ans. (C) |
| 43. Ans. (A)                                                                                                   | 44. Ans. (C)    | 45. Ans. (D)    | 46. Ans. (C)     | 47. Ans. (B)  | 48. Ans. (A) |
| 49. Ans. (C)                                                                                                   | 50. Ans. (B)    | 51. Ans. (C)    | 52. Ans. (A)     | 53. Ans. (B)  | 54. Ans. (A) |
| 55. Ans. (A)                                                                                                   | 56. Ans. (C)    | 57. Ans. (B)    | 58. Ans. (C)     | 59. Ans. (C)  | 60. Ans. (A) |
| 61. Ans. (C)                                                                                                   | 62. Ans. (A)    | 63. Ans. (D)    | 64. Ans. (B)     | 65. Ans. (B)  | 66. Ans. (A) |
| 67. Ans. (A,C,D)                                                                                               | 68. Ans. (B, C) | 69. Ans. (A,D)  | 70. Ans. (A,B,C) |               |              |
| 71. Ans. (B,C)                                                                                                 | 72. Ans. (B,C)  | 73. Ans. (B,D)  | 74. Ans. (A,C)   |               |              |
| 75. Ans. (B,C,D)                                                                                               | 76. Ans. (A,C)  | 77. Ans. (B,C)  | 78. Ans. (A,B)   |               |              |
| 79. Ans. (A,B,C,D)                                                                                             | 80. Ans. (A,D)  | 81. Ans. (A, C) | 82. Ans. (A, D)  |               |              |
| 83. Ans. (B)                                                                                                   | 84. Ans. (B)    | 85. Ans. (C)    | 86. Ans. (B)     |               |              |
| 87. Ans. (D)                                                                                                   | 88. Ans. (D)    | 89. Ans. (D)    | 90. Ans. (C)     |               |              |
| 91. Ans. (A) $\rightarrow$ (Q), (B) $\rightarrow$ (P), (C) $\rightarrow$ (S), (D) $\rightarrow$ (R)            |                 |                 |                  |               |              |
| 92. Ans. (A) $\rightarrow$ (P,S,T); (B) $\rightarrow$ (Q,S); (C) $\rightarrow$ (Q, R) (D) $\rightarrow$ (Q, R) |                 |                 |                  |               |              |
| 93. Ans. (A)-Q; (B)-P; (C)-S; (D)-R                                                                            |                 |                 |                  |               |              |
| 94. Ans. (A)-P; (B)-Q; (C)-RT; (D)-PT                                                                          |                 |                 |                  |               |              |
| 95. Ans. (B)                                                                                                   |                 |                 |                  |               |              |
| 96. Ans. (A)-Q,S; (B)-P,Q,R, S; (C)-P,Q,S (D)-S                                                                |                 |                 |                  |               |              |
| 97. Ans. (A)                                                                                                   | 98. Ans. (A)    | 99. Ans. (C)    | 100. Ans. (B)    | 101. Ans. (A) |              |
| 102. Ans. (C)                                                                                                  | 103. Ans. (A)   | 104. Ans. (A)   | 105. Ans. (B)    | 106. Ans. (A) |              |
| 107. Ans. (C)                                                                                                  | 108. Ans. (D)   | 109. Ans. (A)   | 110. Ans. (B)    | 111. Ans. (A) |              |
| 112. Ans. (C)                                                                                                  | 113. Ans. (D)   |                 |                  |               |              |

**EXERCISE (O-2)**

- |                |                |                  |                  |                  |                    |
|----------------|----------------|------------------|------------------|------------------|--------------------|
| 1. Ans. (A)    | 2. Ans. (C)    | 3. Ans. (D)      | 4. Ans. (A)      | 5. Ans. (D)      | 6. Ans. (C)        |
| 7. Ans. (D)    | 8. Ans. (A)    | 9. Ans. (C)      | 10. Ans. (A)     | 11. Ans. (D)     | 12. Ans. (A)       |
| 13. Ans. (A)   | 14. Ans. (D)   | 15. Ans. (A,D)   | 16. Ans. (A,B,C) | 17. Ans. (A,D)   |                    |
| 18. Ans. (B,D) | 19. Ans. (A,C) | 20. Ans. (B,C,D) | 21. Ans. (A, B)  | 22. Ans. (A,B,C) | 23. Ans. (A,B,C,D) |
| 24. Ans. (A)   | 25. Ans. (B)   |                  |                  |                  |                    |

**EXERCISE - (JM)**

- |              |              |             |              |              |              |
|--------------|--------------|-------------|--------------|--------------|--------------|
| 1. Ans. (3)  | 2. Ans. (2)  | 3. Ans. (4) | 4. Ans. (2)  | 5. Ans. (3)  | 6. Ans. (1)  |
| 7. Ans. (3)  | 8. Ans. (3)  | 9. Ans. (4) | 10. Ans. (4) | 11. Ans. (3) | 12. Ans. (1) |
| 13. Ans. (2) | 14. Ans. (3) |             |              |              |              |

**EXERCISE-(JA)**

- |                               |                                              |                                                             |
|-------------------------------|----------------------------------------------|-------------------------------------------------------------|
| 1. Ans. (A)                   | 2. Ans. (B)                                  | 3. Ans. (A) p, q, r, s (B) q, (C) p, q, r, s (D) p, q, r, s |
| 4. Ans. (C)                   | 5. Ans. (C, D)                               | 6. Ans. (B)                                                 |
| 7. Ans. (A,B,C)               | 8. Ans. (A) -pr, (B) -qst, (C) -prt, (D) -qs | 9. Ans. 6                                                   |
| 10. Ans. 3                    | 11. Ans. 6                                   | 12. Ans. (C)                                                |
| 13. Ans. 2                    | 14. Ans. (B)                                 | 15. Ans. (C)                                                |
| 16. Ans. (B)                  | 17. Ans. (A)                                 | 18. Ans. (C)                                                |
| 19. Ans. (D)                  | 20. Ans. (A,C)                               | 21. Ans. (C)                                                |
| 22. Ans. (B)                  | 23. Ans. 7                                   | 24. Ans. (B)                                                |
| 25. Ans. 2                    | 26. Ans. (A,C)                               | 27. Ans. (D)                                                |
| 28. Ans. (A)                  | 29. Ans. (A,B,D)                             | 30. Ans. (A,D)                                              |
| 31. Ans. (A)                  | 32. Ans. (A,C,D)                             | 33. Ans. 8                                                  |
| 34. Ans. 130 [129.95, 130.05] | 35. Ans. (D)                                 |                                                             |